

Biobuz lessons

Release

Catholic Central Spires Robotics

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This guide walks you through making the robot do things, one task at a time. Each task page says what you will have when it is done, what has to be true before you start, and the steps to get there. Every step tells you what you will see when it worked and what to try when it did not.

How to use it

- **Start with the task you need, not with page one.** Every task lists what it needs finished first. If you missed a session, find the task you are up to and read its *Before you start*.
- **Do the step, then read the check.** The check tells you whether to go on. If what you see is not what the check describes, the *if it didn't* line under it is the first thing to try.
- **The answers are in the back.** They are the real code from the solutions branch. Use them when you are stuck, not instead of being stuck.

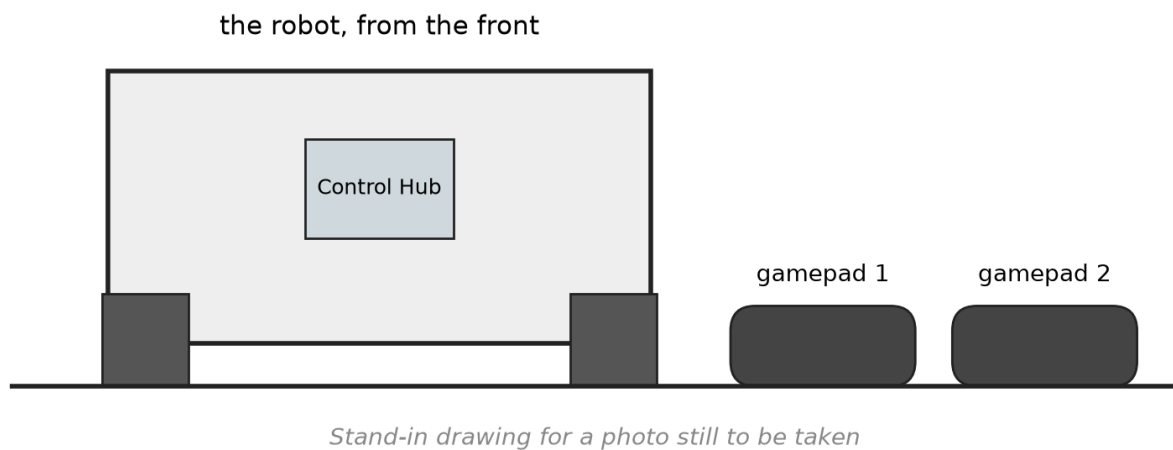


Figure 1.1. The robot from the front, on the floor, with both gamepads beside it, so a student can match what is in front of them to what the guide calls each part.

The tasks

Getting the robot and the laptop ready, L010, has no page yet.

2.1 L020: read the sticks

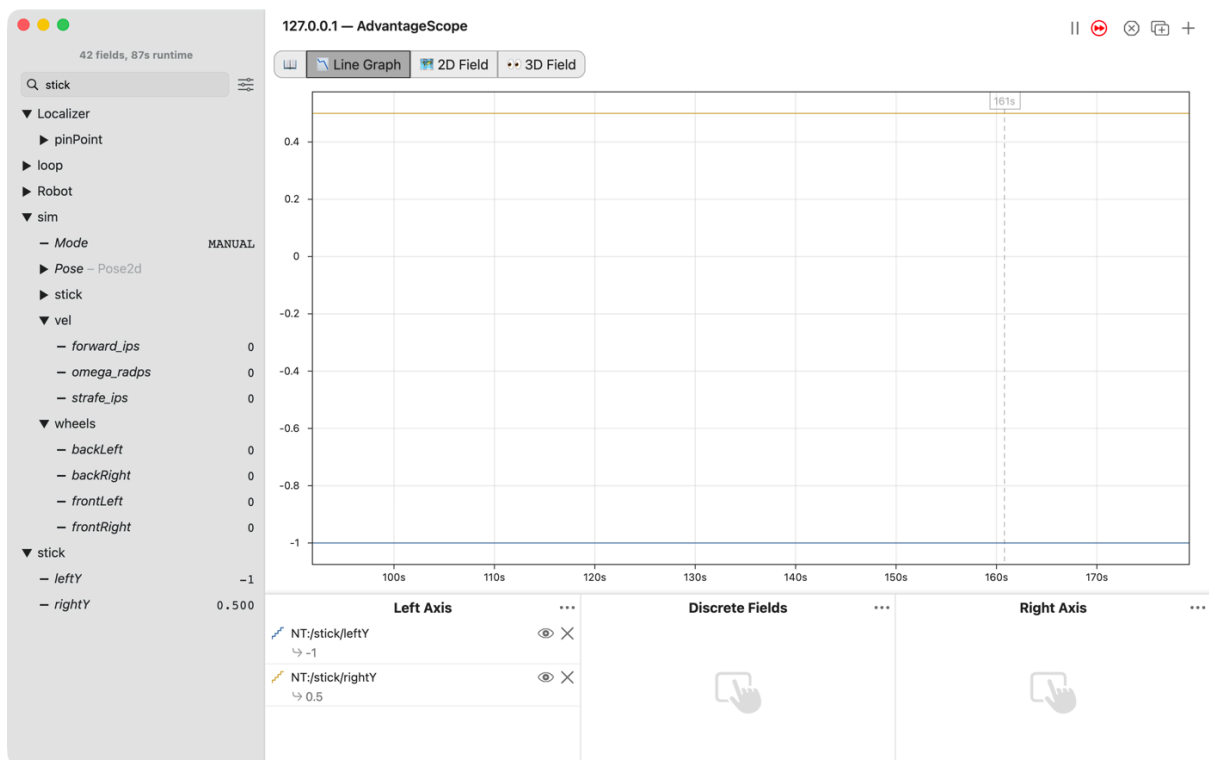


Figure 2.1. AdvantageScope's Line Graph connected to the simulator, with NT:/stick/leftY drawn at -1 and NT:/stick/rightY drawn at 0.5, and both names in Left Axis.

Push the gamepad's sticks and watch the numbers they send, live on the laptop. Nothing on the robot moves yet.

2.1.1 The steps

L020S010: copy the starter into mytry

Copy. Open `TeamCode/java/org.firstinspires.ftc.teamcode/lessons`. Scroll to the bottom of it, click `L020SticksOpMode` once, and copy it: ⌘C on a Mac, Ctrl+C on Windows.

Click `org.firstinspires.ftc.teamcode` once, and paste.

A box called **Copy Class** opens. Click at the end of **Destination package**, type `.mytry`, and click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

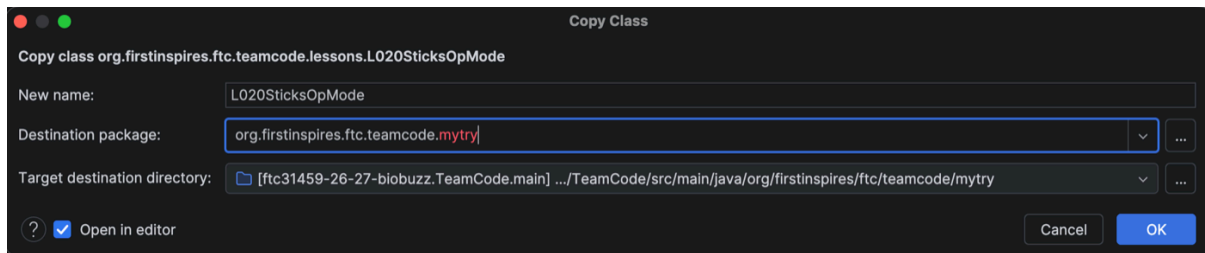


Figure 2.2. The Copy Class box with New name `L020SticksOpMode` and Destination package `org.firstinspires.ftc.teamcode.mytry`, with `mytry` in red.

Check. A new `mytry` folder holds `L020SticksOpMode`, the file is open, and its first line reads `package org.firstinspires.ftc.teamcode.mytry;`.

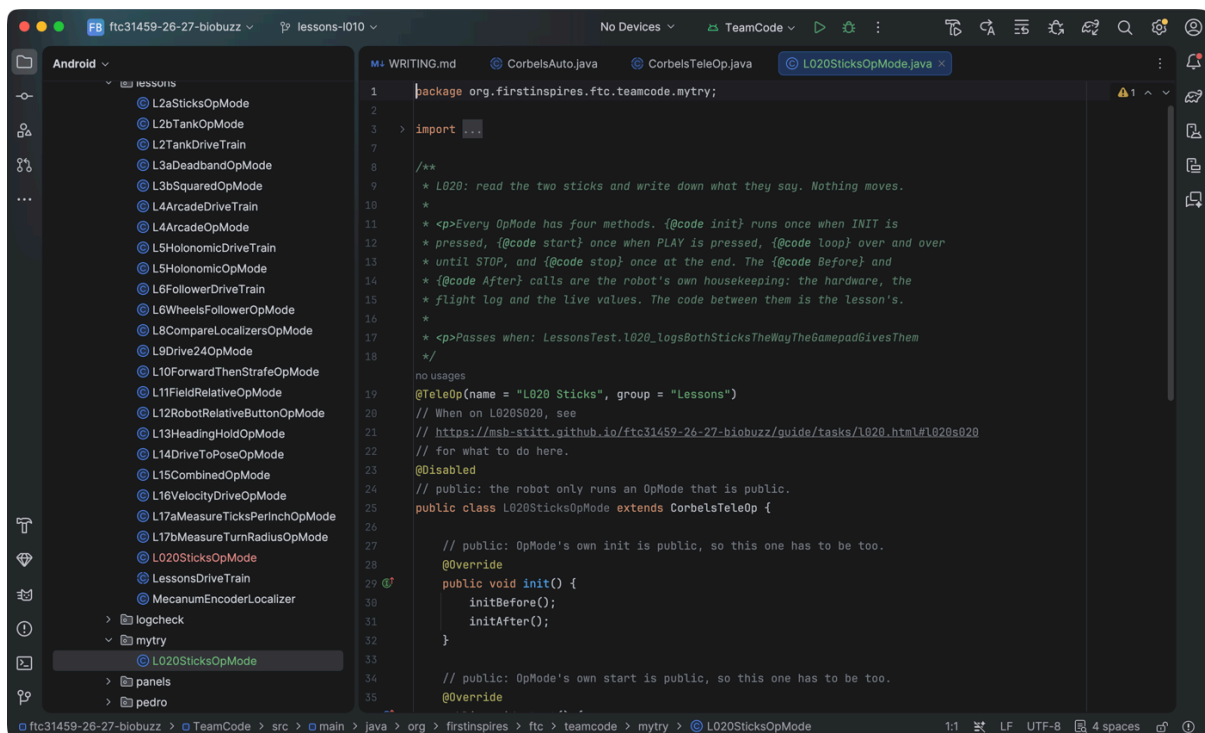


Figure 2.3. The Project window with `lessons` and the new `mytry` folder both open, `L020SticksOpMode` in each, and the copy open in the editor with its first line ending in `mytry;`.

L020S020: switch your copy on

Change. Find the line `@Disabled`, just under the comment that names this step. Click on it and delete the whole line: `⌘?` on a Mac, `Ctrl+Y` on Windows.

`@Disabled` hides an `OpMode` from the robot's list. The starter in `lessons` keeps it, so the robot only ever lists your copy.

Check. Under the step's comment, the next lines read `// public: the robot only runs an OpMode that is public.` and `public class L020SticksOpMode`.

L020S030: read the left stick

Change. Scroll down to `loop()` and find the comment that names this step. Click at the end of its last line and press Enter. Type this much:

```
double leftStickY = gamepad1.
```

A list opens with everything a gamepad has. Type `left_st` and the list shrinks to the left stick. Move the highlight to `left_stick_y` with the arrow keys, then press Tab. Type `;` to end the line.

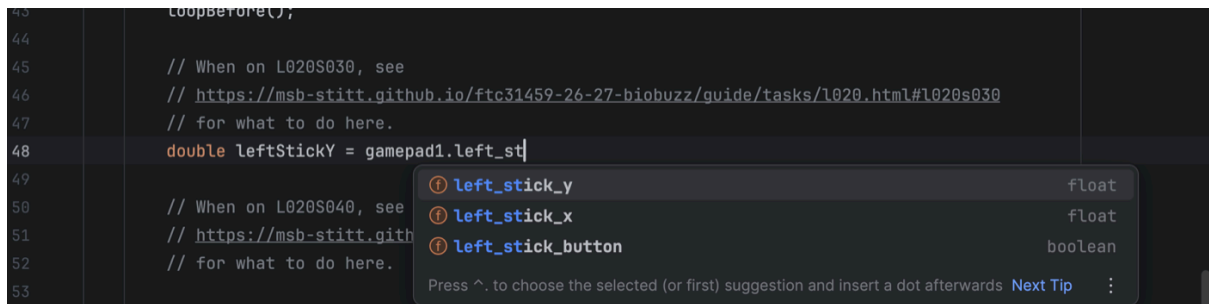


Figure 2.4. The list under `gamepad1.` after typing `left_st`, with `left_stick_y` highlighted and marked float.

```
// When on L020S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/1020.html#1020s030
// for what to do here.
double leftStickY = gamepad1.left_stick_y;
```

`double` is Java's word for a number with a decimal point, like 0.5 or -1. `leftStickY` is a name you chose for one. The list said `float`, which is a smaller kind of decimal number. A `double` holds any `float`.

Now hold the pointer still over `left_stick_y`. A box opens and says what it is: *left analog stick vertical axis*. You can do this with any name you are not sure about.

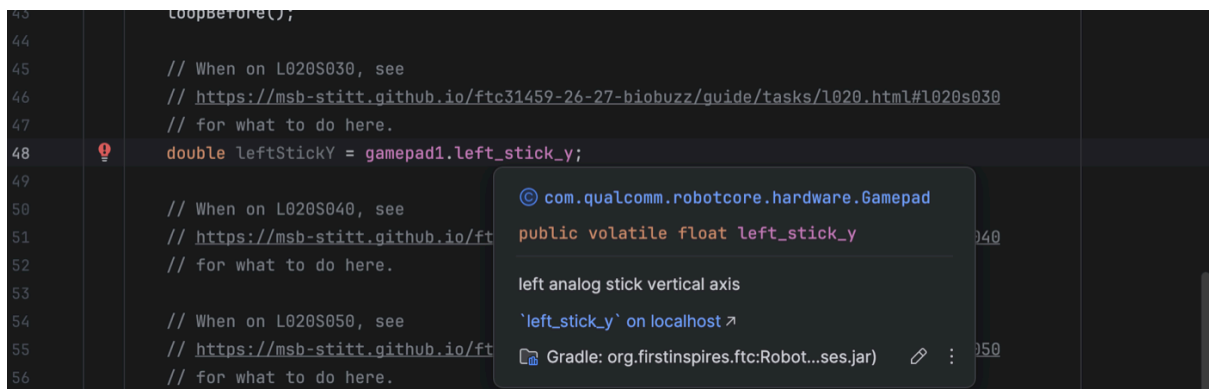
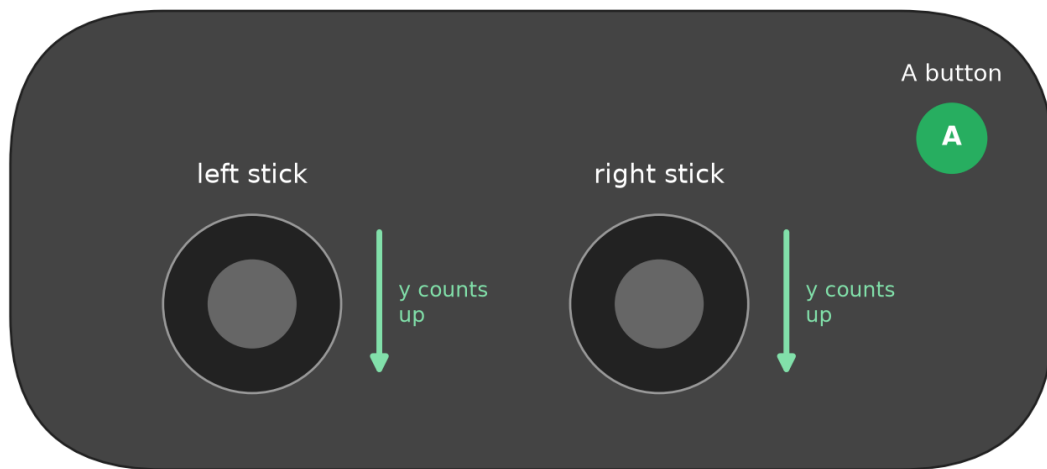


Figure 2.5. The box over `left_stick_y` reading `public volatile float left_stick_y` and *left analog stick vertical axis*.

a stick pushed away from the driver reads -1, pulled back +1



the driver holds it from this side

Stand-in drawing for a photo still to be taken

Figure 2.6. Gamepad 1 from above, with the left stick, the right stick and the A button labelled, and an arrow showing which way each y axis counts up.

Check. The line has no red underline.

L020S040: write the left stick down

Change. Under the comment that names this step, type `Tracker`. The list shows more than one `Tracker`. Pick the one from `org.firstinspires.ftc.teamcode.base` and press `Tab`. Then finish the line:

```
// When on L020S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1020.html#l020s040
// for what to do here.
Tracker.publish("stick/leftY", leftStickY);
```

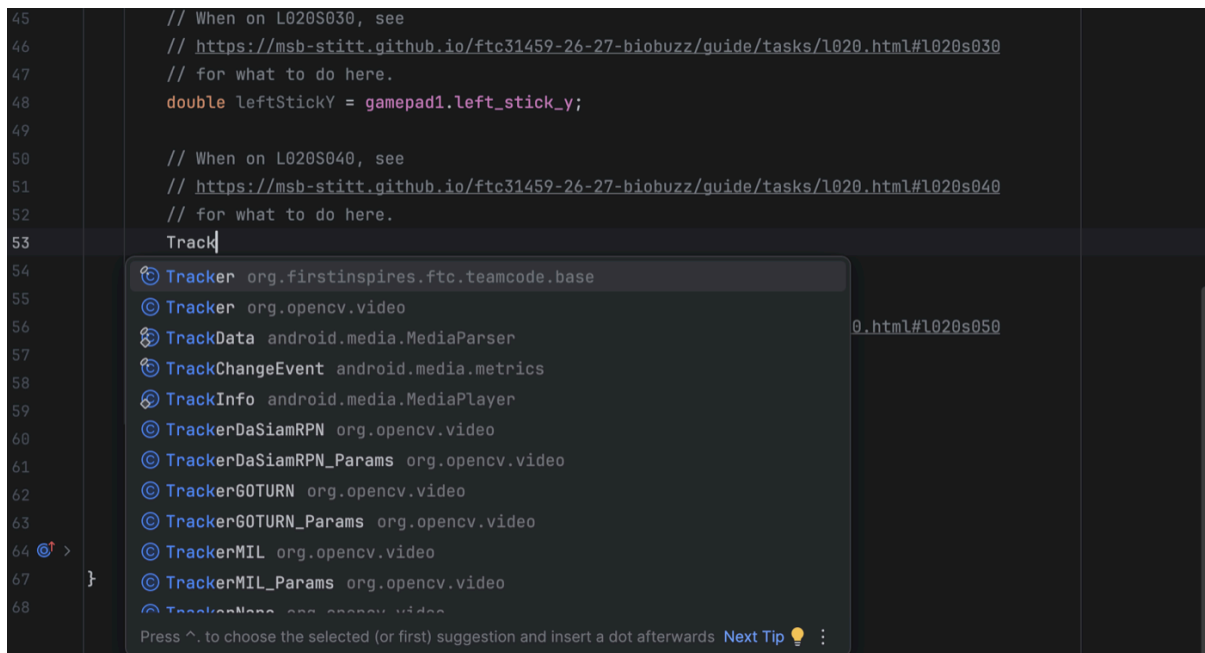


Figure 2.7. The list under `Tracker`, with `Tracker` from `org.firstinspires.ftc.teamcode.base` highlighted at the top and `Tracker` from `org.opencv.video` under it.

`Tracker.publish` takes a name and a value. It sends both to `AdvantageScope` while the code runs, and writes them in the flight log. Android Studio added an `import` line for `Tracker` near the top of the file.

Check. `Tracker` is not red, and there is an `import org.firstinspires.ftc.teamcode.base.Tracker;` line near the top.

Simulator. Click the **Terminal** button near the bottom of the left edge. Type this and press Enter:

```
./gradlew :TeamCode:simRun --args="mytry.L020SticksOpMode --pad"
```

Check. It prints `Flight log:` and a folder, then `mytry.L020SticksOpMode running.`, and stays running. The folder is the project's top folder, the one that holds `README.md`.

AdvantageScope. Open **File > Connect to Simulator > NetworkTables 4**, and click **Line Graph**. Type `stick` in the search box at the top left, and click `NT:/stick/leftY`. Drag `leftY` from the `stick` folder down into **Left Axis**.

Before you push the stick, guess. When you push the left stick all the way away from you, will the line go to 1 or to -1? Now push it and see.

Check. The line moves when you push. All the way forward it reads -1, and all the way back it reads 1.

L020S050: the right stick too

Change. Click in the Terminal and press `Ctrl+C` to stop the simulator. Under the comment that names this step, read and write down the right stick the same way:

```
// When on L020S050, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1020.html#1020s050
// for what to do here.
double rightStickY = gamepad1.right_stick_y;
Tracker.publish("stick/rightY", rightStickY);
```

Simulator. Run the same command again.

AdvantageScope. Drag `rightY` from the `stick` folder into **Left Axis** beside `leftY`.

Check. Each line follows its own stick. Push each stick all the way both ways. The numbers never go past 1 or below -1, and forward is -1 on both.

Log. Press Ctrl+C in the Terminal to stop the simulator. In AdvantageScope, open **File > Open Log(s)...**, go to the project's top folder, and open the newest file whose name starts `L020SticksOpMode`. Drag `leftY` and `rightY` from `stick` into **Left Axis**.

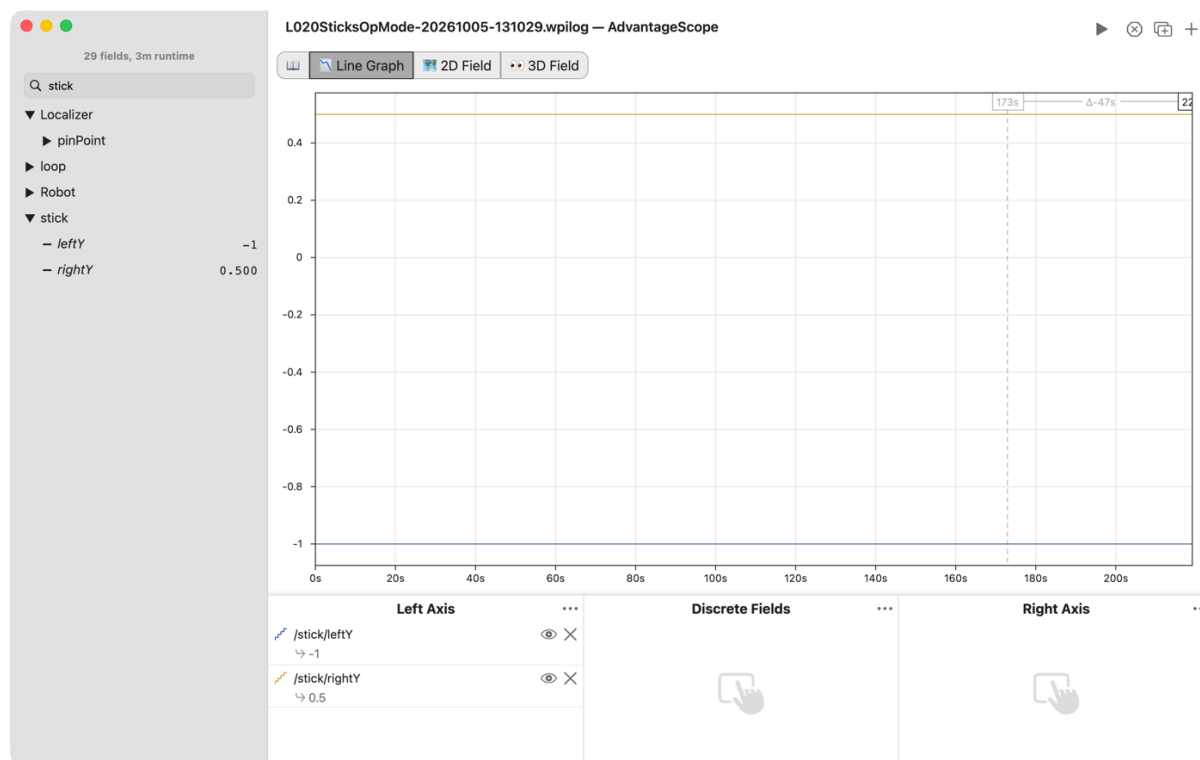


Figure 2.8. AdvantageScope's Line Graph of the flight log `L020SticksOpMode-...wpilog`, with `/stick/leftY` at -1 and `/stick/rightY` at 0.5 for the whole run.

Check. The lines show what the sticks did during the run. It is the same `stick/leftY` and `stick/rightY` you watched live, kept in a file you can open again later.

2.2 L030: turn the sticks into speeds

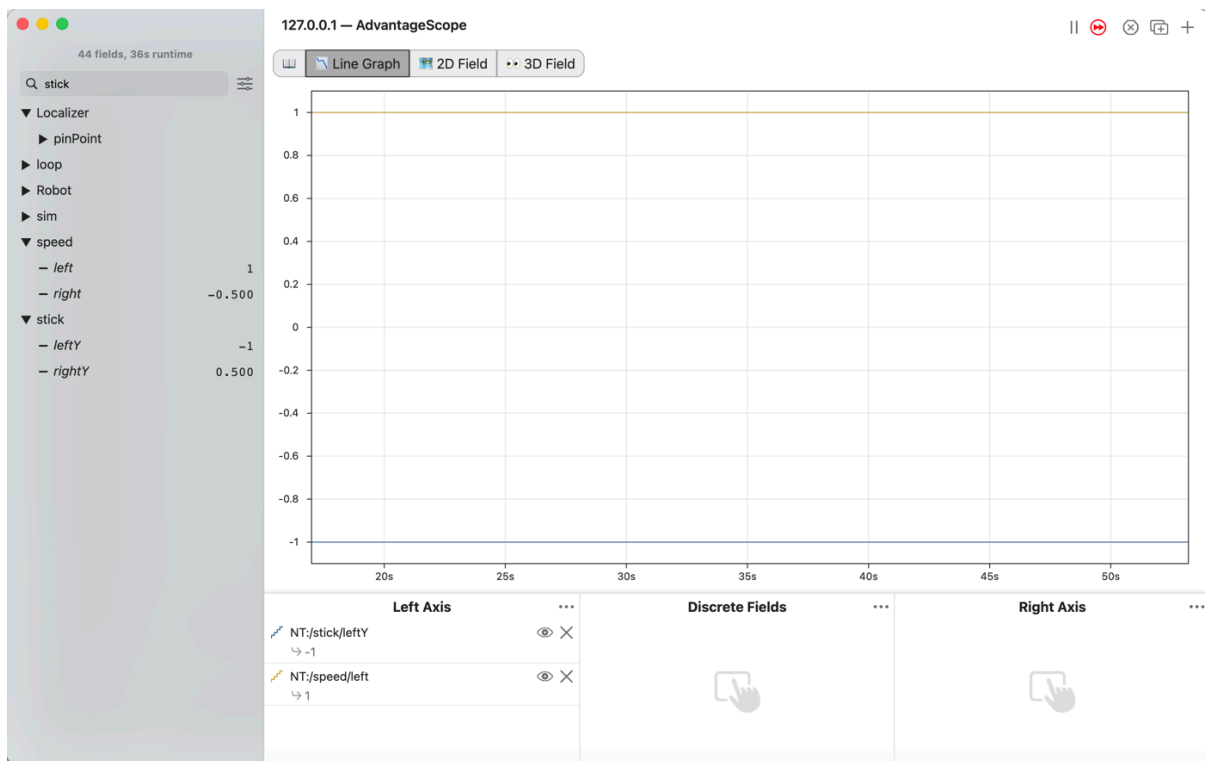


Figure 2.9. AdvantageScope’s Line Graph connected to the simulator, with NT:/stick/leftY drawn at -1 and NT:/speed/left drawn at 1, and speed/right reading -0.500 beside stick/rightY at 0.500 in the list.

Turn each stick into a wheel speed, where forward is positive. Nothing on the robot moves yet.

2.2.1 The steps

L030S010: copy the starter into mytry

Copy. Open `TeamCode/java/org.firstinspires.ftc.teamcode/lessons`. Click `L030SpeedsOpMode` once, and copy it: `⌘C` on a Mac, `Ctrl+C` on Windows. Click the `mytry` folder once, and paste.

The **Copy Class** box opens with **Destination package** already ending in `mytry`. Click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check. `mytry` holds `L020SticksOpMode` and `L030SpeedsOpMode`, and the new file is open.

L030S020: switch your copy on

Change. Find the line `@Disabled`, just under the comment that names this step, and delete the whole line, the way you did in L020.

Check. Under the step’s comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L030S030: the left speed

We are working toward tank drive, where the left stick drives the left wheels and the right stick drives the right wheels. Push a stick away from you and its wheels should turn forward. We call forward speeds positive and backward speeds negative. But in L020 a stick pushed away read -1, so its number

has to be turned round to make a speed.

The starter already has the four lines you wrote in L020. They read the sticks and write them down.

Change. Under the comment that names this step, type `double leftSpeed = -leftS`. The list offers `leftStickY`, the name from the lines above. Press Tab, then finish the two lines:

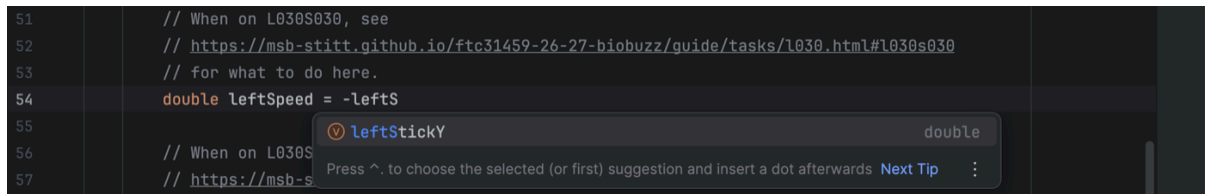


Figure 2.10. The list under `-leftS`, offering the variable `leftStickY`, marked `double`.

```
// When on L030S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/1030.html#1030s030
// for what to do here.
double leftSpeed = -leftStickY;
Tracker.publish("speed/left", leftSpeed);
```

The minus sign turns the number round: -1 becomes 1, and 0.5 becomes -0.5. `leftStickY` is still what the stick says, and `leftSpeed` is how fast the wheels should go. Keeping two names for the two ideas means each name always says what it holds.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L030SpeedsOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020. Type `stick` in the search box. Drag `leftY` from the stick folder, and `left` from the speed folder, into **Left Axis**.

Check. Push the left stick away from you. `stick/leftY` goes to -1, and `speed/left` goes to 1. The two lines are always mirror images of each other.

L030S040: the right speed

Change. Press Ctrl+C in the Terminal to stop the simulator. Under the comment that names this step, do the same for the right stick:

```
// When on L030S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/1030.html#1030s040
// for what to do here.
double rightSpeed = -rightStickY;
Tracker.publish("speed/right", rightSpeed);
```

Simulator. Run the same command again.

AdvantageScope. Drag `rightY` from stick and `right` from speed into **Left Axis**.

Check. Push both sticks away from you. Both speeds read 1. Pull both back, and both read -1.

2.3 L040: drive like a tank

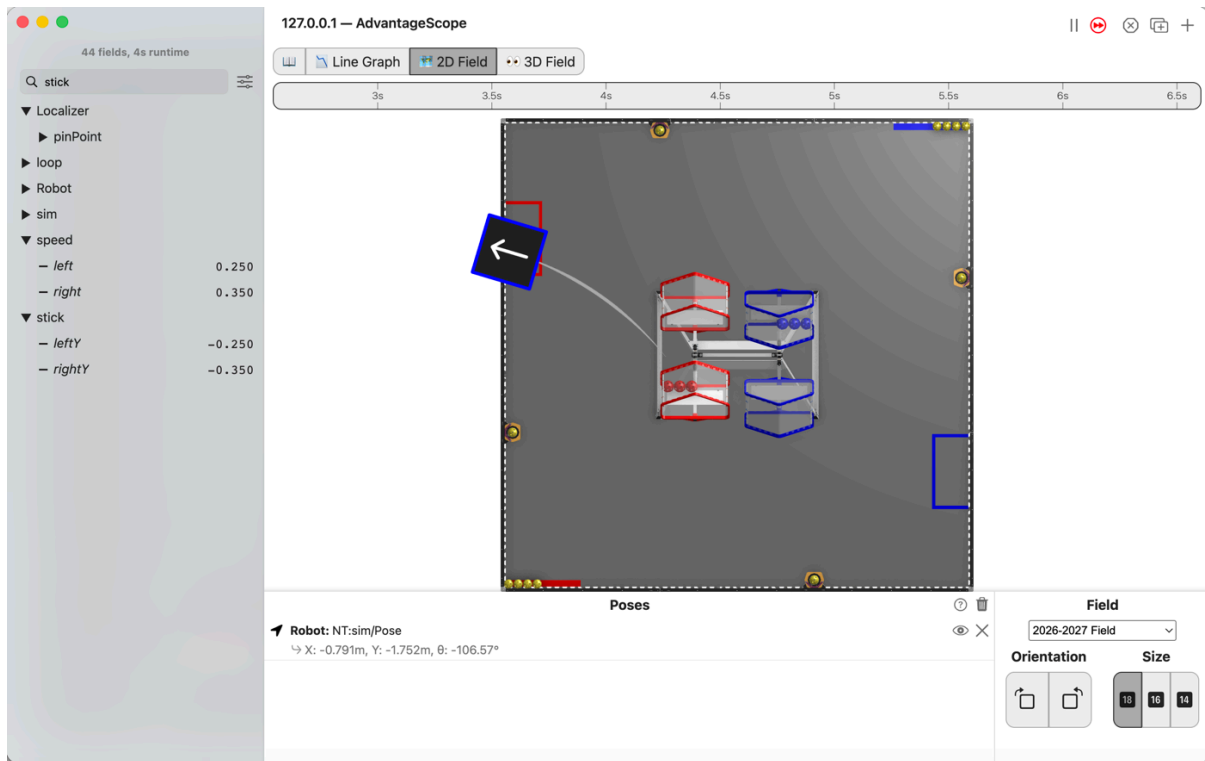


Figure 2.11. AdvantageScope's 2D Field connected to the simulator, with the robot NT:sim/Pose near the left wall after curving left from its start, speed/left reading 0.250 and speed/right 0.350.

The two speeds drive the wheels, and the robot drives in the simulator: tank drive.

2.3.1 The steps

L040S010: copy the starter into mytry

Copy. In lessons, click L040TankOpMode once and copy it. Click mytry once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check. mytry holds L040TankOpMode, and the new file is open.

L040S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check. Under the step's comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L040S030: the left wheels

The starter already reads the sticks and turns them into `leftSpeed` and `rightSpeed`, as you did in L030.

Change. The robot has four motors, one on each wheel. `hardware` holds all four. Under the comment that names this step, type `hardware.front`. The list offers `frontLeft` and `frontRight`. Pick `frontLeft` and press `Tab`.

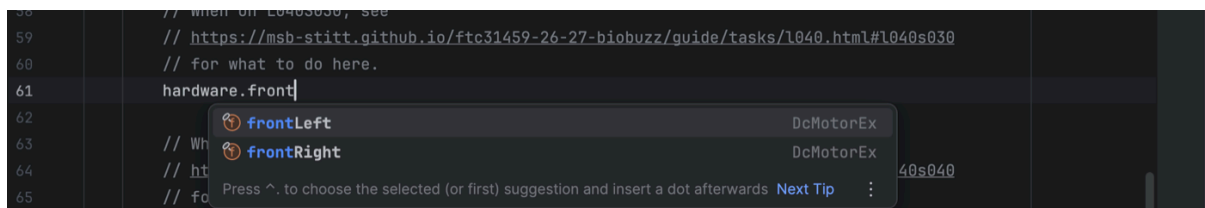


Figure 2.12. The list under `hardware.front`, offering `frontLeft` and `frontRight`, each marked `DcMotorEx`.

Type `.setP`. The list shows what a motor can be told to do. The one we want is `setPower` (double power).

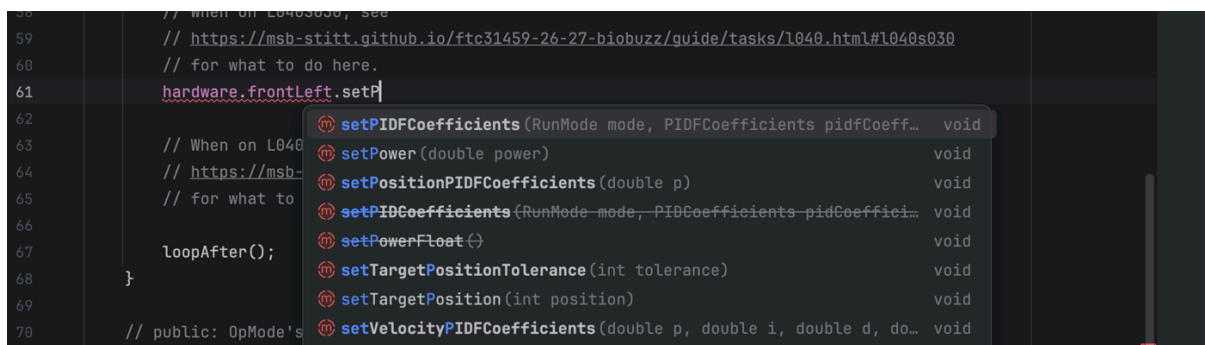


Figure 2.13. The list under `hardware.frontLeft.setP`, with `setPower` (double power) second.

Finish the line, and add the same for the back left wheel:

```
// When on L040S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1040.html#1040s030
// for what to do here.
hardware.frontLeft.setPower(leftSpeed);
hardware.backLeft.setPower(leftSpeed);
```

A power is a number from -1 to 1. It is how hard the motor pushes: 1 is full forward, -1 is full backward, and 0 is stop.

Check. Neither line has a red underline.

L040S040: the right wheels

Change. Under the comment that names this step, do the same for the right side:

```
// When on L040S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1040.html#1040s040
// for what to do here.
hardware.frontRight.setPower(rightSpeed);
hardware.backRight.setPower(rightSpeed);
```

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L040TankOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020, and click **2D Field**. Open the `sim` folder and drag `Pose` into **Poses**. The robot appears near the bottom edge of the field.

Check. Push both sticks gently forward, and the robot drives forward. Pull both back, and it comes back. Push one forward and pull the other back, and it spins in place. Push the right stick further than the left, and it curves to the left.

2.4 L050: drive the robot on blocks

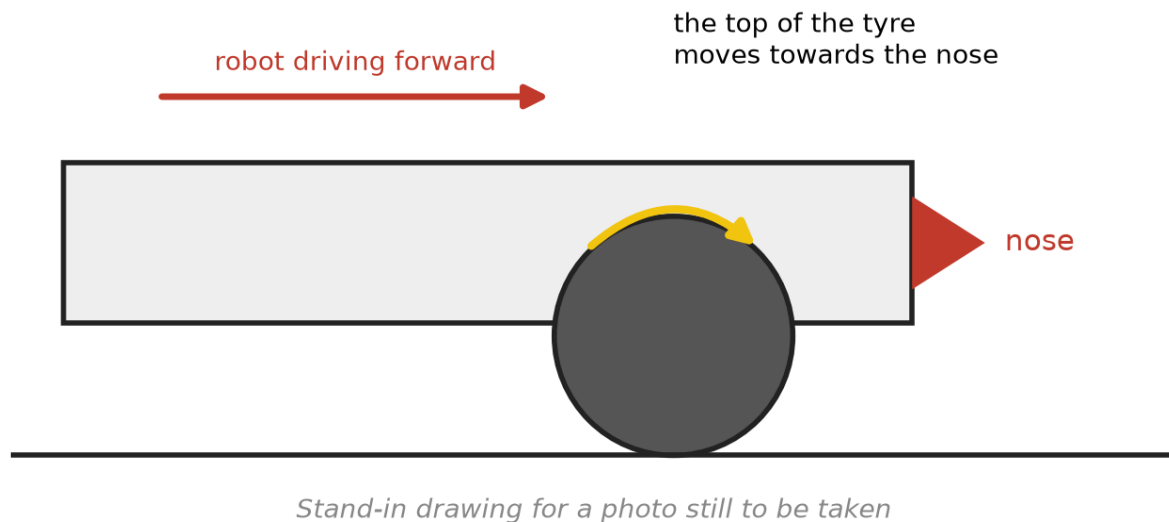


Figure 2.14. One wheel from the side, with an arrow on the top of the tyre showing which way it travels when that wheel is driving the robot forward.

Tell each motor which way it is mounted. Then run tank drive on the robot, up on blocks, and check that every wheel turns the way its stick says.

2.4.1 The steps

L050S010: copy the starter into mytry

Copy. In `lessons`, click `L050BlocksOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check. `mytry` holds `L050BlocksOpMode`, and the new file is open.

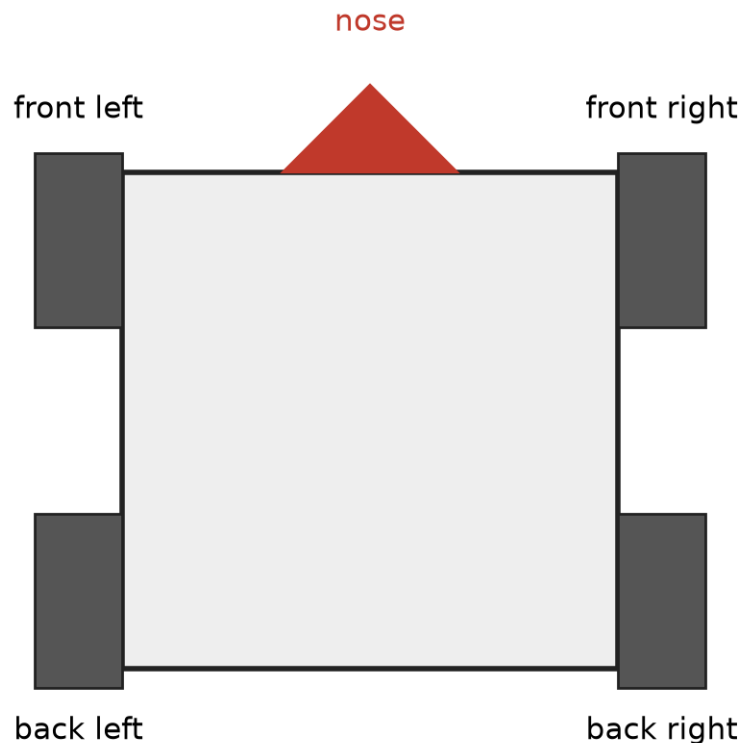
L050S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check. Under the step's comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L050S030: label the robot

Label. Put a strip of tape on the robot's front, and mark it **FRONT**. Mark the left side **LEFT** and the right side **RIGHT**, as the robot sees them when it drives forward.



Stand-in drawing for a photo still to be taken

Figure 2.15. The robot from above with its nose marked, and each of the four wheels labelled front left, front right, back left and back right.

Check. Stand behind the robot, facing the way FRONT faces. LEFT is on your left.

L050S040: tell each motor which way it is mounted

The two sides of the robot face opposite ways. So the same power turns the left motors one way and the right motors the other. In L040 the simulator did not care. A real robot does.

Constants.java already says which way each motor is mounted, and hardware.mecanumConfig holds what it says.

Change. Scroll up to `init()` and find the comment that names this step. Type `hardware.frontLeft.setD`, and pick `setDirection`. Inside the brackets, type `hardware.mecanumConfig.frontLeftD`. The list offers `frontLeftDirection.get()`, marked `Direction`. Pick it.



Figure 2.16. The list under `hardware.mecanumConfig.frontLeftD`, offering `frontLeftDirection` and `frontLeftDirection.get()`, the second marked `Direction`.

Finish the line, and do the same for the other three motors:

```
// When on L050S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
// /1050.html#1050s040
// for what to do here.

hardware.frontLeft.setDirection(hardware.mecanumConfig.frontLeftDirection.get());

hardware.frontRight.setDirection(hardware.mecanumConfig.frontRightDirection.get());
hardware.backLeft.setDirection(hardware.mecanumConfig.backLeftDirection.get());

hardware.backRight.setDirection(hardware.mecanumConfig.backRightDirection.get());
```

init runs once, when INIT is pressed, so the motors know their directions before PLAY.

Check. Each line ends `.get() ;`, and none has a red underline.

L050S050: send the code to the robot

Robot. Rest the robot on the blocks, so all four wheels are off the ground. Turn it on, and plug the laptop into the Control Hub with the USB cable.

Send. In Android Studio, the device list left of **TeamCode** reads **No Devices** until the robot is plugged in. Pick the Control Hub from it, and click the green triangle right of **TeamCode**.

Check. Android Studio says the install worked, and the Driver Station shows the robot.

L050S060: watch the wheels

Gamepad. On the gamepad, hold Start and press A. That makes it gamepad 1.

Run. On the Driver Station, pick L050 Blocks from the Lessons group. Press INIT, then PLAY.

Panels. Join the laptop to the robot's Wi-Fi, and open `http://192.168.43.1:8001`.

Check. Push the left stick a little way forward. In Panels, `speed/left` goes above 0. Both left wheels turn, with the top of each tyre moving toward FRONT. Let go, and do the same with the right stick and the right wheels.

Write down any wheel that turns the other way, or that turns for the wrong stick.

L050S070: fix a wheel that is wrong

Wrong way. A wheel that turns backward has the wrong direction in `TeamCode/src/main/java/org/firstinspires/ftc/teamcode/pedro/Constants.java`. Find its line, such as `c.frontLeftDirection.set(DcMotorSimple.Direction.FORWARD);`, and swap FORWARD for REVERSE, or REVERSE for FORWARD.

Wrong stick. A wheel that turns for the other side's stick is plugged into a port the Robot Configuration gives a different name. Ask a mentor to check the names on the Driver Station.

Send. Send the code again, as in L050S050, and repeat L050S060.

Check. All four wheels pass L050S060.

L050S080: drive it on the floor

Drive. Press STOP. Lift the robot off the blocks and put it on the floor, with room in front of it. Press INIT, then PLAY.

Check. Push both sticks gently forward, and the robot drives forward. Pull both back, and it comes back. Push one forward and pull the other back, and it spins in place.

2.5 L060: shape the sticks

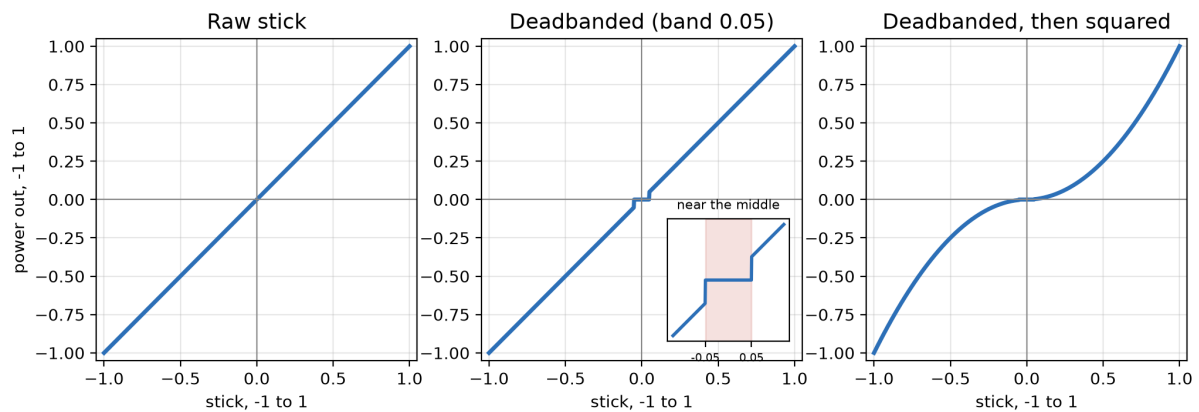


Figure 2.17. Three graphs side by side, each with the stick from -1 to 1 along the bottom and the power out from -1 to 1 up the side: the raw stick as a straight line, the deadbanded stick as a line with a flat step at the middle, and the squared stick as an S through the middle.

Make the robot sit still when the sticks are let go, and make a gentle push mean gently.

2.5.1 The steps

L060S010: copy the starter into mytry

Copy. In lessons, click L060ShapingOpMode once and copy it. Click mytry once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check. mytry holds L060ShapingOpMode, and the new file is open.

L060S020: switch your copy on

Change. Delete the line @Disabled, under the comment that names this step.

Check. Under the step's comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L060S030: write a deadband

A stick that has been let go does not always read 0. It can read a few hundredths, and a few hundredths of power is enough to make a robot crawl across the floor on its own. A **deadband** fixes that: any reading closer to the middle than the band counts as let go.

Change. Scroll to the bottom of the file and find the comment that names this step. Under it, write a method of your own:

```
// When on L060S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1060.html#1060s030
// for what to do here.
double deadband(double value, double band) {
    if (Math.abs(value) < band) {
        return 0;
    }
    return value;
}
```

A **method** is a named piece of code you can use again. This one takes two numbers, `value` and `band`,

and gives back one number. `return` is the number it gives back.

`Math.abs` takes the minus sign off: `Math.abs(-0.03)` is 0.03. A stick pulled a long way back reads -0.5, and -0.5 is less than 0.05. But it is far from the middle, so it has to be compared without its minus sign.

Check. The method has no red underline. Android Studio shows deadband in grey, with *no usages* above it, because nothing uses it yet.

L060S040: use the deadband

Change. In `loop()`, under the comment that names this step, make a power for each side:

```
// When on L060S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1060.html#1060s040
// for what to do here.
double band = 0.05;
double leftPower = deadband(leftSpeed, band);
Tracker.publish("power/left", leftPower);
double rightPower = deadband(rightSpeed, band);
Tracker.publish("power/right", rightPower);
```

Then change the four `setPower` lines below, so each one gets the power and not the speed:

```
hardware.frontLeft.setPower(leftPower);
hardware.backLeft.setPower(leftPower);

hardware.frontRight.setPower(rightPower);
hardware.backRight.setPower(rightPower);
```

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L060ShapingOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020. Drag `left` from the speed folder and `left` from the power folder into **Left Axis**.

Check. Let go of the sticks. `speed/left` may sit a little off 0, and `power/left` reads 0. The wheels in `sim/wheels` all read 0.

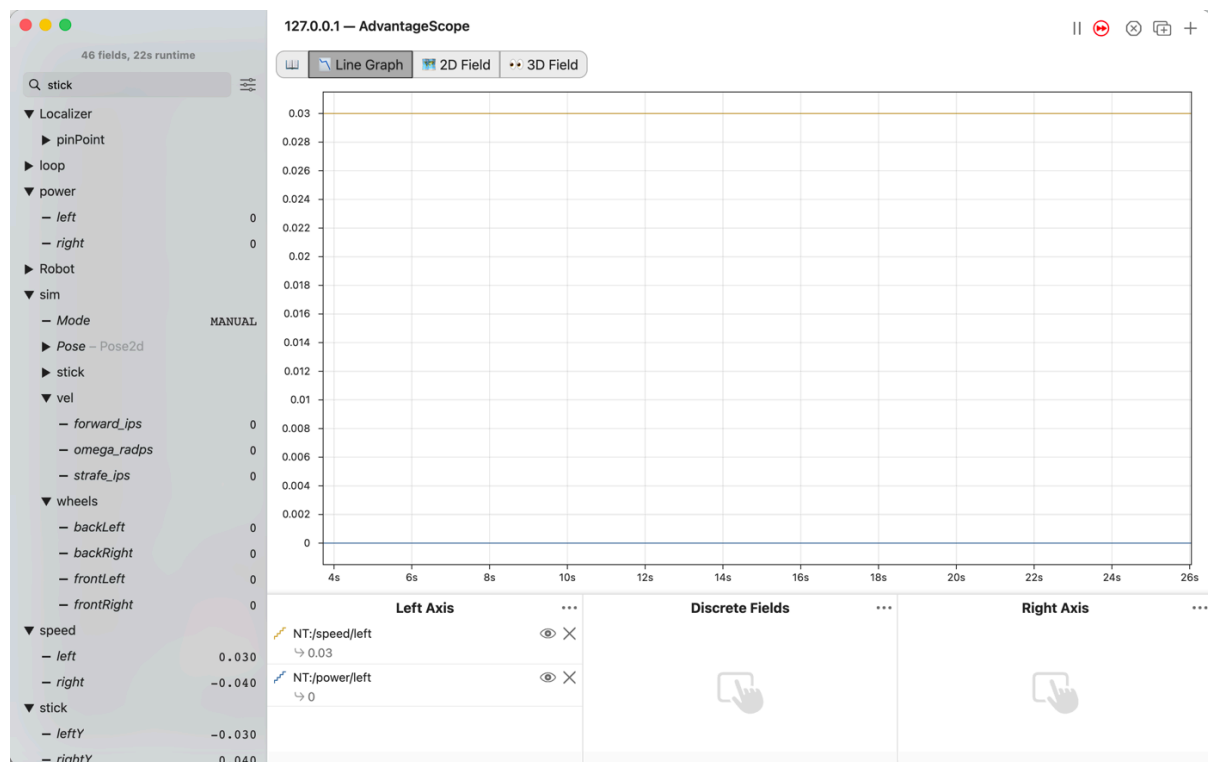


Figure 2.18. AdvantageScope’s Line Graph with NT : /speed/left at 0.03 and NT : /power/left at 0, and the list showing power/left and power/right at 0, all four sim/wheels at 0, and speed/right at -0.040.

L060S050: write squaring

With the deadband in, a tenth of a push is still a tenth of the power. Near the middle that is more robot than a driver wants. **Squaring** fixes it: half a push becomes a quarter of the power, and a full push is still full power.

Change. Press Ctrl+C in the Terminal to stop the simulator. Find the comment that names this step. Under it, write a second method:

```
// When on L060S050, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l060.html#l060s050
// for what to do here.
double squared(double value) {
    return value * Math.abs(value);
}
```

value * value would lose the minus sign: -0.5 times -0.5 is 0.25, and backward would turn into forward. value * Math.abs(value) keeps it: -0.5 times 0.5 is -0.25.

Check. The method has no red underline.

L060S060: use squaring

Change. Go back to the two deadband lines from L060S040. Put squared(in front of deadband, and one more) before the ;:

```
double leftPower = squared(deadband(leftSpeed, band));
```

```
double rightPower = squared(deadband(rightSpeed, band));
```

squared works on what deadband gives back.

Simulator. Run the same command again, and connect AdvantageScope.

Check. Push the left stick halfway forward. `speed/left` reads 0.5, and `power/left` reads 0.25. Push it all the way, and both read 1. Pull it back, and the power goes below 0.

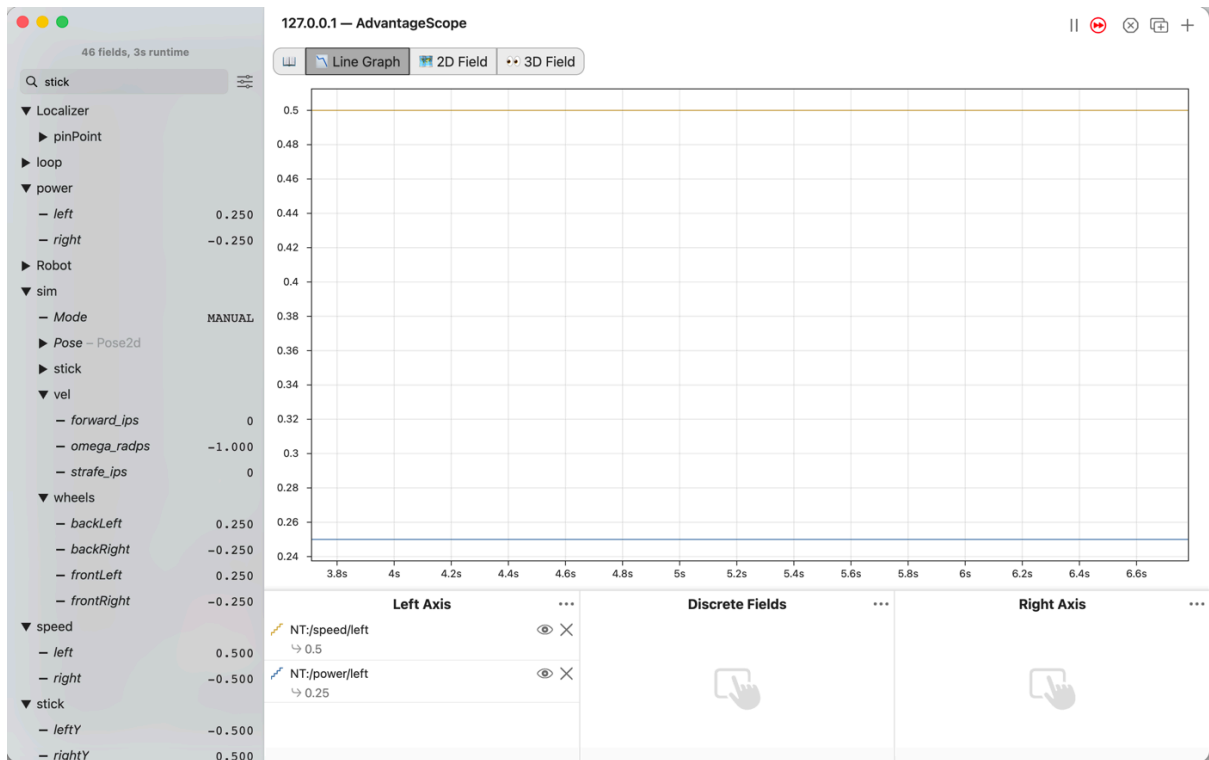


Figure 2.19. AdvantageScope's Line Graph with `NT:/speed/left` at 0.5 and `NT:/power/left` at 0.25, and the list showing `power/right` at -0.250 beside `speed/right` at -0.500.

L060S070: drive it on the robot

Send. Send the code to the robot, as in L050. On the Driver Station, pick L060 Shaping from the Lessons group. Press INIT, then PLAY.

Check. Let go of both sticks, and the robot sits still. Push a stick about a third of the way, and the robot creeps along under control. Push it all the way, and the robot goes full speed.

2.6 L070: read the buttons

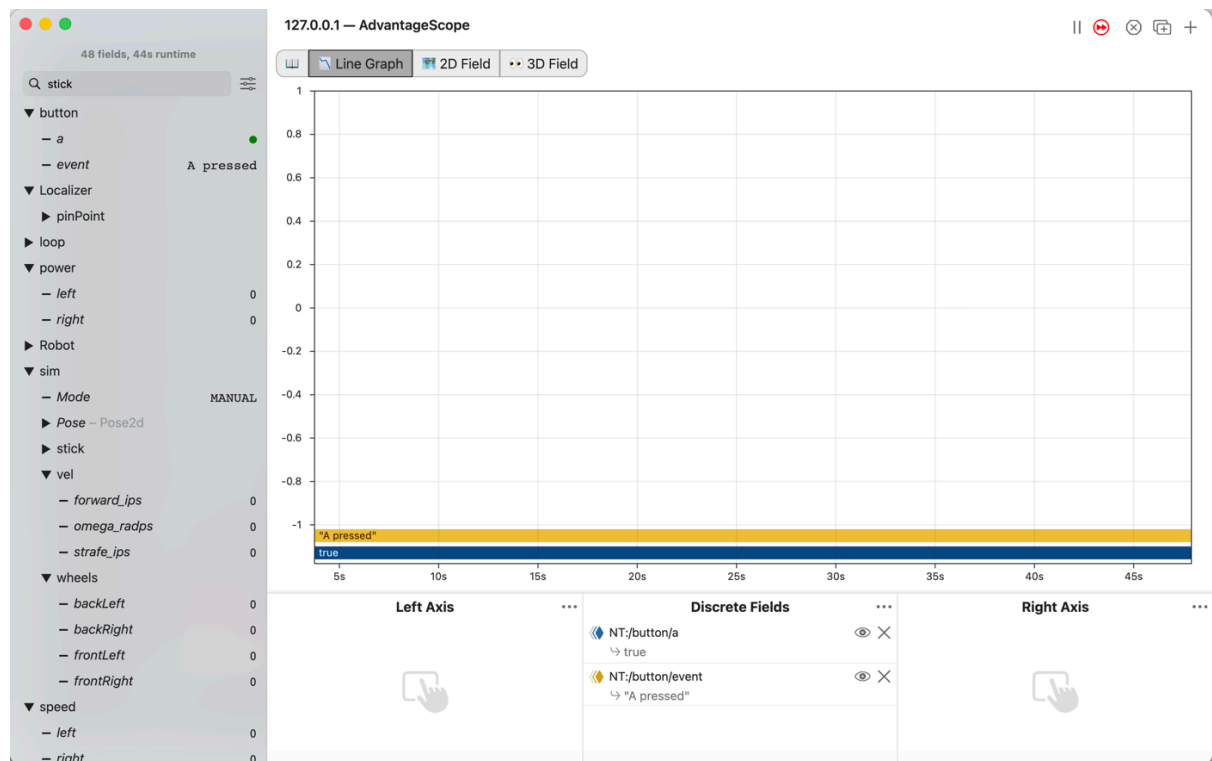


Figure 2.20. AdvantageScope’s Line Graph with NT : /button/a at true and NT : /button/event at “A pressed” in **Discrete Fields**, drawn as two bars along the bottom of the graph, and the list showing button/a as a green dot and button/event as A pressed.

Read the A button, say the moment it goes down and the moment it comes up, and read both sticks sideways. The robot drives as it did in L060 the whole time.

2.6.1 The steps

L070S010: copy the starter into mytry

Copy. In `lessons`, click `L070ButtonsOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check. `mytry` holds `L070ButtonsOpMode`, and the new file is open.

L070S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check. Under the step’s comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L070S030: read the A button

A stick is a number. A button is only down or up, so Java keeps it as a **boolean**: `true` when it is down, `false` when it is up.

Change. In `loop()`, find the comment that names this step, below the four `setPower` lines. Under it, type `boolean buttonA = gamepad1.a`. The list offers `a`, marked `boolean`.

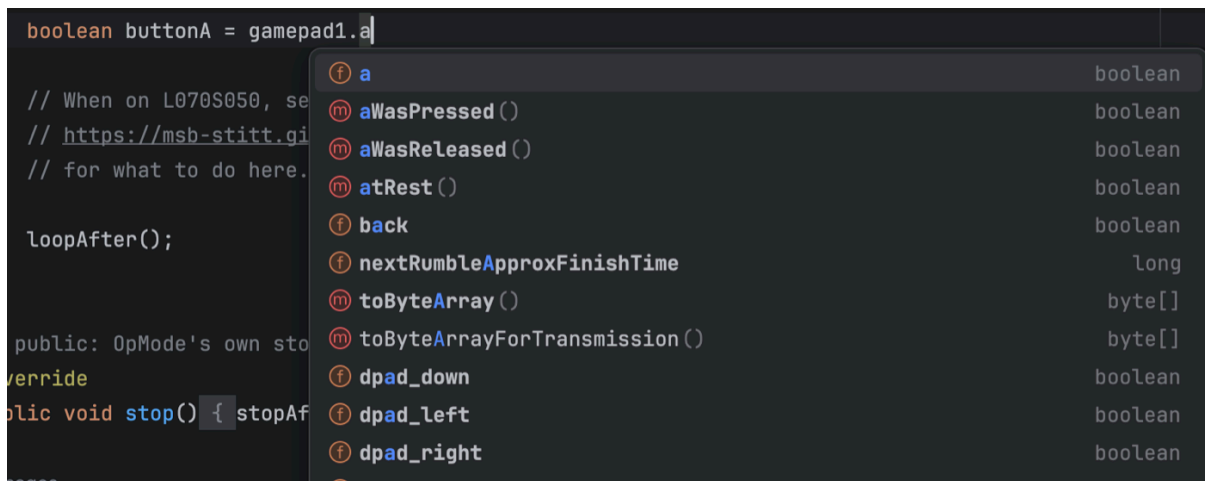


Figure 2.21. The list under `gamepad1.a`, offering a marked `boolean` first, then `aWasPressed()` and `aWasReleased()`.

Pick `a`, and finish the lines:

```
// When on L070S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1070.html#1070s030
// for what to do here.
boolean buttonA = gamepad1.a;
Tracker.publish("button/a", buttonA);
```

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L070ButtonsOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020. Open the `button` folder, and drag `a` into **Discrete Fields**.

Check. `button/a` is a red dot. Press A and hold it, and the dot turns green. Let go, and it turns red again.

L070S040: remember last loop's button

`buttonA` says whether A is down in this loop. To know the moment it changed, the code also needs to know whether it was down in the loop before. A variable made inside `loop()` is gone when the loop ends, so this one goes outside it.

Change. Press Ctrl+C in the Terminal to stop the simulator. Scroll to the top of the class, and find the comment that names this step. Under it, type:

```
// When on L070S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1070.html#1070s040
// for what to do here.
boolean previousButtonA;
```

A variable made here is a **field**. It keeps its value from one loop to the next. A `boolean` field starts out `false`.

Check. The line has no red underline.

L070S050: say when the button changes

Change. Back in `loop()`, find the comment that names this step. Under it, type:

```
// When on L070S050, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1070.html#1070s050
// for what to do here.
if (buttonA != previousButtonA) {
    if (buttonA) {
        Tracker.publish("button/event", "A pressed");
    } else {
        Tracker.publish("button/event", "A released");
    }
}
previousButtonA = buttonA;
```

!= means “is not the same as”. The first if is true only on the loop where the button changed. The second one says which way it changed. else is what runs when the if is false. The last line saves this loop’s button for the next loop.

The text in quotes is a **String**. Tracker writes it down the same way it writes a number.

Simulator. Run the same command again, and connect AdvantageScope.

AdvantageScope. Drag event from the button folder into **Discrete Fields**. Until A has been pressed once, event is crossed out, because nothing has been written to it yet.

Check. Press A and hold it. button/a turns into a green dot, and event reads A pressed. Let go. The dot turns red, and event reads A released. Holding A down does not write A pressed again; it only changes on the loop the button changed.

L070S060: read the sticks sideways

Each stick also moves left and right. Pushed right, it reads above 0. Pushed left, it reads below 0. Tank drive does not use these yet. Later lessons drive sideways with them.

Change. Press Ctrl+C in the Terminal to stop the simulator. Near the top of loop(), find the comment that names this step, under the lines that read the sticks up and down. Under it, type:

```
// When on L070S060, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1070.html#1070s060
// for what to do here.
double leftStickX = gamepad1.left_stick_x;
Tracker.publish("stick/leftX", leftStickX);
double rightStickX = gamepad1.right_stick_x;
Tracker.publish("stick/rightX", rightStickX);
```

Simulator. Run the same command again, and connect AdvantageScope.

Check. Push the left stick to the right, and stick/leftX goes above 0. Push the right stick to the left, and stick/rightX goes below 0. speed and power stay at 0, and the robot does not move.



Figure 2.22. AdvantageScope's list with the `stick` folder open, showing `leftX` at 0.500 and `rightX` at -0.250, `leftY` and `rightY` at 0, and `speed` and `power` at 0.

L070S070: watch the buttons on the robot

Send. Send the code to the robot, as in L050. On the Driver Station, pick L070 Buttons from the Lessons group. Press INIT, then PLAY.

Panels. Join the laptop to the robot's Wi-Fi, and open <http://192.168.43.1:8001>.

Check. Press A, and in Panels `button/event` reads A pressed. Let go, and it reads A released. The robot still drives with the sticks as it did in L060.

2.7 L080: drive like an arcade game

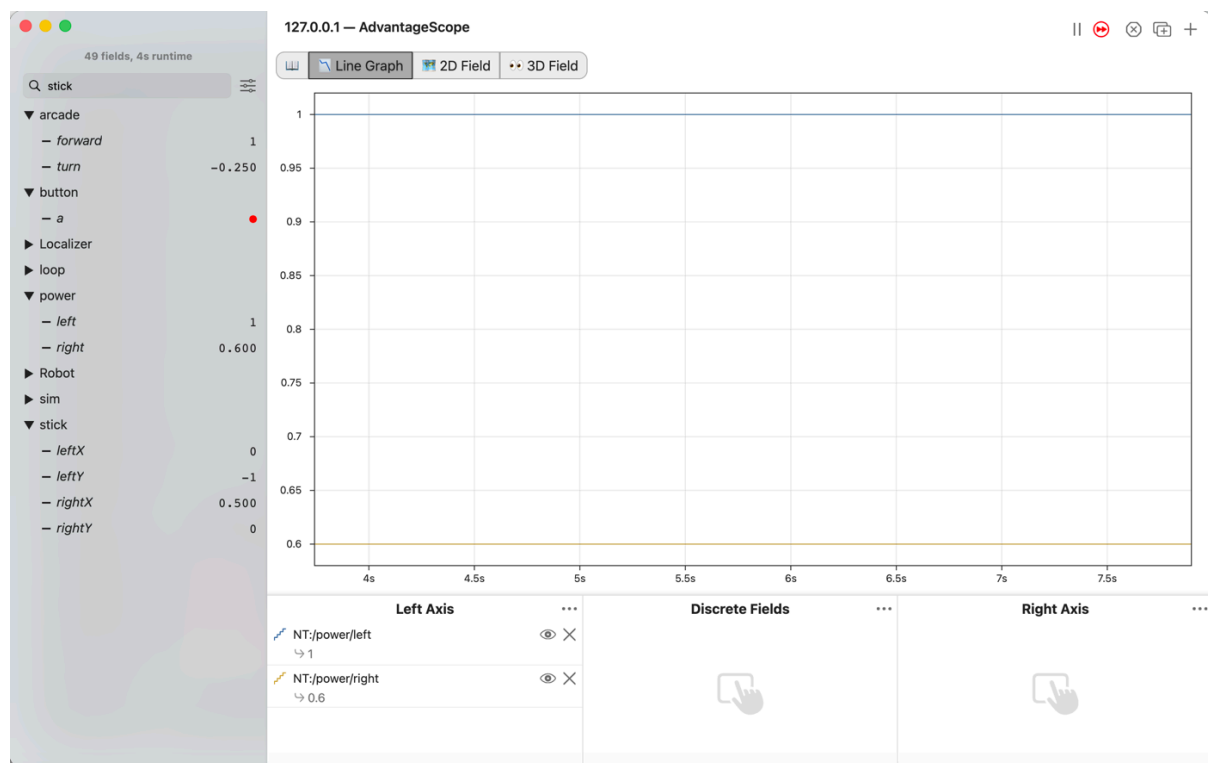


Figure 2.23. AdvantageScope’s Line Graph with NT:/power/left at 1 and NT:/power/right at 0.6 on **Left Axis**, and the list showing arcade/forward at 1 and arcade/turn at -0.250.

Drive with one stick for going and one for turning: arcade drive. When the sticks ask for more power than a motor has, scale every wheel down together, so the robot still drives the curve the driver asked for.

2.7.1 The steps

L080S010: copy the starter into mytry

Copy. In `lessons`, click `L080ArcadeOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check. `mytry` holds `L080ArcadeOpMode`, and the new file is open.

L080S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check. Under the step’s comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L080S030: forward and turn

Tank drive asks the driver to think in left wheels and right wheels. Arcade drive asks for forward and turn instead, which is how a driver already thinks. The code works out the two sides.

The starter has the sticks, the buttons, deadband and `squared` from the lessons before. It does not drive the wheels yet.

Change. In `loop()`, find the comment that names this step. Under it, type:

```
// When on L080S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
// /1080.html#l080s030
// for what to do here.
double band = 0.05;
double forward = squared(deadband(-leftStickY, band));
Tracker.publish("arcade/forward", forward);
double turn = squared(deadband(-rightStickX, band));
Tracker.publish("arcade/turn", turn);
```

forward comes from the left stick, up and down. turn comes from the right stick, sideways. Both get a minus sign. For forward, that makes pushing away from you positive, as in L030. For turn, it makes a turn to the left positive. That is the way robots measure turning: left, or counter-clockwise, is positive.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L080ArcadeOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020, and open the arcade folder.

Check. Push the left stick halfway forward, and forward reads 0.25. Push the right stick halfway right, and turn reads -0.25. The robot does not move yet.

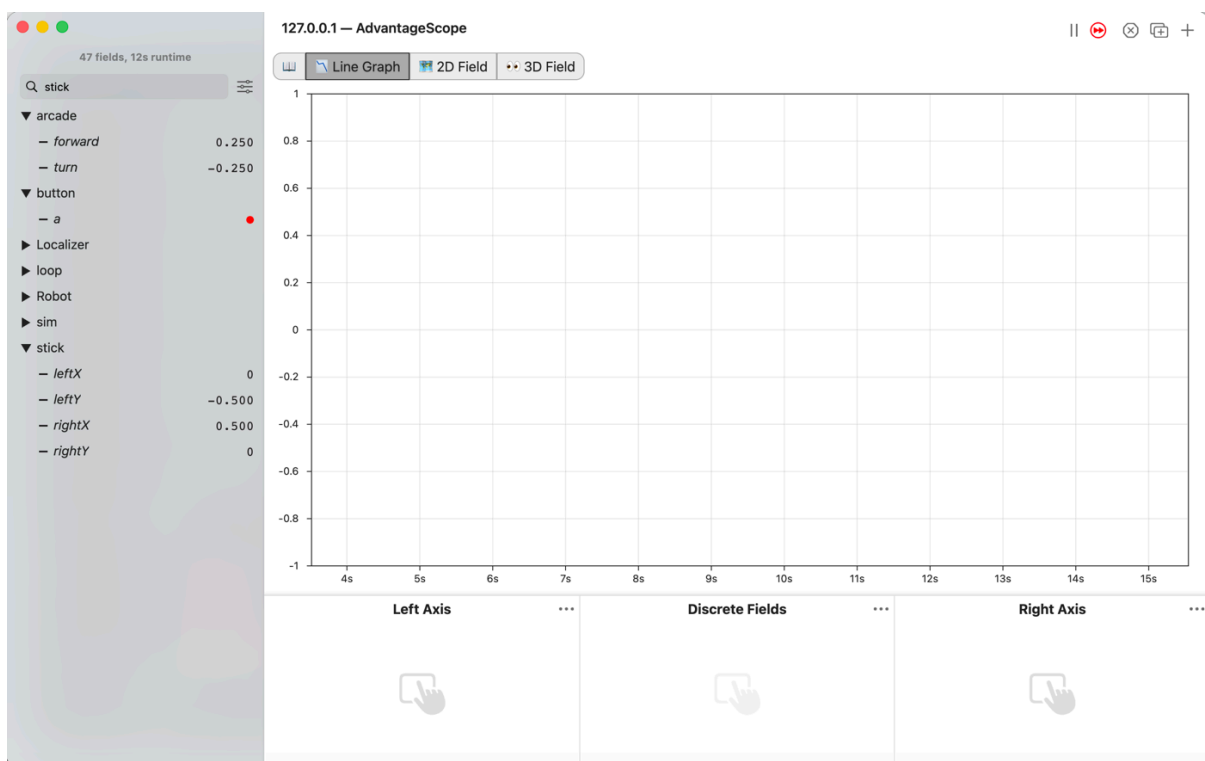


Figure 2.24. AdvantageScope's list with the arcade folder open, showing forward at 0.250 and turn at -0.250, beside stick/leftY at -0.500 and stick/rightX at 0.500.

L080S040: mix them into two sides

To turn right, the left side goes faster and the right side goes slower. A turn to the right is below 0, so taking turn away from the left side speeds it up, and adding it to the right side slows that side down.

Change. Press Ctrl+C in the Terminal to stop the simulator. Find the comment that names this step. Under it, type:

```
// When on L080S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
// /1080.html#1080s040
// for what to do here.
double leftPower = forward - turn;
double rightPower = forward + turn;
Tracker.publish("power/left", leftPower);
Tracker.publish("power/right", rightPower);

hardware.frontLeft.setPower(leftPower);
hardware.backLeft.setPower(leftPower);
hardware.frontRight.setPower(rightPower);
hardware.backRight.setPower(rightPower);
```

Simulator. Run the same command again, and connect AdvantageScope. Click **2D Field**. The robot from L040 is still in **Poses**.

Check. Push the left stick forward, and the robot drives forward. Push the right stick right, and it turns right. Push both, and it drives a curve.

Now push the left stick all the way forward and the right stick halfway right. Click **Line Graph**, and drag left and right from the power folder into **Left Axis**. power/left reads 1.25.

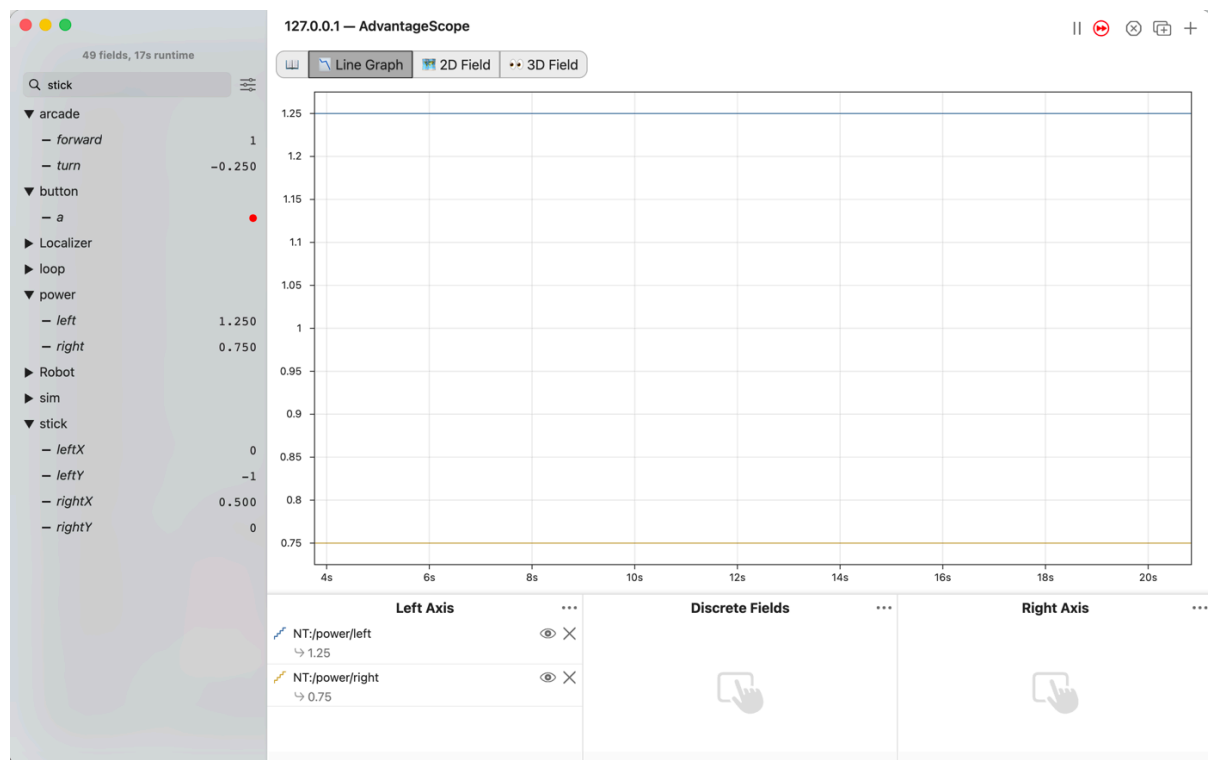


Figure 2.25. AdvantageScope's Line Graph with NT:/power/left at 1.25 and NT:/power/right at 0.75 on **Left Axis**, and the list showing arcade/forward at 1 and arcade/turn at -0.250.

A motor's power only goes from -1 to 1. On the robot, the Control Hub cuts 1.25 down to 1 and leaves 0.75 alone. The left side loses some of what it asked for and the right side loses none, so the robot drives a wider curve than the sticks asked for. The simulator does not cut it, so the curve there looks right. The number shows the problem.

L080S050: scale, don't cut

The fix is to divide both sides by the bigger one. 1.25 and 0.75 become 1 and 0.6. The robot goes a little slower, and the curve stays the one the driver asked for.

Change. Press Ctrl+C to stop the simulator. Go back to the two lines from L080S040 that make `leftPower` and `rightPower`. Under them, type:

```
double leftPower = forward - turn;
double rightPower = forward + turn;
double biggest = Math.max(1, Math.max(Math.abs(leftPower),
Math.abs(rightPower)));
leftPower = leftPower / biggest;
rightPower = rightPower / biggest;
```

`Math.max` gives back the bigger of two numbers. The inner one finds the bigger power, with its minus sign taken off. The outer one makes sure `biggest` is never less than 1. Without it, a gentle 0.25 would be divided by 0.25 and become full power.

The last two lines have no `double` in front. `leftPower` already exists, so they change it rather than making a new one.

Simulator. Run the same command again, and connect AdvantageScope.

Check. Push the left stick all the way forward and the right stick halfway right. `power/left` reads 1, and `power/right` reads 0.6. Push the left stick halfway forward alone, and both read 0.25, the same as before this step.

L080S060: drive it on the robot

Send. Send the code to the robot, as in L050. On the Driver Station, pick L080 Arcade from the Lessons group. Press INIT, then PLAY.

Check. Push the left stick halfway forward and the right stick a quarter right, and the robot drives a circle to the right. Push both sticks twice as far, and it drives a circle the same size, only faster.

2.8 L090: slide it sideways

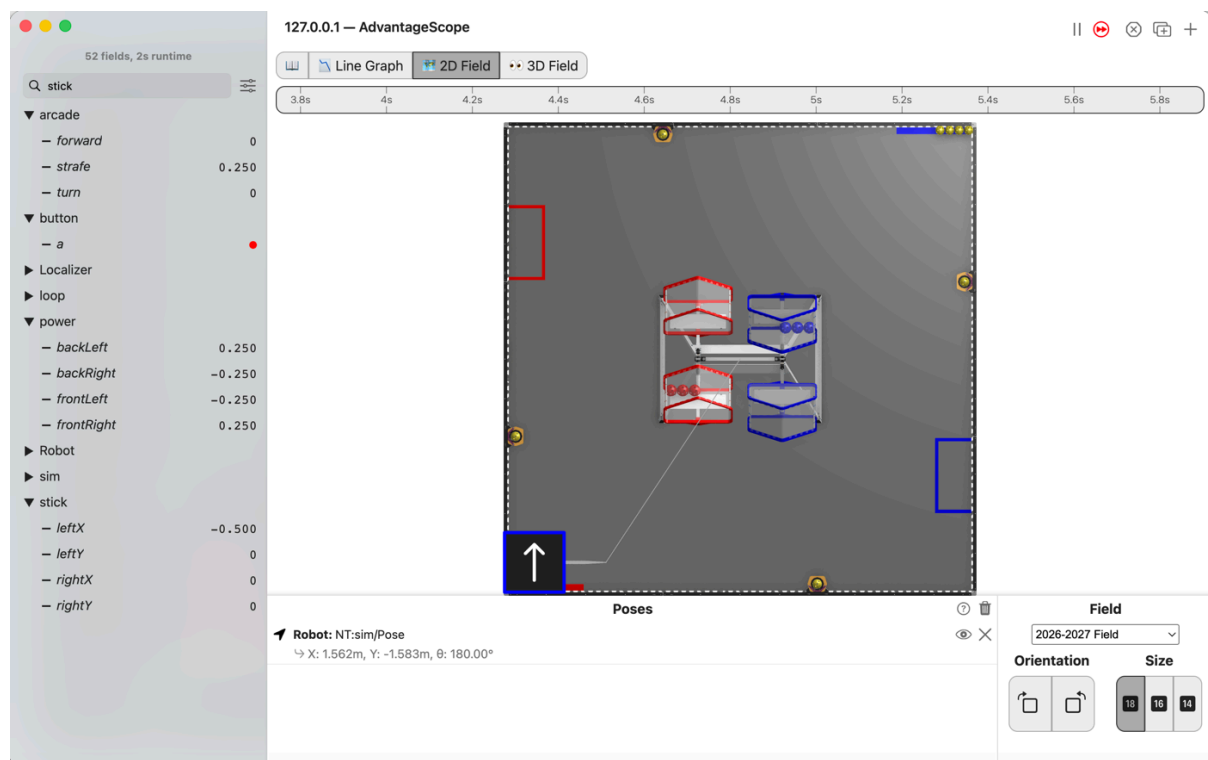


Figure 2.26. AdvantageScope's 2D Field with the robot `NT:sim/Pose` slid to its own left along the bottom edge, its nose still pointing the same way at 180.00° , and the list showing `arcade/strafe` at 0.250 and the four wheel

powers at -0.250, 0.250, 0.250 and -0.250.

Drive a mecanum robot: the left stick drives and slides, the right stick turns, and all three can happen at once. Each wheel gets its own power.

2.8.1 The steps

L090S010: copy the starter into mytry

Copy. In `lessons`, click `L090MecanumOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check. `mytry` holds `L090MecanumOpMode`, and the new file is open.

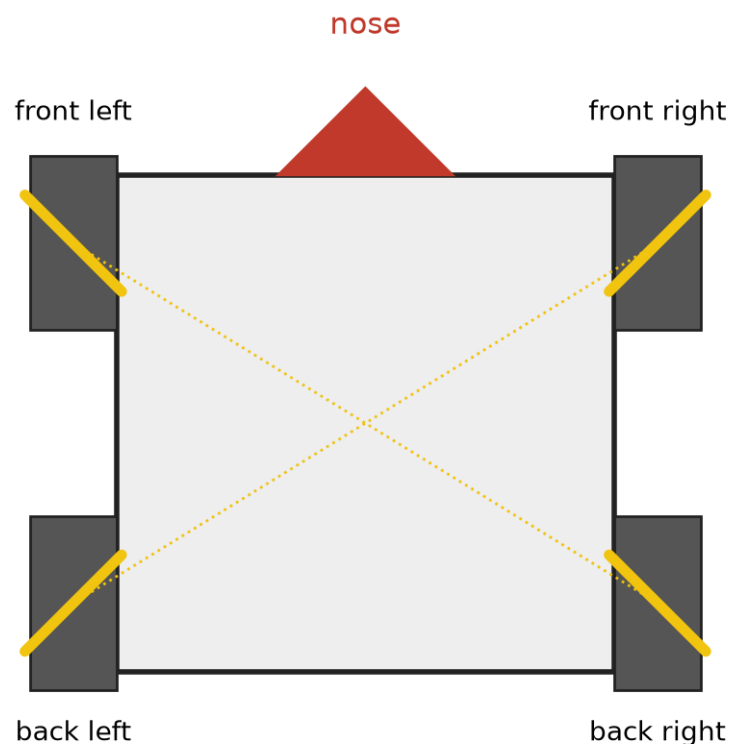
L090S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check. Under the step's comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L090S030: read the slide

A mecanum wheel has small rollers around it, set at 45 degrees. Spinning the wheel pushes the robot along and sideways at the same time. With four of them, the robot can drive, slide and turn.



the top roller of each wheel, at 45 degrees: together they make an X

Figure 2.27. The robot from above with its nose marked, and the top roller of each of the four wheels drawn as a line at 45 degrees, so the four lines make an X across the robot.

Seen from above, the top rollers make an X. Game Manual 0 shows it on [this page](#). A mecanum wheel is a left wheel or a right wheel, and the two are mirror images. goBILDA's page shows both: <https://www.gobilda.com/96mm-mecanum-wheel-set-70a-durometer-bearing-supported-rollers/>

The starter has forward and turn from L080. The slide is the third number, and it comes from the left stick, sideways.

Change. In `loop()`, find the comment that names this step. Under it, type:

```
// When on L090S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1090.html#1090s030
// for what to do here.
double strafe = squared(deadband(-leftStickX, band));
Tracker.publish("arcade/strafe", strafe);
```

Strafe is the robot word for sliding sideways. The stick reads above 0 to the right, and the minus sign makes a slide to the left positive, the same way a turn to the left is.

Check. Neither line has a red underline.

L090S040: give each wheel its power

Each wheel adds up the three numbers its own way:

Wheel	forward	strafe	turn
front left	+	-	-
front right	+	+	+
back left	+	+	-
back right	+	-	+

Change. Find the comment that names this step. Under it, type:

```
// When on L090S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1090.html#1090s040
// for what to do here.
double frontLeftPower = forward - strafe - turn;
double frontRightPower = forward + strafe + turn;
double backLeftPower = forward + strafe - turn;
double backRightPower = forward - strafe + turn;
Tracker.publish("power/frontLeft", frontLeftPower);
Tracker.publish("power/frontRight", frontRightPower);
Tracker.publish("power/backLeft", backLeftPower);
Tracker.publish("power/backRight", backRightPower);

hardware.frontLeft.setPower(frontLeftPower);
hardware.frontRight.setPower(frontRightPower);
hardware.backLeft.setPower(backLeftPower);
hardware.backRight.setPower(backRightPower);
```

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L090MecanumOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020, and click **2D Field**. Open the arcade and power folders.

Check. Push the left stick halfway left. `strafe` reads 0.25. The front right and back left wheels read 0.25, and the other two read -0.25. The robot slides to its own left, and its nose keeps pointing the same way. Forward and turning work as they did in L080.

L090S050: scale all four

Push the left stick into a corner, all the way forward and all the way left. `forward` and `strafe` are

both 1, so the front right and back left wheels ask for 2. That is twice what a motor has.

Change. Press Ctrl+C in the Terminal to stop the simulator. Go back to the four lines that make the wheel powers. Under them, type:

```
double frontLeftPower = forward - strafe - turn;
double frontRightPower = forward + strafe + turn;
double backLeftPower = forward + strafe - turn;
double backRightPower = forward - strafe + turn;
double biggest = Math.max(1, Math.max(Math.max(Math.abs(frontLeftPower),
Math.abs(frontRightPower)),
    Math.max(Math.abs(backLeftPower), Math.abs(backRightPower))));
frontLeftPower = frontLeftPower / biggest;
frontRightPower = frontRightPower / biggest;
backLeftPower = backLeftPower / biggest;
backRightPower = backRightPower / biggest;
```

This is L080's scaling, for four wheels. The two inner `Math.max` calls find the bigger of each pair, and the next one finds the biggest of all four. Type it as two lines, pressing Enter where the first one ends, and Android Studio moves the second one in for you.

Simulator. Run the same command again, and connect AdvantageScope.

Check. Push the left stick into the forward-left corner. The front right and back left wheels read 1, and the other two read 0. The robot drives diagonally, forward and to its left, with its nose still pointing forward.

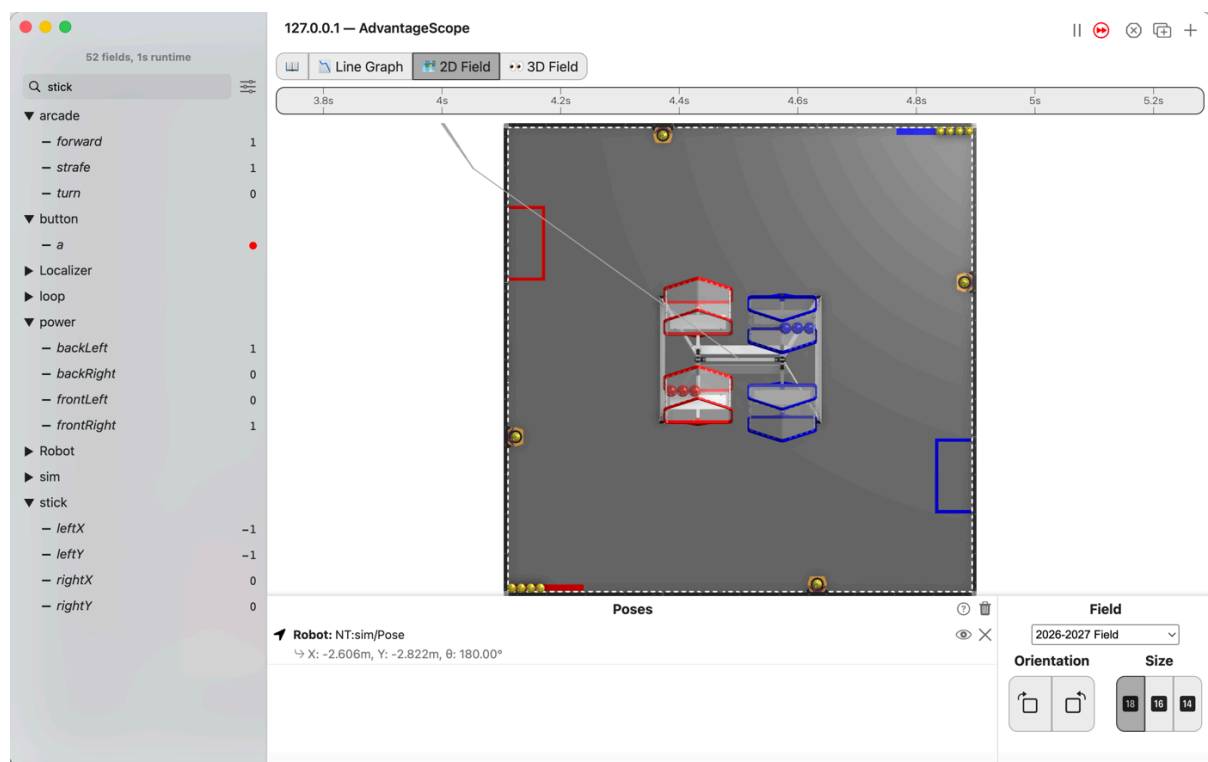
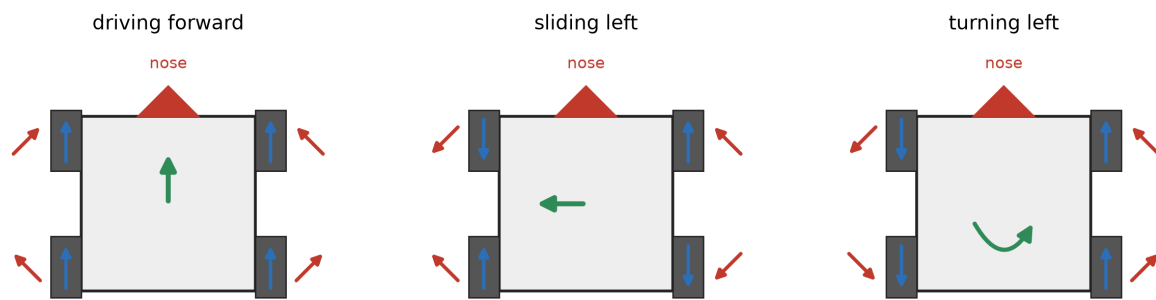


Figure 2.28. AdvantageScope's 2D Field and list with the left stick in the forward-left corner: arcade/forward and arcade/strafe at 1, power/frontRight and power/backLeft at 1, power/frontLeft and power/backRight at 0, and the robot's heading still 180.00°.

L090S060: check every wheel on blocks

A mecanum wheel spinning forward pushes the robot forward and sideways at once. The front left and back right wheels push forward and to the right. The front right and back left wheels push forward and to the left. Spinning backward, each pushes the opposite way. The robot moves the way the four pushes add up.



blue: which way each wheel spins red: which way it pushes the robot green: what the four pushes add up to

Figure 2.29. The robot from above, three times: driving forward, sliding left and turning left. On each wheel, an arrow for the way it spins and an arrow at 45 degrees for the way it pushes the robot, and in the middle one arrow for where the four pushes add up to.

Robot. Rest the robot on the blocks, so all four wheels are off the ground. Send the code to the robot, as in L050. On the Driver Station, pick L090 Mecanum from the Lessons group. Press INIT, then PLAY.

Check. Push each way gently and watch the top of each tyre:

Push	front left	front right	back left	back right
left stick forward	forward	forward	forward	forward
left stick left	backward	forward	forward	backward
right stick left	backward	forward	backward	forward

Left stick forward: all four push forward, and their sideways pushes cancel. Left stick left: the forward and backward pushes cancel, and all four push left. Right stick left: the left side pushes backward and the right side forward, so the robot turns left.

Write down any wheel that turns the other way. A wheel that is wrong for every push has the wrong direction, and L050S070 fixes it. A wheel that is right for forward but wrong for a slide is on the wrong corner. Check the X from above.

L090S070: drive it on the floor

Drive. Press STOP. Lift the robot off the blocks and put it on the floor. Press INIT, then PLAY.

Check. Push the left stick left, and the robot slides to its own left with its nose still pointing the same way. Push it into a corner, and the robot drives diagonally. Push the right stick while sliding, and it turns as it slides.

2.9 L100: let AutoTune tune the robot

The robot measures itself. It finds which way each motor turns and where the pods sit. It finds how fast it can drive and how hard it can brake. Pedro Pathing's AutoTune does the measuring. You paste what it finds into `Constants.java`.

2.9.1 The steps

L100S010: open AutoTune

AutoTune is a web page the robot serves. It is not in the Driver Station's list of OpModes. It starts its own OpMode, called `Pedro Tuning`, when a step needs the robot to move.

Send. Send the code to the robot, as in L050. Join the laptop to the robot's Wi-Fi. In a browser, open `http://192.168.43.1:10158`.

Check. The page lists four tuners: Mecanum Tuner, Tests, Pinpoint Tuner and Foresight Tuner.

L100S020: find which way each motor turns

Put the robot on blocks, so its wheels spin freely.

Robot. Pick Mecanum Tuner. Type the four motor names as the Robot Configuration spells them: Front Left, Front Right, Back Left and Back Right. Each motor then spins on its own, at half power, beside a picture of which wheel it is. Say whether it spun, and whether it spun forward or reversed.

Change. At the end, AutoTune shows Java that starts `public static MecanumConfig drivetrainConfig`. Open `Constants`, in the `pedro` folder. Find the eight lines that set the names and directions. Put the eight AutoTune shows in their place. Keep `manualBrakeMode` and `powerThreshold`. AutoTune does not print them.

Check. The directions match what L050 found on blocks.

L100S030: measure the pods

The Pinpoint needs four more numbers: which way each pod counts, and how far each one sits from the middle of the robot. The x offset is how far sideways the forward pod is from the point the robot turns about, and left counts up. The y offset is how far forward the strafe pod is, and forward counts up. This robot's are about 1.06 in and 3.36 in.

Put the robot on the floor.

Robot. Pick Pinpoint Tuner. Keep the name `pinpoint` and the pod type it offers. Then do the three pushes it asks for, stopping each one when you are done:

- Push the robot forward.
- Push the robot to the left.
- Spin the robot half a turn on the spot, counterclockwise.

Change. Replace `localizerConfig` in `Constants` with the `PinpointConfig` AutoTune shows.

Check. In `Tests`, pick `Localization Test` and drive with the gamepad. Driving forward raises x, and strafing left raises y.

L100S040: let the robot find its speed and its brakes

This one drives itself. Clear the floor first.

Robot. Pick Foresight Tuner. Keep the distances and the speed it offers: 48 in, 30 in/s and 36 in. It drives forward and left as fast as it can, coasts to a stop, turns back and forth, spins, brakes, and steps 12 in to 24 in each way.

Change. Replace `foresightConfig` in `Constants` with the `ForesightConfig` AutoTune shows. Keep the five lines it does not print, from `translationalConstraint` to `holdPointHeadingScaling`.

Three of its numbers are kept twice in `Constants`, for the lessons' own drivetrain. Copy each one up into its own line:

- the number in `c.coast.set(...)` goes in `powerPerInchPerSecond`;
- `maxAchievableForwardVelocity` goes in `maxForwardInchesPerSecond`;
- `maxAchievableStrafeVelocity` goes in `maxStrafeInchesPerSecond`.

Check. Each of the three pairs holds the same number.

L100S050: watch it drive a line

Send. Send the code again, so the new Constants is on the robot.

Robot. In Tests, pick Line Test, with the distance at 48.

Check. The robot drives 48 in and stops on the line.

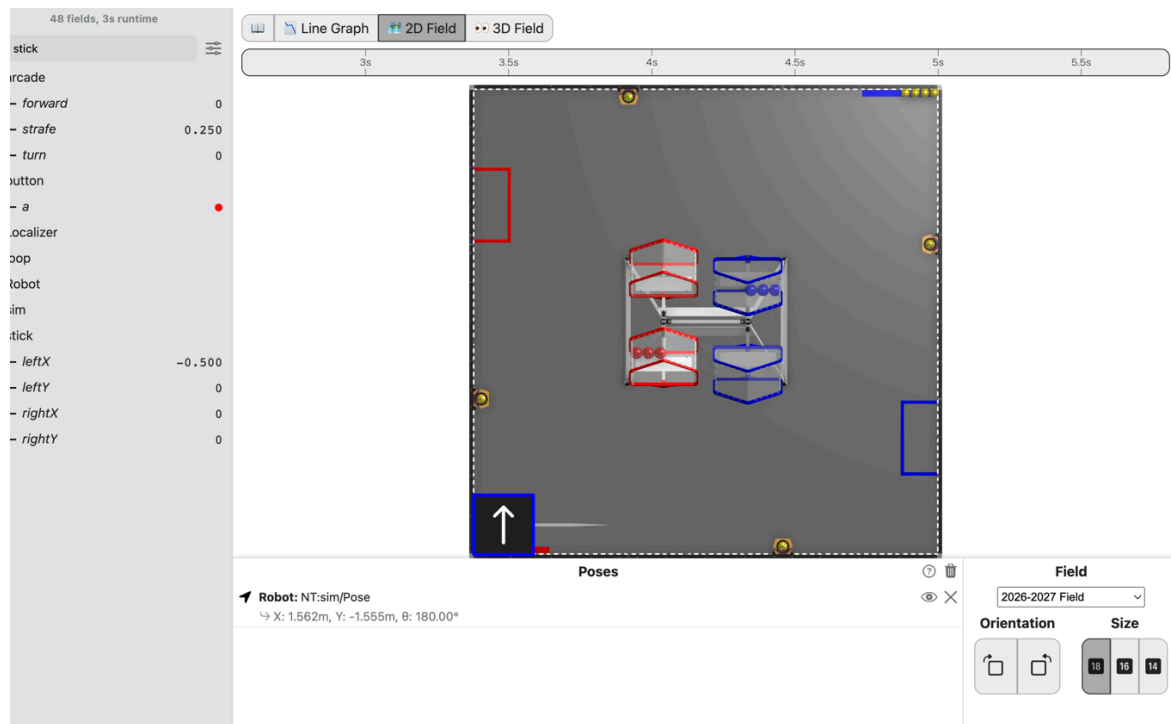
2.10 L110: hand the wheels to the path follower

Figure 2.30. AdvantageScope's 2D Field with the robot NT:sim/Pose slid to its own left along the bottom edge, its nose still pointing the same way at 180.00°, and the list showing arcade/strafe at 0.250, as in L090, with the path follower holding the drivetrain.

The robot drives on the sticks as it did in L090, and the path follower is holding the drivetrain. The sticks still decide where the robot goes.

2.10.1 The steps**L110S010: copy the two starters into mytry**

This lesson has two files. One is the OpMode. The other is the drivetrain: the code that turns three numbers into four wheel powers.

Copy. In lessons, click L110FollowerDriveTrain once and copy it. Click mytry once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**. Do the same for L110FollowerOpMode.

Check. mytry holds L110FollowerDriveTrain and L110FollowerOpMode.

L110S020: switch your copy on

Change. In L110FollowerOpMode, delete the line @Disabled, under the comment that names this step.

Check. Under the step's comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L110S030: write the mix

Open `L110FollowerDriveTrain`. It extends `CorbelsDriveTrain`, which means it gets everything `CorbelsDriveTrain` already has. That class sets each motor's direction, scales the powers as L090 did, and sends them to the motors. The one part it leaves to you is `mix`.

`wheels` is an **array**: four numbers in a row, in one variable. `wheels[FL]` is the front left wheel's slot, and `FR`, `BL` and `BR` name the other three.

Change. In `mix`, find the comment that names this step. Under it, type L090's four lines, each into its own slot:

```
// When on L110S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/l110.html#l110s030
// for what to do here.
wheels[FL] = forward - strafe - turn;
wheels[FR] = forward + strafe + turn;
wheels[BL] = forward + strafe - turn;
wheels[BR] = forward - strafe + turn;
```

Check. No line has a red underline, and `FL`, `FR`, `BL` and `BR` turn purple.

L110S040: make room for the drivetrain

Change. Open `L110FollowerOpMode`. Near the top of the class, find the comment that names this step. Under it, type:

```
// When on L110S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/l110.html#l110s040
// for what to do here.
L110FollowerDriveTrain drivetrain;
```

This is a field, like `previousButtonA` in L070. It holds the drivetrain from `init` until the `OpMode` stops.

Check. The line has no red underline.

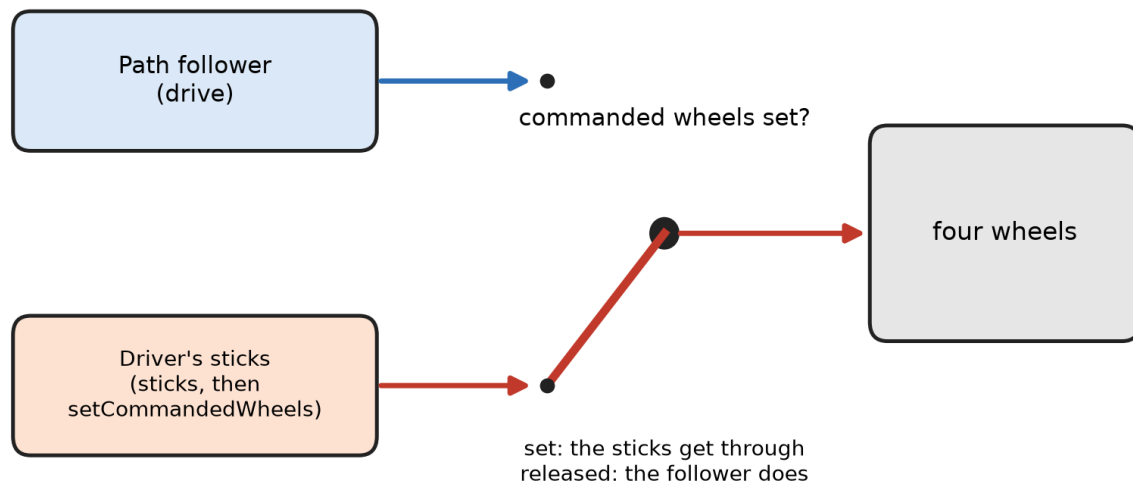
L110S050: hand the drivetrain to the follower

Figure 2.31. Two arrows reaching the same four wheels, one from the path follower and one from the driver's sticks, with the commanded-wheels switch between them showing which one gets through.

Change. In `init()`, find the comment that names this step. Under it, type the first line below. Then change `initAfter();` to `initAfter(drivetrain);`:

```

// When on L110S050, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l110.html#l110s050
// for what to do here.
drivetrain = new L110FollowerDriveTrain(hardware);

initAfter(drivetrain);

```

`new` makes the drivetrain. `initAfter` makes the path follower, and the follower holds what it is handed. Up to L090, `initAfter()` was handed nothing, because the lesson drove its own wheels.

The four `setDirection` lines from L050 are gone from `init`. The drivetrain sets each motor's direction itself, from the same config.

Check. Neither line has a red underline.

L110S060: drive with the sticks

Change. In `loop()`, find the comment that names this step, under the `strafe` lines. Under it, type:

```

// When on L110S060, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l110.html#l110s060
// for what to do here.
drivetrain.sticks(forward, strafe, turn);

```

`sticks` sends the three numbers through your mix. Then it scales them and saves them. At the end of the loop, `loopAfter()` updates the follower, and the follower drives the wheels.

Simulator. In the Terminal, run:

```

./gradlew :TeamCode:simRun --args="mytry.L110FollowerOpMode --pad"

```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the arcade folder.

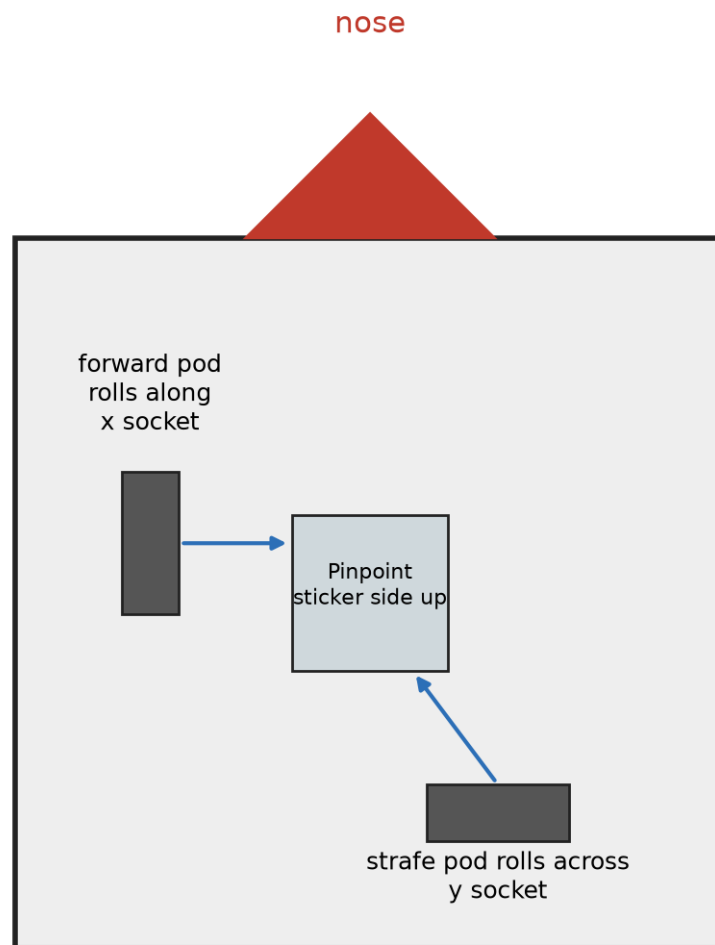
Check. Push the left stick halfway left. `strafe` reads 0.25, and the robot slides to its own left with its nose pointing the same way. Forward, turning and the corners all work as they did in L090.

L110S070: drive it on the robot

Send. Send the code to the robot, as in L050. On the Driver Station, pick L110 `Follower` from the `Lessons` group. Press INIT, then PLAY.

Check. It drives the way L090 did: forward, sideways, turning, and the diagonals.

2.11 L115: mount the Pinpoint and plug it in



Stand-in drawing for a photo still to be taken

Figure 2.32. The Pinpoint on the robot from above, with the forward pod and the strafe pod drawn at right angles to each other, each labelled with the socket it plugs into, and the sticker side marked facing up.

The Pinpoint is bolted on, its two pods are in the right sockets, and the Control Hub finds it when you press INIT. The robot does not yet know how far the pods sit from its middle. L100 measures that.

2.11.1 The steps

L115S010: put one pod along the robot and one across it

The Pinpoint counts two wheels that nothing drives. One rolls along the length of the robot, and that one is the **forward** pod. The other rolls across it, and that one is the **strafe** pod.

Both have to touch the floor the whole time, and neither may be able to lift. A pod that skips loses count, and nothing after it can tell that it did.

Mount. Bolt the two pods on, one along the robot and one across it. Plug the forward pod into the Pinpoint's x socket and the strafe pod into its y socket.

Check. Push the robot forward by hand. The forward pod's wheel turns. The strafe pod's wheel drags sideways.

L115S020: plug it into an I2C port that is not port 0

The Pinpoint talks to the Control Hub over I2C. Port 0 already has the Control Hub's own IMU on it, so the Pinpoint goes on another port.

Mount. Bolt the Pinpoint on with its sticker side, the side the ports are on, facing **up**. Plug its cable into I2C port 1, 2 or 3 on the Control Hub.

Configure. On the Driver Station, open the robot's configuration. Add the Pinpoint on that port, and name it `pinpoint`, spelled exactly that way. That is the name the code asks for:

```
c.name.set("pinpoint");
```

Check. Pick L110 Follower from the Lessons group and press INIT. INIT finishes with no error. INIT looks for the Pinpoint by its name, so if INIT finishes, it was found.

L115S030: hold the robot still while it inits

Pressing INIT builds the follower, and the follower builds the Pinpoint. The Pinpoint then resets its own gyro, which takes about a quarter of a second. The robot has to sit still for it.

Robot. Put the robot on the floor and take your hands off it. Press INIT.

Panels. Join the laptop to the robot's Wi-Fi, and open `http://192.168.43.1:8001`, as in L070.

Check. `Localizer/pinPoint/heading_deg` holds steady while the robot sits still. Press PLAY and drive a slow square, and the heading comes back to where it started.

2.12 L120: work out where the robot is from the drive wheels

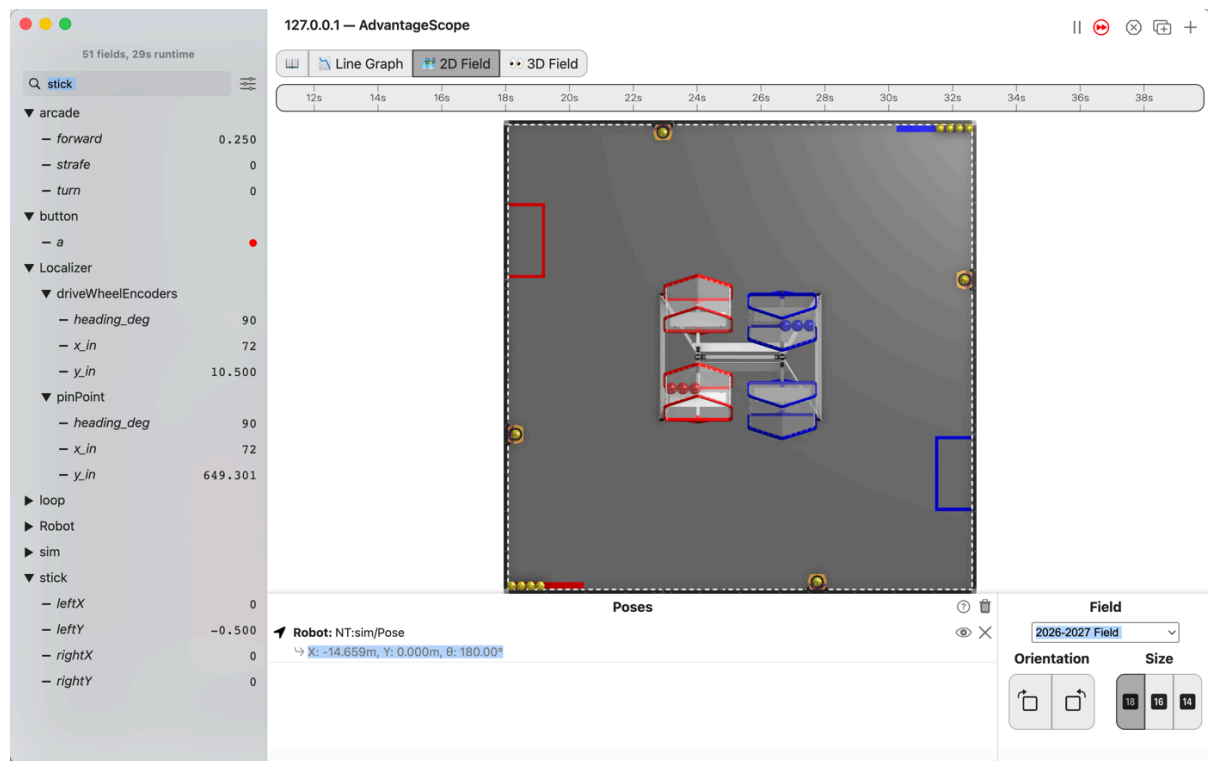


Figure 2.33. AdvantageScope's list with the Localizer folder open: pinPoint/y_in climbing past 600 while the robot drives forward, and driveWheelEncoders still at x 72, y 10.5 and heading 90, where it started.

Your own localizer runs beside the robot's own, and both are published so they can be graphed against each other. Yours counts how far the four drive wheels roll. The robot's own counts the Pinpoint's two pods. Neither is the truth. A tape measure is.

2.12.1 The steps

L120S010: copy the two starters into mytry

This lesson has two files. One is the OpMode. The other is the **localizer**: the code that works out where the robot is.

Copy. In lessons, click L120CompareLocalizersOpMode once and copy it. Click mytry once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**. Do the same for L120EncoderLocalizer.

Check. mytry holds L120CompareLocalizersOpMode and L120EncoderLocalizer.

L120S020: switch your copy on

Change. In L120CompareLocalizersOpMode, delete the line @Disabled, under the comment that names this step.

Check. Under the step's comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L120S030: run your localizer beside the robot's own

The robot's own localizer reads the Pinpoint. A **shadow** localizer runs beside it. It is updated and

published every loop, but it never steers the robot.

Change. In `L120CompareLocalizersOpMode`, find `shadowLocalizers()` and the comment that names this step. Under it, type:

```
// When on L120S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l120.html#l120s030
// for what to do here.
shadowLocalizers.add("driveWheelEncoders",
    new L120EncoderLocalizer(new HardwareWheelSource(hardware)));
```

`HardwareWheelSource` turns red, because this file does not know where it lives yet. Click it and press `Option+Enter`. Android Studio adds an import line at the top of the file, and the red goes away.

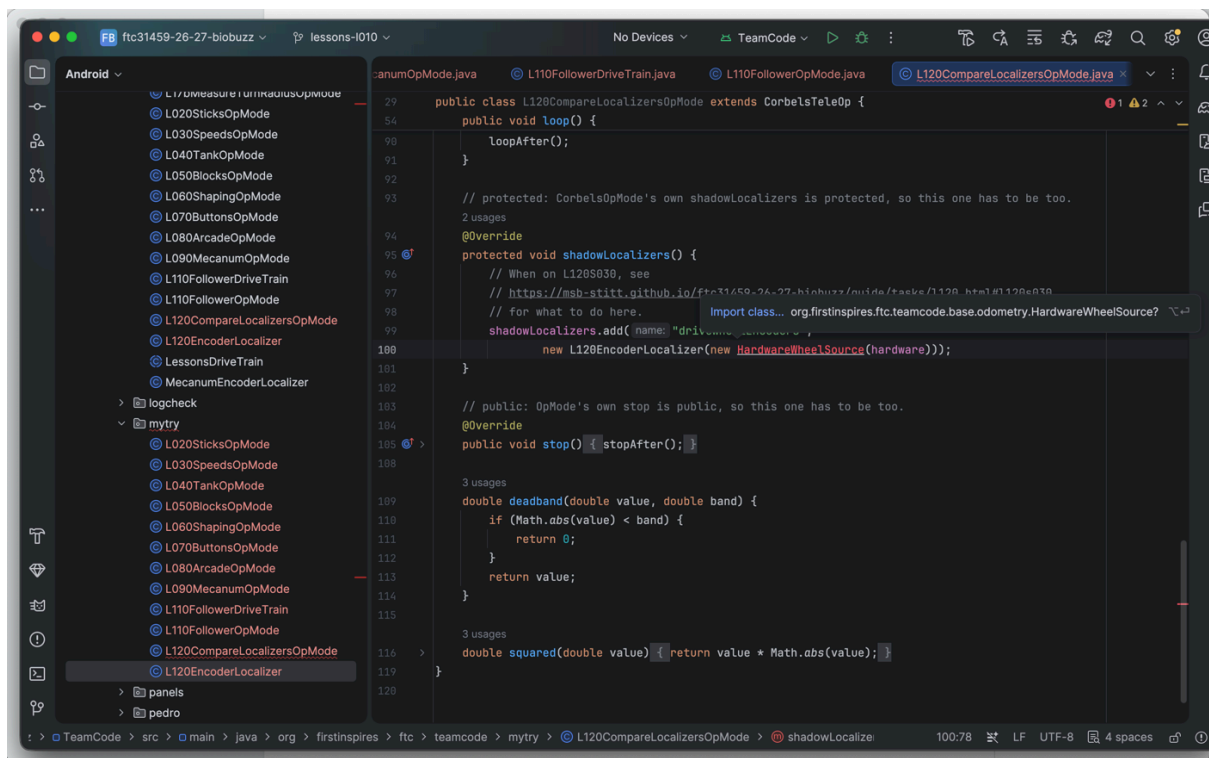


Figure 2.34. Android Studio with `HardwareWheelSource` underlined in red and the popup offering `Import class... org.firstinspires.ftc.teamcode.base.odometry.HardwareWheelSource?` with `\n↵`.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L120CompareLocalizersOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020. Open the `Localizer` folder, then `driveWheelEncoders` and `pinPoint` inside it.

Check. Both start at `x_in` 72, `y_in` 10.5 and `heading_deg` 90. Push the left stick forward. `pinPoint` / `y_in` climbs, and `driveWheelEncoders` stays where it started. Yours does not count anything yet.

L120S040: add up how far the wheels rolled

`update` runs once a loop. Each time, it can only see how far each wheel has rolled in total, so it remembers last loop's numbers and takes them away.

The first loop has nothing to take away from, so it only remembers.

Change. Press `Ctrl+C` in the Terminal to stop the simulator. Open `L120EncoderLocalizer`. In `update`, find the comment that names this step. Under it, type:

```

// When on L120S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l120.html#l120s040
// for what to do here.
if (lastWheels == null) {
    lastWheels = wheels.clone();
    lastRawHeading = rawHeading;
    return;
}

double frontLeft = wheels[0] - lastWheels[0];
double frontRight = wheels[1] - lastWheels[1];
double backLeft = wheels[2] - lastWheels[2];
double backRight = wheels[3] - lastWheels[3];

double forward = (frontLeft + frontRight + backLeft + backRight) / 4;
double strafe = (-frontLeft + frontRight + backLeft - backRight) / 4;

x += forward;
y += strafe;

```

Android Studio types the `}` for you when you press Enter after the `{`. Type the rest below it.

`null` means “nothing yet”. `return` leaves update early, until next loop.

The two long lines are L090’s table read backwards. There, three numbers made four wheel powers. Here, four wheel moves make forward and strafe again, in inches.

Check. In the Terminal, run:

```
./gradlew :TeamCode:testDebugUnitTest --tests '*L120EncoderLocalizerTest*'
```

It ends with 8 tests completed, 2 failed. The two that fail are `appliesTheHeadingSoSidewaysIsRelativeToTheRobot` and `setPoseMovesTheEstimateWithoutMovingTheRobot`. Both are about which way the robot faces. That is the next step.

L120S050: turn the step to face the way the robot faces

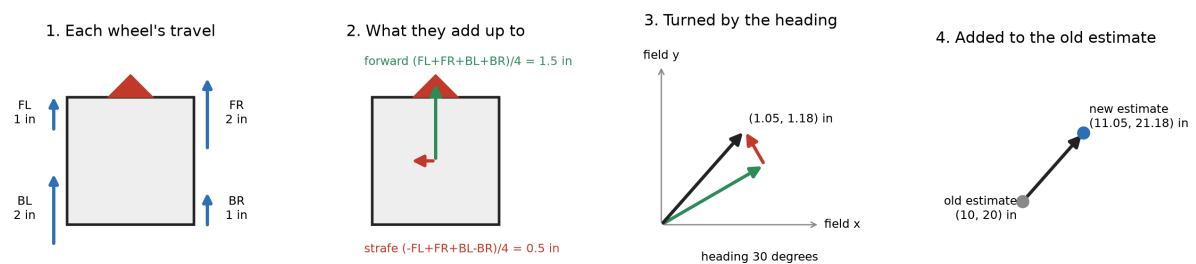


Figure 2.35. One loop of dead reckoning: the four wheel travels since the last loop, the forward and left arrows those add up to, the same pair turned to point along the robot’s heading, and the old estimate with that pair added on the end.

`forward` is forward for the robot, not for the field. The robot starts facing along the field’s `y`, so its forward is the field’s `y`. L120S040 added it to `x`.

Change. Delete the two lines `x += forward;` and `y += strafe;`. Find the comment that names this step. Under it, type:

```

// When on L120S050, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l120.html#l120s050

```

```
// for what to do here.
double heading = rawHeading + headingOffset;
x += forward * Math.cos(heading) - strafe * Math.sin(heading);
y += forward * Math.sin(heading) + strafe * Math.cos(heading);
```

`Math.cos` and `Math.sin` share the step out between `x` and `y`. Facing along `x`, `cos` is 1 and `sin` is 0, so forward all goes on `x`. Facing along `y`, `cos` is 0 and `sin` is 1, so forward all goes on `y`. In between, each gets a part.

Check. Run the same test command again. It ends with `BUILD SUCCESSFUL`.

L120S060: drive it, then measure it

Send. Send the code to the robot, as in L050. On the Driver Station, pick L120 Compare Localizers from the Lessons group. Press INIT, then PLAY.

Panels. Join the laptop to the robot's Wi-Fi, and open `http://192.168.43.1:8001`, as in L070.

Check. Drive about 2 m straight ahead and stop. Both `y_in` numbers moved, and they do not quite agree. Measure how far the robot really went with the tape measure.

Then put the robot on the blocks and push the left stick forward. The wheels spin, and the robot does not go anywhere. `driveWheelEncoders` walks off across the field. `pinPoint` stays put, because its pods are off the floor too.

2.13 L125: the first autonomous

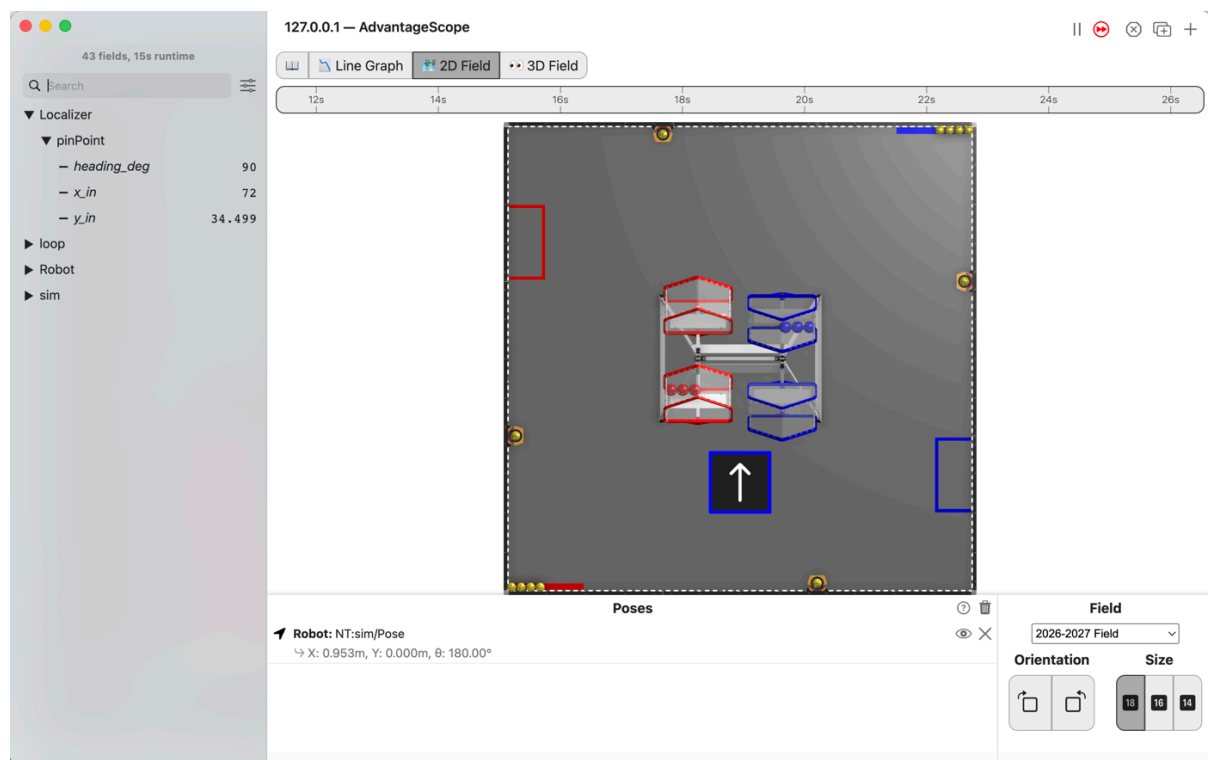


Figure 2.36. AdvantageScope's 2D Field with the robot stopped 24 inches out from the back wall, its arrow pointing up the field, and the list showing `pinPoint/y_in` at 34.499, `x_in` at 72 and `heading_deg` at 90.

The robot drives 24 inches forward on its own and stops there. Nobody touches a stick. The path follower does the driving, and you write one line to tell it where to go.

2.13.1 The steps

L125S010: copy the starter into mytry

Copy. In `lessons`, click `L125Drive24OpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check. `mytry` holds `L125Drive24OpMode`, and the new file is open.

L125S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check. Under the step's comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L125S030: say where to go

An autonomous has no sticks to read. It has a `routine`, and the robot runs it after `PLAY`. The routine hands back a **command**: a list of things to do, in order.

`start` and `end` are already near the top of the class, 24 inches apart along the field's `y`. The robot starts facing along `y`, so the line from one to the other is straight ahead. `startPose()` tells the follower where the robot is at the start.

Change. In `routine()`, find the comment that names this step. Delete the line `return Command.NOOP;` under it, and type:

```
// When on L125S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l125.html#l125s030
// for what to do here.
return sequential(follow(follower, line(start, end).constant(start)));
```

Read it from the inside out. `line(start, end)` is the path. `.constant(start)` holds the heading `start` has the whole way. `follow` drives the path. `sequential` makes the list, and this list has one thing in it.

`sequential`, `follow` and `line` turn red. Click each one and press `Option+Enter`. This time the popup says **Import static method...**, because each is a method, not a class. Android Studio adds an `import static` line at the top of the file for each one.

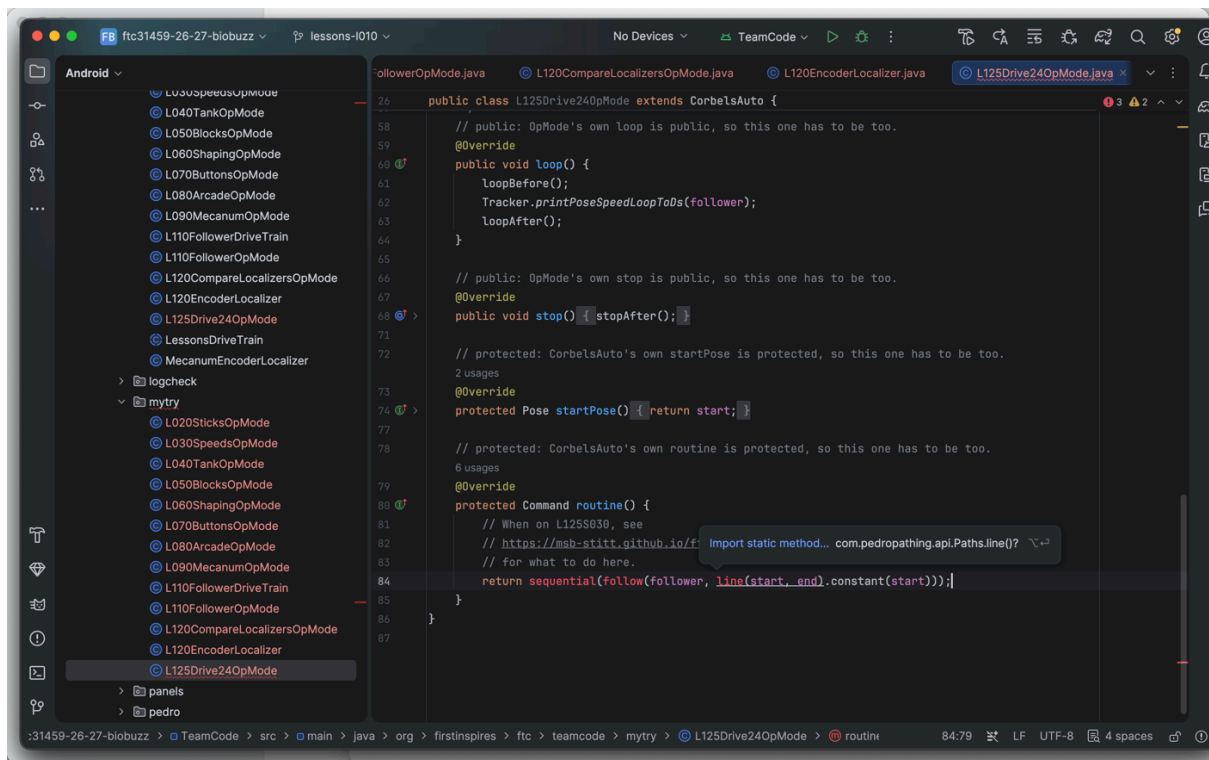


Figure 2.37. Android Studio with `sequential`, `follow` and `line` underlined in red and the popup offering `Import static method... com.pedropathing.api.Paths.line()?` with `\↵`.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L125Drive24OpMode"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the Localizer folder, then `pinPoint`.

Check. The robot has driven up the field and stopped. `y_in` reads 34.5, which is 24 more than the 10.5 it started at. `x_in` stays at 72 and `heading_deg` at 90.

L125S040: drive it on the floor

Send. Press `Ctrl+C` in the Terminal to stop the simulator. Put the robot on the floor with a piece of tape at its front edge. Send the code to the robot, as in L050. On the Driver Station, pick L125 Drive 24 from the autonomous list. Press `INIT`, then `PLAY`, and keep your hands off the sticks.

Check. The robot drives forward, slows down, and stops. Measure from the tape to the front edge. It is close to 24 inches.

2.14 L130: two legs, forward then sideways

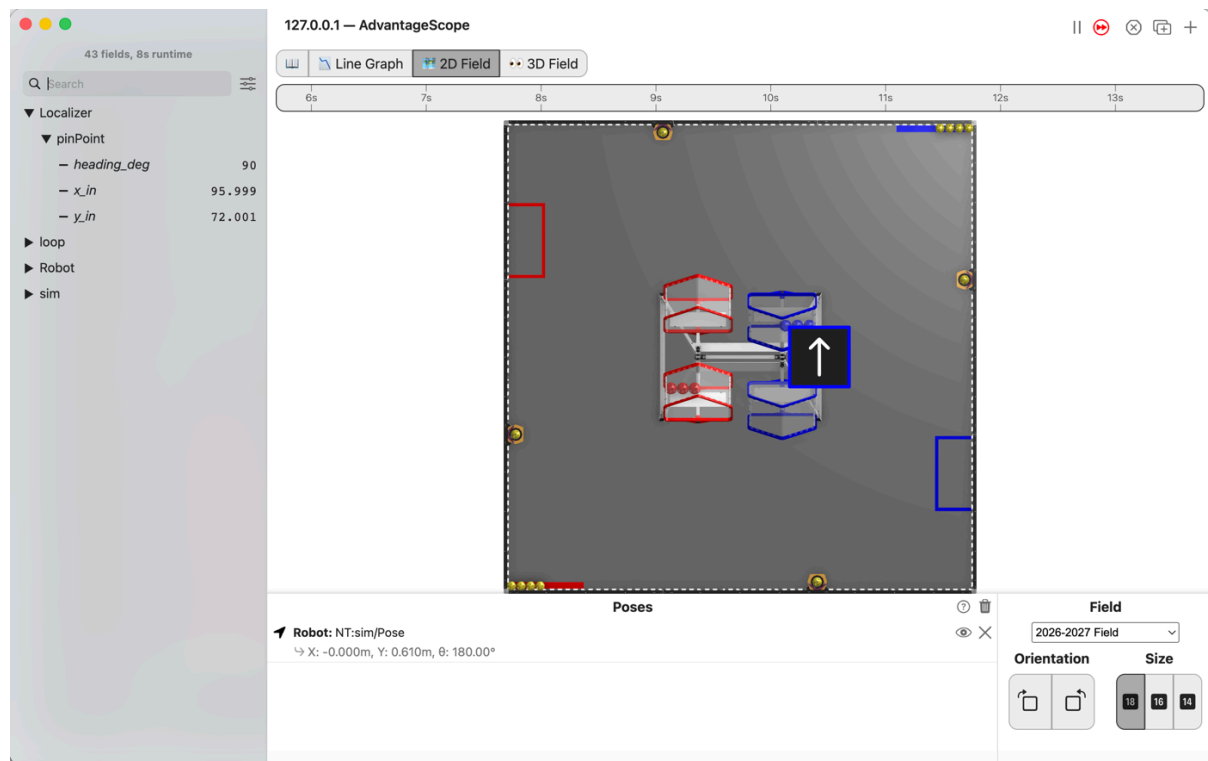


Figure 2.38. AdvantageScope's 2D Field with the robot stopped right of the middle of the field, its arrow still pointing up the field, and the list showing pinPoint/x_in at 95.999, y_in at 72.001 and heading_deg at 90.

The robot drives forward, then slides 24 inches to its right without turning, and holds where it ends up. It is the first autonomous that does more than one thing.

2.14.1 The steps

L130S010: copy the starter into mytry

Copy. In `lessons`, click `L130ForwardThenStrafeOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check. `mytry` holds `L130ForwardThenStrafeOpMode`, and the new file is open.

L130S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check. Under the step's comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L130S030: add a second leg and a hold

Near the top of the class are three poses: `start`, `corner` and `end`. All three face 90 degrees. `corner` is straight up the field from `start`, and `end` is 24 inches to the right of `corner`.

`routine()` already holds the first leg, from `start` to `corner`. It is L125's line, with one item on each line so there is room for more.

Change. In `routine()`, type a comma at the end of the first leg. Find the comment that names this

step, under it. Under the comment, type:

```

    return sequential(
        follow(follower, line(start, corner).constant(start)),
        // When on L130S030, see
        // https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
        // /l130.html#l130s030
        // for what to do here.
        follow(follower, line(corner, end).constant(start)),
        hold(follower, end)
    );

```

Each item but the last ends with a comma. The second leg holds `start`'s heading too, so the robot never turns. It slides sideways instead, which is what mecanum wheels are for. `hold` keeps the robot at end once it gets there.

`hold` turns red. Click it and press Option+Enter, as in L125.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L130ForwardThenStrafeOpMode"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the `Localizer` folder, then `pinPoint`.

Check. The robot has stopped right of the middle of the field. `x_in` reads 96 and `y_in` 72. `heading_deg` is still 90.

L130S040: drive it on the floor

Send. Press Ctrl+C in the Terminal to stop the simulator. Put the robot on the floor with a piece of tape at its front edge. Send the code to the robot, as in L050. On the Driver Station, pick L130 Forward Then Strafe from the autonomous list. Press INIT, then PLAY.

Check. The robot drives straight forward and stops, then slides to its right and stops again. Its nose points the same way the whole time.

2.15 L140: make the sticks mean the field

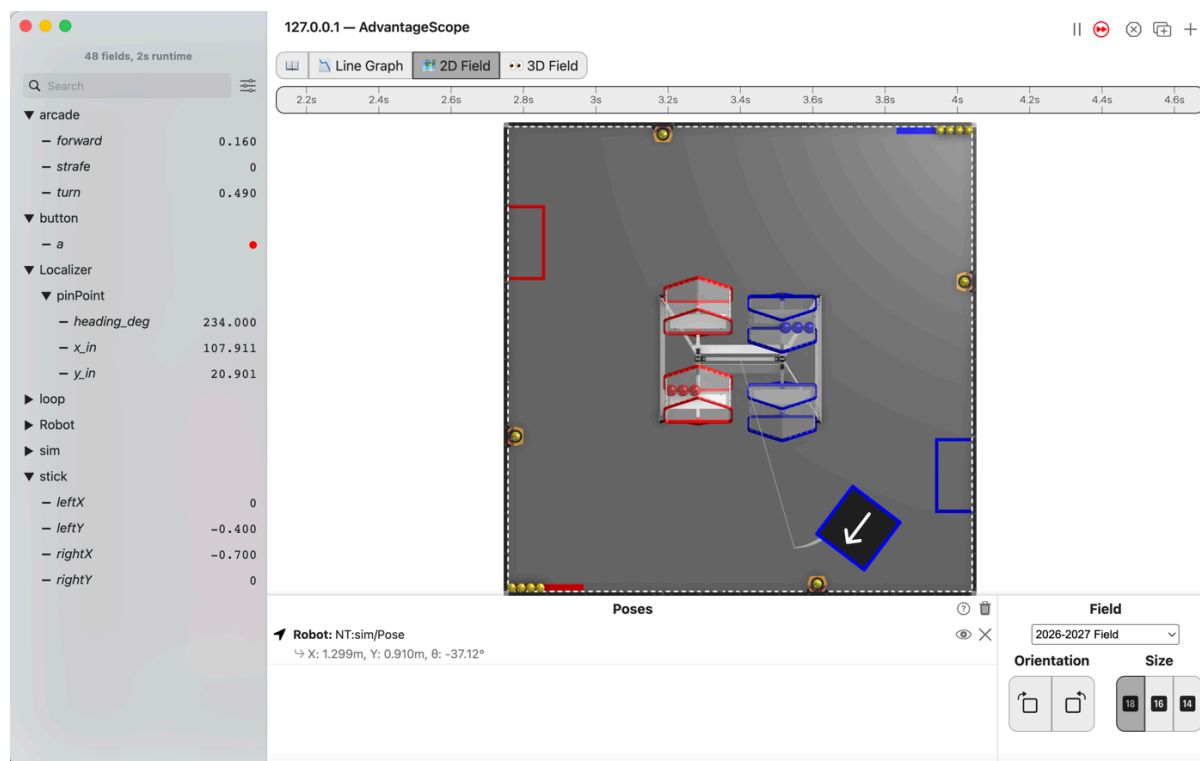


Figure 2.39. AdvantageScope's 2D Field with the robot right of where it started and turned partway round, with the list showing arcade/forward at 0.160, arcade/turn at 0.490, and pinPoint/x_in at 107.911, y_in at 20.901 and heading_deg at 234.

Push the stick away from you and the robot goes away from you. It does that whichever way the robot happens to be facing, so a driver never has to think about which end of the robot is the front.

2.15.1 The steps

L140S010: copy the starter into mytry

Copy. In lessons, click L140FieldRelativeOpMode once and copy it. Click mytry once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check. mytry holds L140FieldRelativeOpMode, and the new file is open.

L140S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check. Under the step's comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L140S030: drive and turn at once, the L110 way

The starter drives the way L110 did. Forward on the stick is forward for the robot.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L140FieldRelativeOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the `Localizer` folder, then `pinPoint`.

Check. Push the left stick a little forward and the right stick left, and hold both. The robot drives in a small circle, and `x_in` and `y_in` stay close to where they started. It goes where its nose points, and its nose keeps turning.

L140S040: turn the stick backwards through the heading

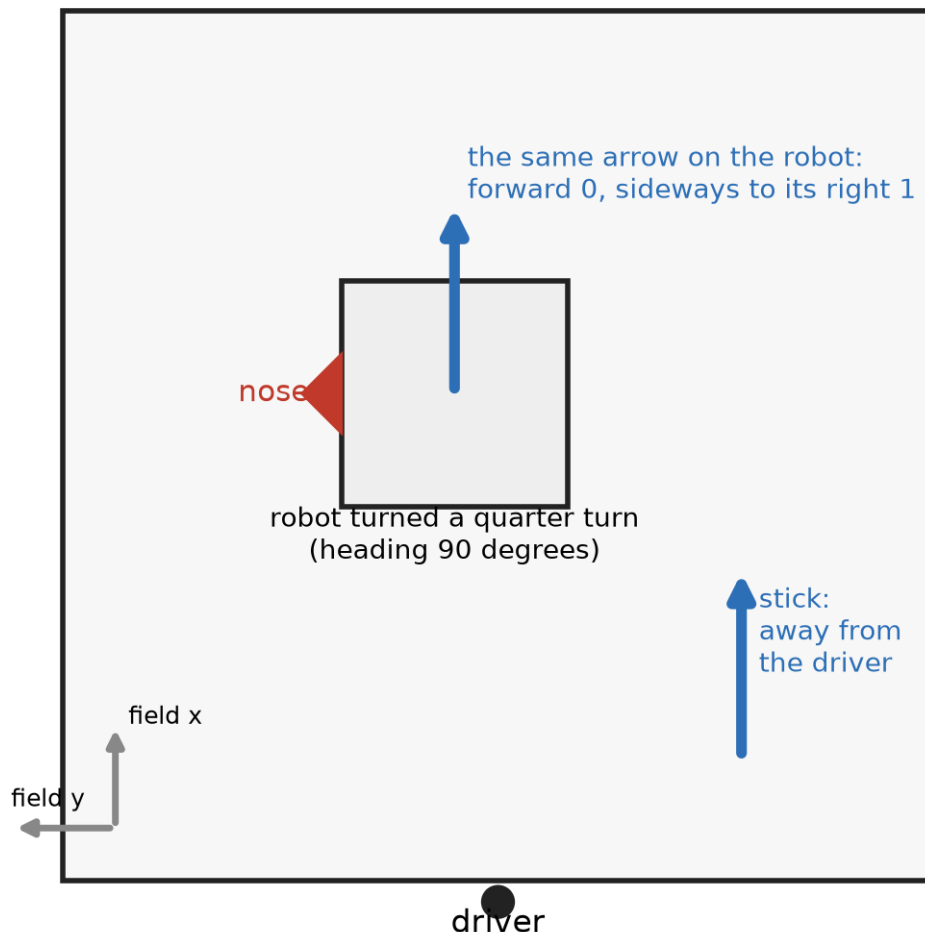


Figure 2.40. The field from above with the driver at the bottom, the robot turned a quarter turn, the stick's direction as an arrow on the field, and the same arrow drawn again on the robot showing what forward and sideways it turns into.

From now on, `forward` on the stick means away from the driver, along the field's `x`. `strafe` means to the driver's left, along the field's `y`. On AdvantageScope's 2D Field, `+x` runs to the right and `+y` runs up.

The wheels only know the robot's own forward and sideways. L120 turned a step on the robot into a step on the field, through the heading. This turns the other way, from the field to the robot, so the signs on `sin` swap.

Change. Press `Ctrl+C` in the Terminal to stop the simulator. In `loop()`, find the comment that names this step. Delete the line `drivetrain.sticks(forward, strafe, turn);` under it, and type:

```
// When on L140S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
```

```

/1140.html#1140s040
// for what to do here.
double heading = follower.pose().heading();
double robotForward = forward * Math.cos(heading) + strafe *
Math.sin(heading);
double robotStrafe = -forward * Math.sin(heading) + strafe *
Math.cos(heading);
drivetrain.sticks(robotForward, robotStrafe, turn);

```

The follower knows which way the robot faces. The drivetrain does not. So the OpMode asks the follower, and does the turn itself.

Simulator. Run the same command again, and connect AdvantageScope.

Check. Hold the same two sticks. The robot spins, and at the same time it travels to the right on the 2D Field, along +x. x_{in} climbs while $heading_{deg}$ goes round and round.

L140S050: drive it facing the wrong way

Send. Press Ctrl+C in the Terminal to stop the simulator. Send the code to the robot, as in L050. On the Driver Station, pick L140 Field Relative from the Lessons group. Stand off the robot's left side, so that away from you is the field's +x. Press INIT, then PLAY.

Check. Push the right stick until the robot's nose points at you. Then push the left stick away from you. The robot drives away from you while it faces you. Turn it a quarter turn and try again, and the stick still means the same way on the field.

2.16 L150: give the driver both

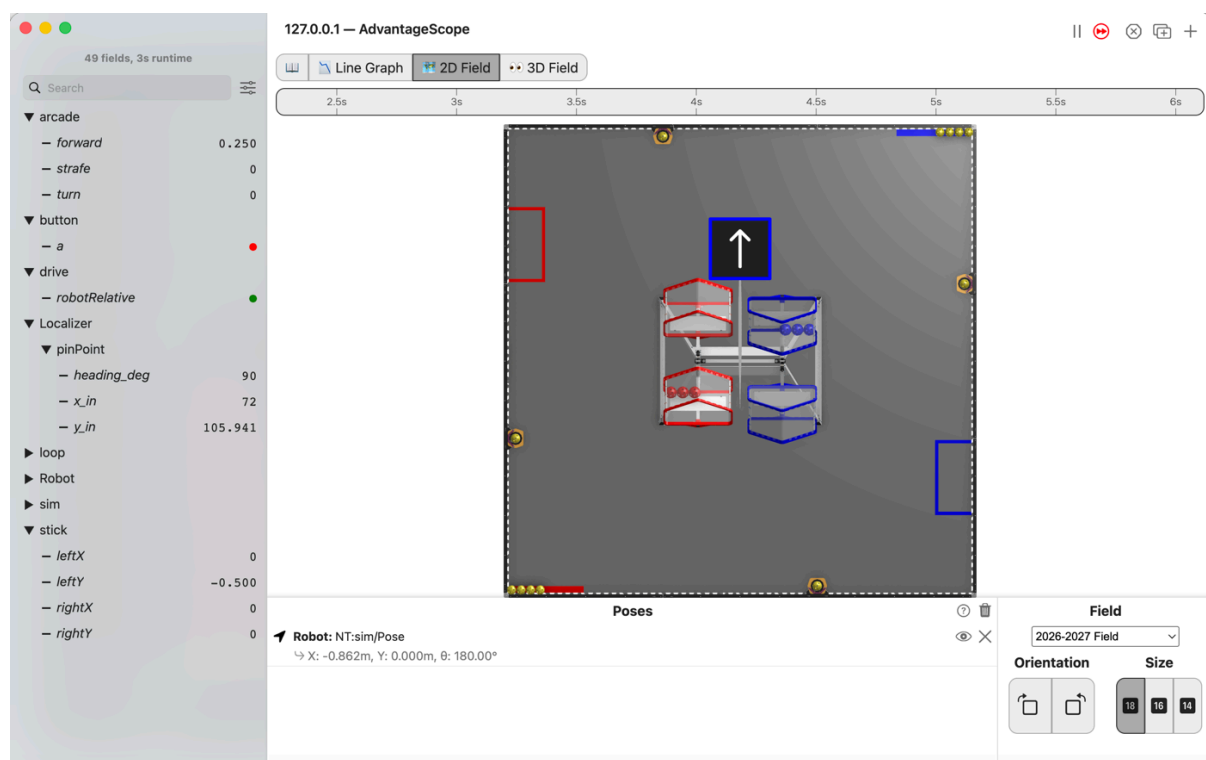


Figure 2.41. AdvantageScope's 2D Field with the robot above the middle of the field, its arrow pointing up the field, and the list showing drive/robotRelative green, arcade/forward at 0.250, and pinPoint x_{in} at 72, y_{in} at 105.941 and $heading_{deg}$ at 90.

The sticks mean the field, the way L140 left them, until the right bumper is held. While it is held they

mean the robot. Let go and they mean the field again.

2.16.1 The steps

L150S010: copy the starter into mytry

Copy. In lessons, click L150RobotRelativeButtonOpMode once and copy it. Click mytry once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check. mytry holds L150RobotRelativeButtonOpMode, and the new file is open.

L150S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check. Under the step's comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L150S030: choose with the bumper

Field relative is the easier way to drive most of the time. It is not the easier way to line up against something. Backing straight out of a corner, or nudging along a wall, is easier when forward means the robot's own forward.

So the driver gets both, and an `if` chooses. The starter already works out `robotForward` and `robotStrafe`, L140's turn. `forward` and `strafe` are the sticks as they are, the robot's own directions.

Change. In `loop()`, find the comment that names this step. Delete the line `drivetrain.sticks(robotForward, robotStrafe, turn);` under it, and type:

```
double heading = follower.pose().heading();
double robotForward = forward * Math.cos(heading) + strafe *
Math.sin(heading);
double robotStrafe = -forward * Math.sin(heading) + strafe *
Math.cos(heading);
// When on L150S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l150.html#l150s030
// for what to do here.
boolean robotRelative = gamepad1.right_bumper;
Tracker.publish("drive/robotRelative", robotRelative);
if (robotRelative) {
    drivetrain.sticks(forward, strafe, turn);
} else {
    drivetrain.sticks(robotForward, robotStrafe, turn);
}
```

When you type `{` and press Enter, Android Studio adds the `}` for you.

The bumper is read fresh every loop. Let go, and the next loop drives field relative again. The driver's thumb tells them which way the sticks work, so there is nothing to remember in a match.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L150RobotRelativeButtonOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the drive folder, then Localizer and `pinPoint`.

Check. Hold the right bumper and push the left stick a little forward. `robotRelative` turns green, and the robot drives up the field, the way its nose points. `y_in` climbs and `x_in` stays at 72. Let go of the bumper and keep pushing. The robot slides to the right, along `+x`, and `x_in` climbs.

L150S040: drive both ways on the floor

Send. Press Ctrl+C in the Terminal to stop the simulator. Send the code to the robot, as in L050. On the Driver Station, pick L150 Robot Relative Button from the Lessons group. Stand off the robot's left side, as in L140. Press INIT, then PLAY.

Check. Turn the robot so its nose points to one side, and push the left stick away from you. The robot drives away from you. Now hold the right bumper and push the same way. The robot drives along its own nose. Let go, and it drives away from you again.

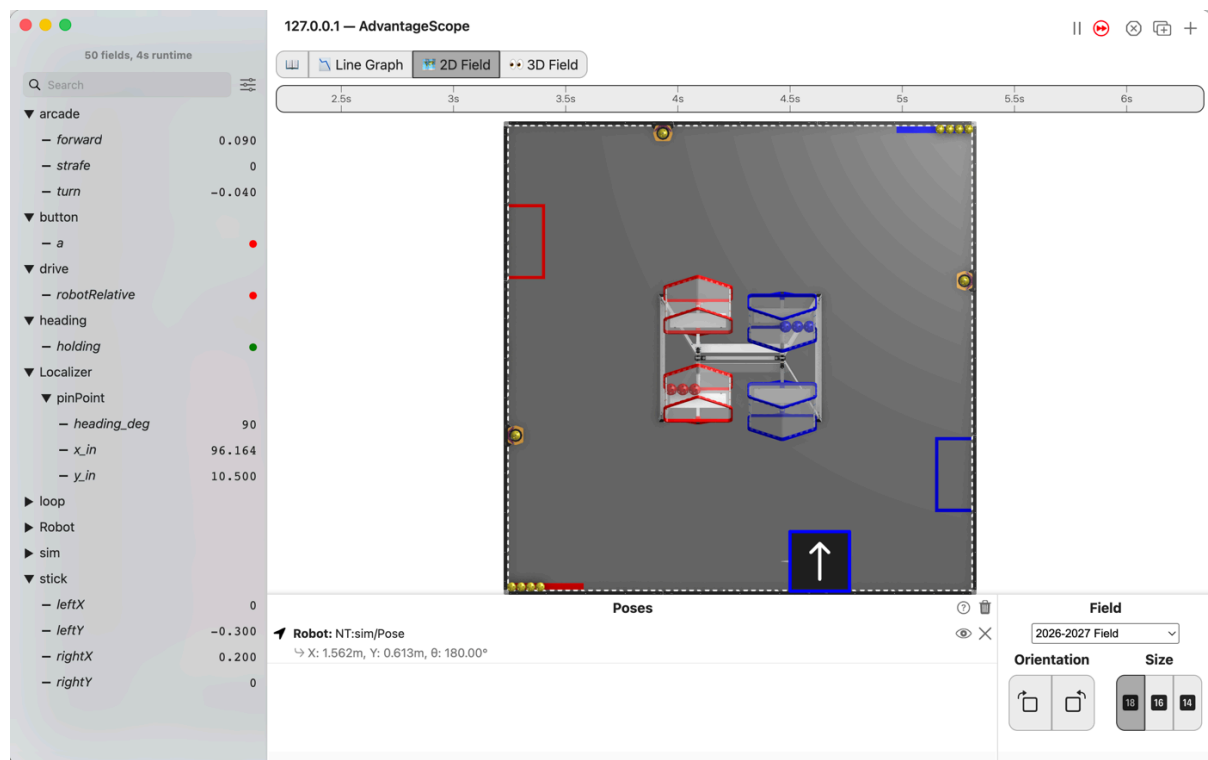
2.17 L160: hold the heading the driver left

Figure 2.42. AdvantageScope's 2D Field with the robot near the bottom of the field, right of the middle, its arrow still pointing up the field, and the list showing arcade/turn at -0.040, heading/holding green, and pinPoint/heading_deg at 90 and x_in at 96.164.

Let go of the turn stick and the robot keeps pointing where it was left. Push the turn stick and the driver has it back at once.

2.17.1 The steps**L160S010: copy the starter into mytry**

Copy. In lessons, click L160HeadingHoldOpMode once and copy it. Click mytry once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check. mytry holds L160HeadingHoldOpMode, and the new file is open.

L160S020: switch your copy on

Change. Delete the line @Disabled, under the comment that names this step.

Check. Under the step's comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L160S030: watch the nose creep

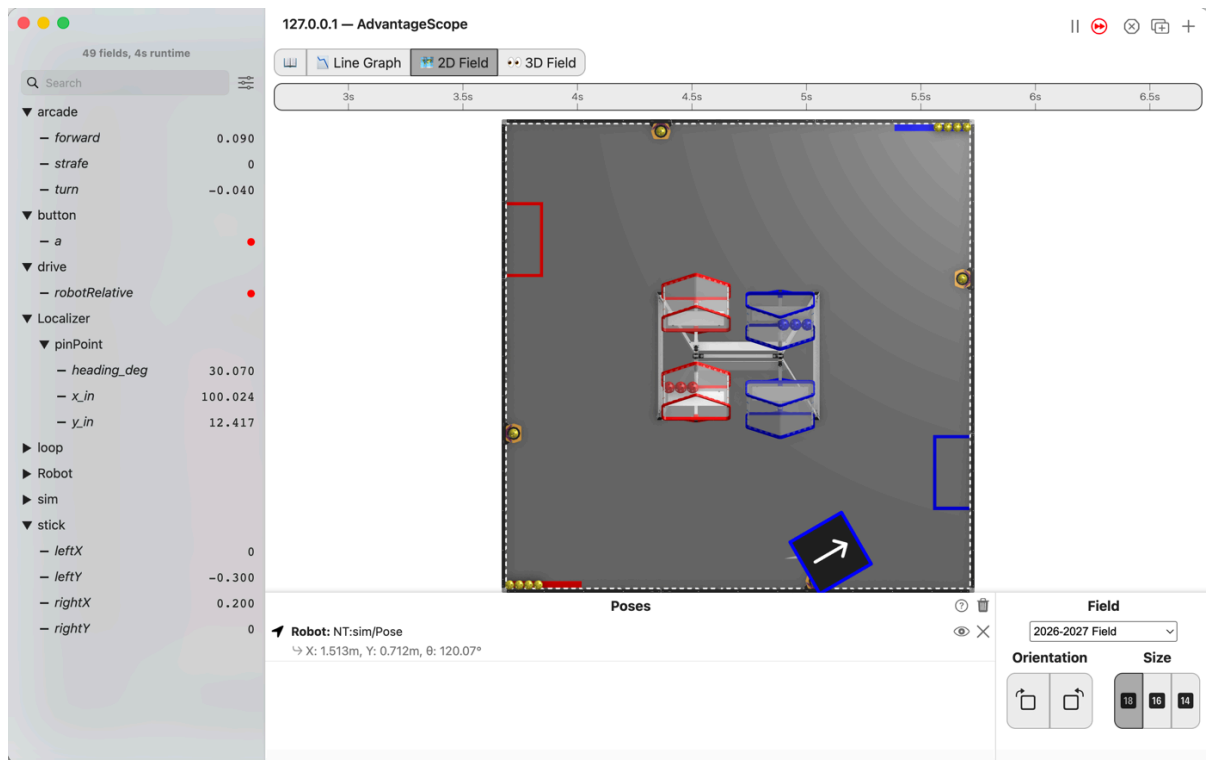


Figure 2.43. AdvantageScope's 2D Field with the robot near the bottom of the field, right of the middle, turned so its arrow points up and to the right, and the list showing arcade/turn at -0.040 and pinPoint/heading_deg at 30.070.

The starter is L150's finished OpMode. This run pretends the turn stick rests a little off centre, the way a worn stick does.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L160HeadingHoldOpMode --left_stick_y=-0.3 --right_stick_x=0.2"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the arcade folder, then Localizer and pinPoint.

Check. turn reads -0.040, which is small. The robot drives to the right, and it turns as it goes. heading_deg started at 90 and keeps dropping.

L160S040: hold the heading

HeadingHold is written for you, in base. While the turn stick is pushed, it hands the stick straight back and the driver steers. The loop the stick is let go, it remembers the heading. After that it hands back the turn that steers the robot back to that heading.

Its own deadband is 0.05, so a stick that is nearly centred counts as let go. turn at -0.040 is inside it.

Change. Press Ctrl+C in the Terminal to stop the simulator. Find the line CorbelsMecanum drivetrain; near the top of the class. Under it, type:

```
CorbelsMecanum drivetrain;
HeadingHold hold = new
HeadingHold(Constants.foresightConfig.headingFeedback.get());
```

Constants.foresightConfig holds the follower's own settings. headingFeedback is the part that steers the heading in autonomous. Tune the follower, and this hold gets the same tuning.

HeadingHold and Constants turn red. Click each one and press Option+Enter, as in L125. The popup for Constants names org.firstinspires.ftc.teamcode.pedro.Constants.

Change. In loop(), find the comment that names this step. Under it, type:

```
double turn = squared(deadband(-rightStickX, band));
Tracker.publish("arcade/turn", turn);
// When on L160S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l160.html#l160s040
// for what to do here.
turn = hold.turn(follower, turn);
Tracker.publish("heading/holding", hold.target() != null);
```

The first new line swaps the stick's turn for the hold's. Both sticks calls below use turn, so both drive with it. Android Studio now underlines turn everywhere. That marks a variable that changes after it is made. It is not an error.

Simulator. Run the same command again, and connect AdvantageScope. Open the heading folder too.

Check. holding is green. The robot drives to the right, its nose still pointing up the field, and heading_deg stays at 90.

L160S050: drive it and then push it

Send. Press Ctrl+C in the Terminal to stop the simulator. Send the code to the robot, as in L050. On the Driver Station, pick L160 Heading Hold from the Lessons group. Press INIT, then PLAY.

Check. Drive a long diagonal with the left stick, and do not touch the turn stick. The nose stays where it was. Then take hold of the robot and twist it a little. It turns back on its own.

2.18 L170: press Y and the robot drives itself there

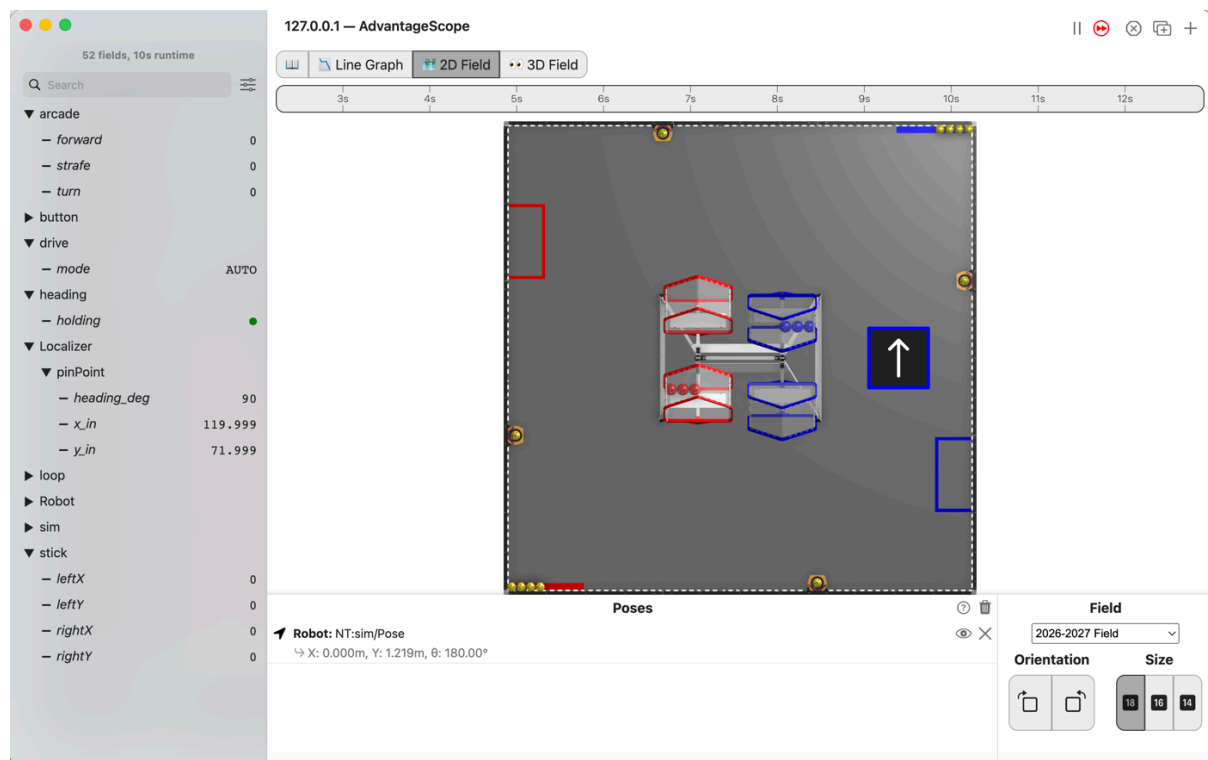


Figure 2.44. AdvantageScope's 2D Field with the robot right of the middle of the field, its arrow pointing up

the field, and the list showing `drive/mode` at `AUTO`, every stick at 0, and `pinPoint/x_in` at 119.999, `y_in` at 71.999 and `heading_deg` at 90.

Press Y and the robot drives itself to a spot on the field. Push any stick and the driver has it back at once.

2.18.1 The steps

L170S010: copy the starter into mytry

Copy. In `lessons`, click `L170DriveToPoseOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check. `mytry` holds `L170DriveToPoseOpMode`, and the new file is open.

L170S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check. Under the step's comment, the next line reads `// public: the robot only runs an OpMode that is public`.

L170S030: give Y a pose

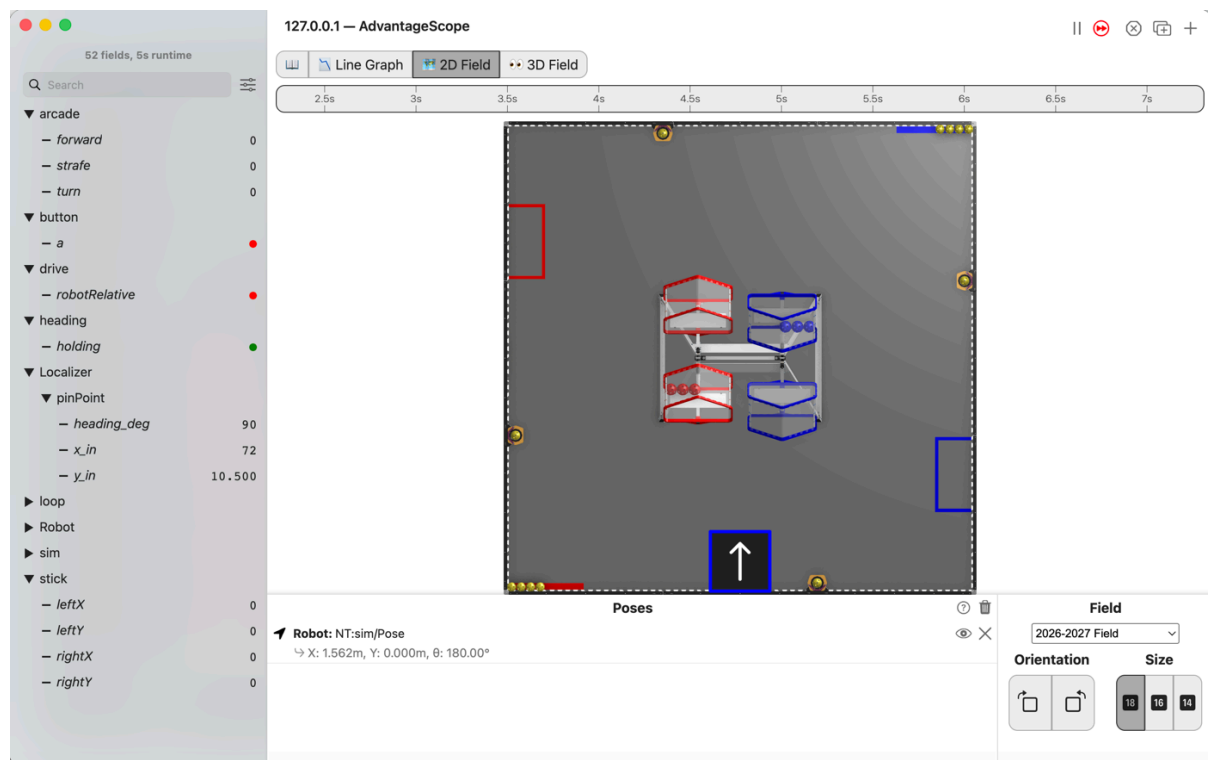


Figure 2.45. AdvantageScope's 2D Field with the robot at the bottom middle of the field, its arrow pointing up, and the list showing every stick at 0 and `pinPoint/x_in` at 72, `y_in` at 10.500 and `heading_deg` at 90.

The starter is L160's finished `OpMode`, moved around a little. All the driving now sits in `driveTheRobot()`, which `loop()` calls. Near the top are three new fields. `target` is the pose, at x 120, y 72 and heading 90. `drivingItself` remembers whether Y sent the robot there.

`bindings()` runs once, when **PLAY** is pressed. It is the place to say what each button does.

Change. In `bindings()`, find the comment that names this step. Under it, type:

```
// When on L170S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
```

```

/1170.html#1170s030
// for what to do here.
buttons.whenPressed(() -> gamepad1.y, Commands.instant(() -> {
    follower.hold(target);
    drivingItself = true;
}));

```

Android Studio closes the brackets for you, as `}})`. Type the `;` after them yourself.

Commands turns red. Click it and press Option+Enter, as in L125. The popup names `com.pedropathing.ivy.commands.Commands`.

`whenPressed` runs its command once, the loop Y goes down. Holding Y does not run it again. The command tells the follower to drive to `target` and stay there.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L170DriveToPoseOpMode --y=true"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the Localizer folder, then `pinPoint`.

Check. The robot does not move. `x_in` stays at 72 and `y_in` at 10.5. The follower wants to go, but every loop the sticks say stop, and since L110 the sticks win.

L170S040: let the follower drive until the driver takes over

`driverWantsControl` is in the starter. It is true when any stick is pushed past the deadband.

Change. Press Ctrl+C in the Terminal to stop the simulator. In `driveTheRobot()`, find the comment that names this step. Under it, type:

```

boolean driverWantsControl = Math.abs(leftStickY) > band
    || Math.abs(leftStickX) > band || Math.abs(rightStickX) > band;
// When on L170S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/1170.html#1170s040
// for what to do here.
if (drivingItself) {
    if (!driverWantsControl) {
        Tracker.publish("drive/mode", "AUTO");
        drivetrain.releaseCommandedWheels();
        return;
    }
    drivingItself = false;
    follower.manual(0, 0, 0);
}
Tracker.publish("drive/mode", "DRIVER");

```

While the robot drives itself and no stick is pushed, `releaseCommandedWheels()` lets go of the wheels, and `return` leaves `driveTheRobot()` before the sticks are read. The follower is left driving. Once a stick is pushed, `drivingItself` goes back to false and `follower.manual` stops the follower. The sticks drive again, from this loop on.

Simulator. Run the same command again, and connect AdvantageScope. Open the drive folder too.

Check. `mode` reads AUTO. The robot drives right and up the field and stops at `x_in` 120 and `y_in` 72, its nose still pointing up the field. It may show 119.999 and 71.999.

L170S050: send it and then take it back

Send. Press Ctrl+C in the Terminal to stop the simulator. Send the code to the robot, as in L050. On the Driver Station, pick L170 Drive To Pose from the Lessons group. Press INIT, then PLAY.

Check. Press Y once and let go. The robot drives itself toward the middle of the field, right of where it started. Press Y again and push the left stick while it is on its way. It stops going there, and drives the way the stick says.

2.19 L180: one OpMode a driver could take to a match

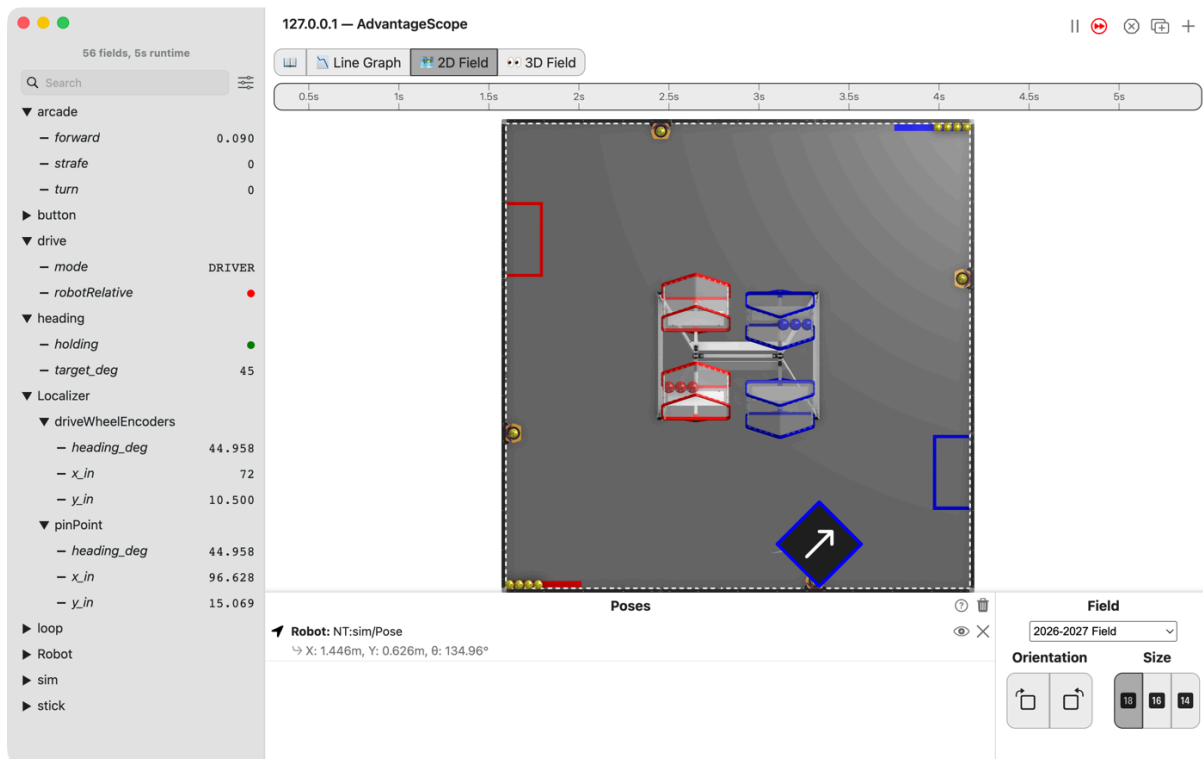


Figure 2.46. AdvantageScope's 2D Field with the robot near the bottom of the field, right of the middle, its arrow pointing up and to the right, and the list showing drive/mode at DRIVER, heading/holding green and target_deg at 45, driveWheelEncoders at x 72, y 10.500 and heading 44.958, and pinPoint at x 96.628, y 15.069 and heading 44.958.

One teleop with all of it in. The sticks drive field relative and hold the heading they were left at. The right bumper means the robot's own directions. A points the robot at 45 degrees while the driver drives. Y sends the robot to its spot until a stick moves. And your own localizer from L120 runs beside the robot's.

2.19.1 The steps

L180S010: copy the starter into mytry

Copy. In lessons, click L180CombinedOpMode once and copy it. Click mytry once and paste. In the Copy Class box, click OK. If Android Studio asks whether to add the file to Git, click Add.

Check. mytry holds L180CombinedOpMode, and the new file is open.

L180S020: switch your copy on

Change. Delete the line @Disabled, under the comment that names this step.

Check. Under the step's comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L180S030: run your localizer again

Change. Find `shadowLocalizers()` and the comment that names this step. Under it, type the same two lines as in L120:

```
// When on L180S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l180.html#l180s030
// for what to do here.
shadowLocalizers.add("driveWheelEncoders",
    new L120EncoderLocalizer(new HardwareWheelSource(hardware)));
```

HardwareWheelSource turns red. Click it and press Option+Enter, as in L120. L120EncoderLocalizer does not turn red. It is your own, from L120, and it sits in mytry too.

Keep the name driveWheelEncoders. AdvantageScope shows each localizer under its name, so a run of this OpMode and a run of L120 line up.

Check. Nothing in the file is red.

L180S040: read what you are about to drive

Nothing else in the starter is new. Read bindings() and driveTheRobot(), and find the lesson each part came from.

- Both drive sticks go through deadband and then squared, from L060.
- forward and strafe are turned through the heading so they mean the field, from L140.
- The right bumper picks the robot's own directions, from L150.
- hold.turn keeps the heading the turn stick was left at, from L160.
- Y sends the robot to target, and any stick takes it back, from L170.

A is the one part that is new here, and it is HeadingHold again. hold.aimAt gives the hold a heading the driver did not leave, 45 degrees. The hold turns the robot there, and the left stick still drives. Push the turn stick and the driver has the turn back.

When a stick takes the robot back from Y, hold.release() forgets the heading the hold had before Y. Read the code without it: the hold would turn the robot back to that old heading.

Simulator. In the Terminal, run this. It holds A and pushes the left stick a little forward:

```
./gradlew :TeamCode:simRun --args="mytry.L180CombinedOpMode --a=true
--left_stick_y=-0.3"
```

AdvantageScope. Connect to the simulator as in L020, and click **2D Field**. Open the heading folder. Then open Localizer and both folders in it.

Check. target_deg reads 45. The robot turns until its arrow points up and to the right, and pinPoint/heading_deg comes to about 45. All the while it drives to the right, and pinPoint/x_in climbs. driveWheelEncoders is there. Its heading comes to about 45 too, because the simulator turns the IMU, but it stays at x 72 and y 10.5. The simulator turns no wheel encoders, so your localizer counts no distance.

L180S050: drive all four

Send. Press Ctrl+C in the Terminal to stop the simulator. Send the code to the robot, as in L050. On the Driver Station, pick L180 Combined from the Lessons group. Press INIT, then PLAY.

Check. Try them in this order: field relative, the bumper held, A, and Y. Each one does what its own lesson did, and switching between them needs nothing but a thumb. On the robot, driveWheelEncoders follows pinPoint as the robot drives.

2.20 L190: ask the wheels for a speed

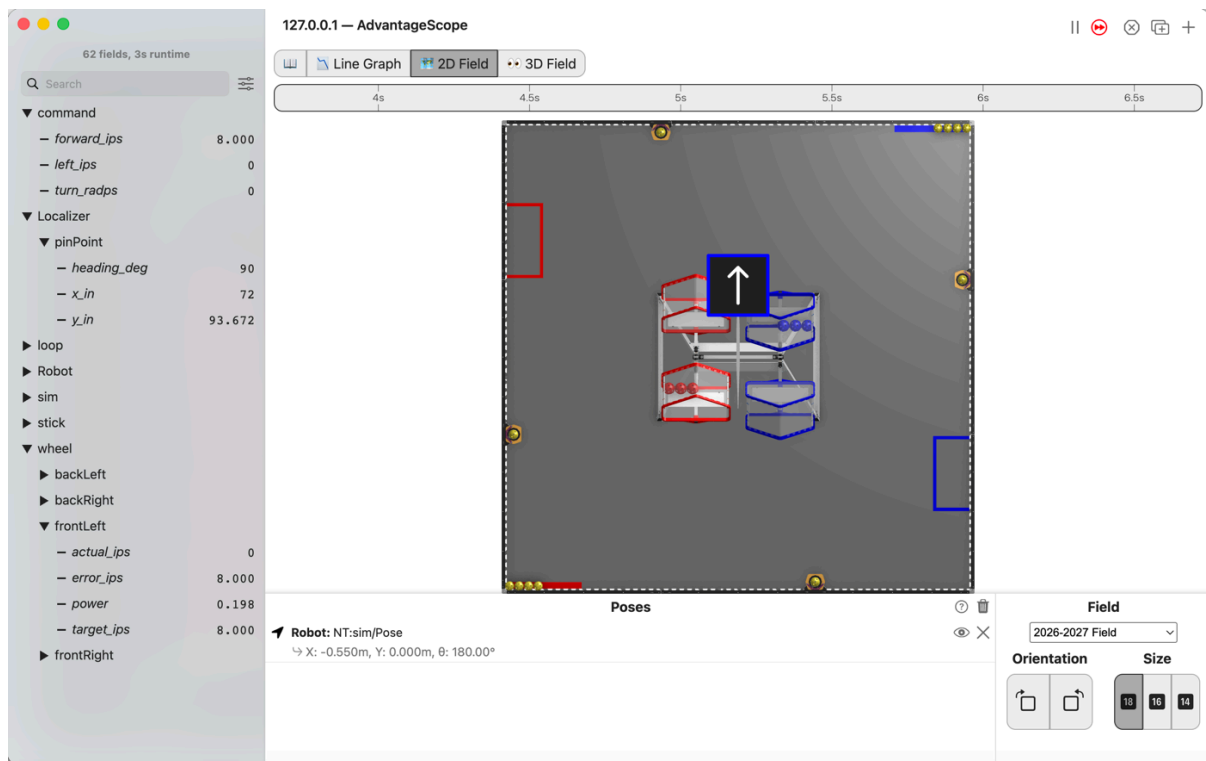


Figure 2.47. AdvantageScope's 2D Field with the robot in the middle of the field, its arrow pointing up, and the list showing command/forward_ips at 8, pinPoint/y_in at 93.672, and wheel/frontLeft with target_ips at 8, actual_ips at 0, error_ips at 8 and power at 0.198.

Full stick means 40 inches per second, not full power. The drivetrain measures how fast each wheel turns and makes up the difference, on a fresh battery or a flat one, on the flat or up a slope.

2.20.1 The steps

L190S010: copy the two starters into mytry

This lesson has two files, as L110 did. One is the OpMode. The other is the drivetrain, which turns a speed into a power.

Copy. In lessons, click L190SpeedDriveTrain once and copy it. Click mytry once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**. Do the same for L190VelocityDriveOpMode.

Check. mytry holds L190SpeedDriveTrain and L190VelocityDriveOpMode.

L190S020: switch your copy on

Change. In L190VelocityDriveOpMode, delete the line @Disabled, under the comment that names this step.

Check. Under the step's comment, the next line reads `// public: the robot only runs an OpMode that is public.`

L190S030: make room for the drivetrain

Change. Near the top of the class, find the comment that names this step. Under it, type:

```
// When on L190S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/l190.html#l190s030
// for what to do here.
L190SpeedDriveTrain drivetrain;
```

Check. The line has no red underline. `L190SpeedDriveTrain` is in `mytry` too, so it needs no import.

L190S040: hand the drivetrain to the follower

Change. In `init()`, find the comment that names this step. Under it, type the first line below. Then change `initAfter();` to `initAfter(drivetrain);`:

```
// When on L190S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/l190.html#l190s040
// for what to do here.
drivetrain = new L190SpeedDriveTrain(hardware);

initAfter(drivetrain);
```

This is L110S050 again, with the new drivetrain.

Check. Neither line has a red underline.

L190S050: turn the sticks into speeds

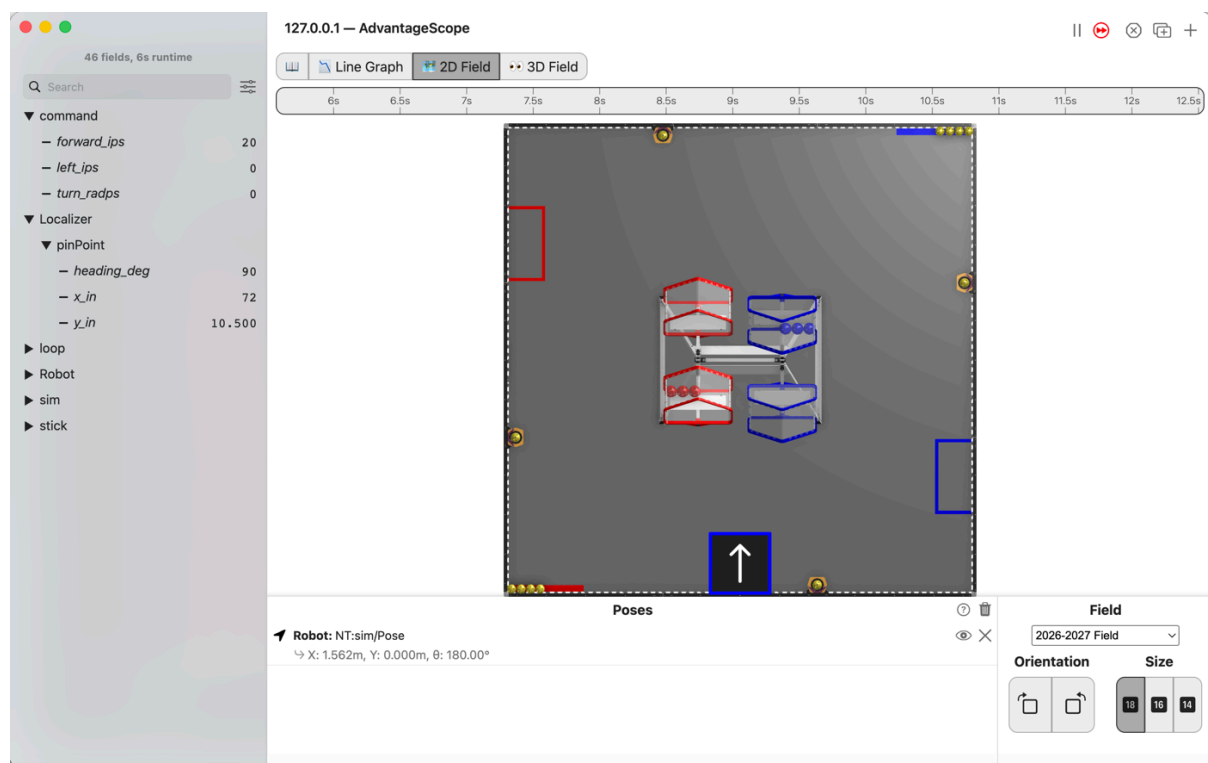


Figure 2.48. AdvantageScope's 2D Field with the robot at the bottom middle of the field, its arrow pointing up, and the list showing `command/forward_ips` at 20, `left_ips` and `turn_radps` at 0, and `pinPoint` at `x 72, y 10.500` and heading 90, with no wheel folder.

`MAX_IPS` is 40, in inches per second. `MAX_TURN_RADPS` is pi radians per second, half a turn each second. Both are near the top of the class.

Change. In `loop()`, find the comment that names this step. Under it are three lines that end in `= 0;`.

Change each 0 as below. Then, under the three publish lines, type the last two lines:

```
// When on L190S050, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
// /1190.html#1190s050
// for what to do here.
double forwardInPerS = deadband(-leftStickY, band) * MAX_IPS;
double strafeInPerS = deadband(-leftStickX, band) * MAX_IPS;
double turnRadPerS = deadband(-rightStickX, band) * MAX_TURN_RADPS;
Tracker.publish("command/forward_ips", forwardInPerS);
Tracker.publish("command/left_ips", strafeInPerS);
Tracker.publish("command/turn_radps", turnRadPerS);
double[] target = WheelTargets.forMecanum(forwardInPerS, strafeInPerS,
turnRadPerS, Constants.turnRadiusInches);
drivetrain.setCommandedWheelSpeeds(target[0], target[1], target[2],
target[3]);
```

The minus signs are the ones from L060. A stick pushed up reads less than 0.

`WheelTargets.forMecanum` works out how fast each wheel has to turn for the robot to move like that. It hands back an array, front left first, then front right, back left and back right. `turnRadiusInches` is how far each wheel sits from the middle of the robot.

`WheelTargets` and `Constants` turn red. Hold the mouse over each one until a box offers **Import class**, and click it. The box for `Constants` names `org.firstinspires.ftc.teamcode.pedro.Constants`.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L190VelocityDriveOpMode
--left_stick_y=-0.5"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the `command` folder. Then open `Localizer` and `pinPoint`.

Check. `forward_ips` reads 20, half of 40. The robot does not move, and `y_in` stays at 10.5. The OpMode asks for a speed, but `setCommandedWheelSpeeds` in your `drivetrain` is still empty.

L190S060: turn a speed into a power

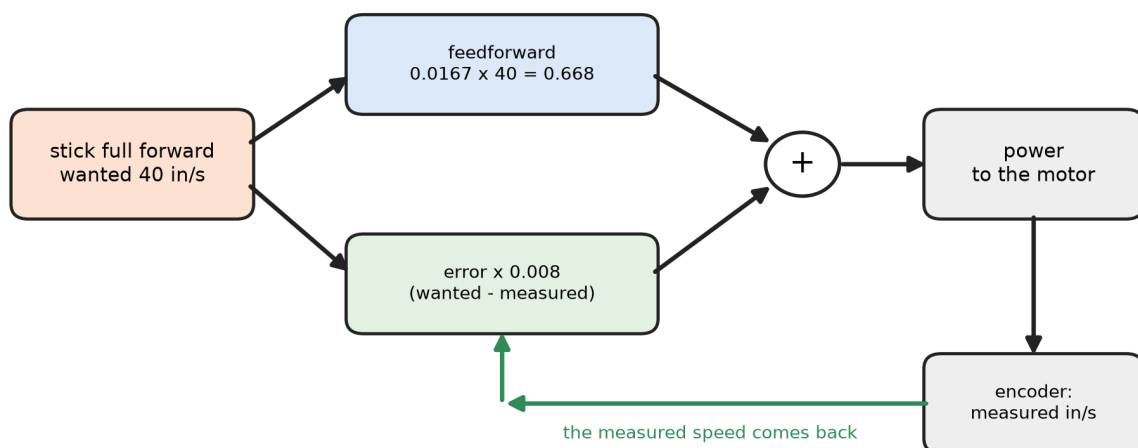


Figure 2.49. The loop drawn once round: the stick asking for 40 in/s, the feedforward guess at the power, the encoder's measured speed coming back, and the error being added in.

Open `L190SpeedDriveTrain`. It extends `CorbelsMecanum`, so it has everything L180's drive-train had. `setCommandedWheelSpeeds` puts the four speeds into the array `wanted`. Each wheel gets its power from two parts added together.

- **Feedforward** is a guess, made before anything is measured. `Constants.powerPerInchPerSecond` is the power one inch per second needs, so the guess gets most of the way there at once.
- **Feedback** is the correction. `measuredSpeeds.all()` reads how fast each wheel really turns. The difference between wanted and measured, times `VELOCITY_KP`, is added on.

The guess is always a little off, and the correction only grows once the wheel is off its speed. Together they get there and stay there.

A for loop runs the lines inside its braces once for each wheel. `i` starts at 0 and goes up by 1 each time, while it is less than 4. So `wanted[i]` is `wanted[FL]` the first time round, then FR, BL and BR.

Change. Press Ctrl+C in the Terminal to stop the simulator. In `setCommandedWheelSpeeds`, find the comment that names this step. Under it, type:

```
// When on L190S060, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l190.html#l190s060
// for what to do here.
double[] measured = measuredSpeeds.all();
double[] powers = new double[4];
for (int i = 0; i < 4; i++) {
    double feedforward = Constants.powerPerInchPerSecond * wanted[i];
    double feedback = VELOCITY_KP * (wanted[i] - measured[i]);
    powers[i] = clampToPower(feedforward + feedback);
}
setCommandedWheels(powers[FL], powers[FR], powers[BL], powers[BR]);
publishWheelSpeeds(wanted, measured, powers);
```

Android Studio types the `}` of the for loop for you, when you press Enter after the `{`.

`clampToPower` keeps each power between -1 and 1. `setCommandedWheels` sends the four powers to the wheels, as in L110. `publishWheelSpeeds` puts each wheel's numbers in the `wheel` folder.

`Constants` turns red. Import it as in L190S050.

Simulator. Run the command again, with the stick pushed less so the robot stays on the field longer:

```
./gradlew :TeamCode:simRun --args="mytry.L190VelocityDriveOpMode
--left_stick_y=-0.2"
```

AdvantageScope. Connect, and open wheel, then frontLeft.

Check. `forward_ips` and `target_ips` read 8. The robot drives up the field. `power` reads 0.198: the guess, 0.134, plus the correction.

`actual_ips` stays at 0, and `error_ips` stays at 8. The simulator moves the robot, but it turns no wheel encoders, so the correction never shrinks. On the robot it does.

L190S070: drive it up something

Send. Press Ctrl+C in the Terminal to stop the simulator. Send the code to the robot, as in L050. On the Driver Station, pick L190 Velocity Drive from the Lessons group. Press INIT, then PLAY.

Check. Drive on the flat first. Then drive straight up the mat, with the stick held in the same place. The robot slows a little at the foot of the slope and then picks back up. In AdvantageScope, `actual_ips` dips below `target_ips`, then climbs back, and `power` rises to do it.

2.21 L195: measure the two numbers L190 trusted

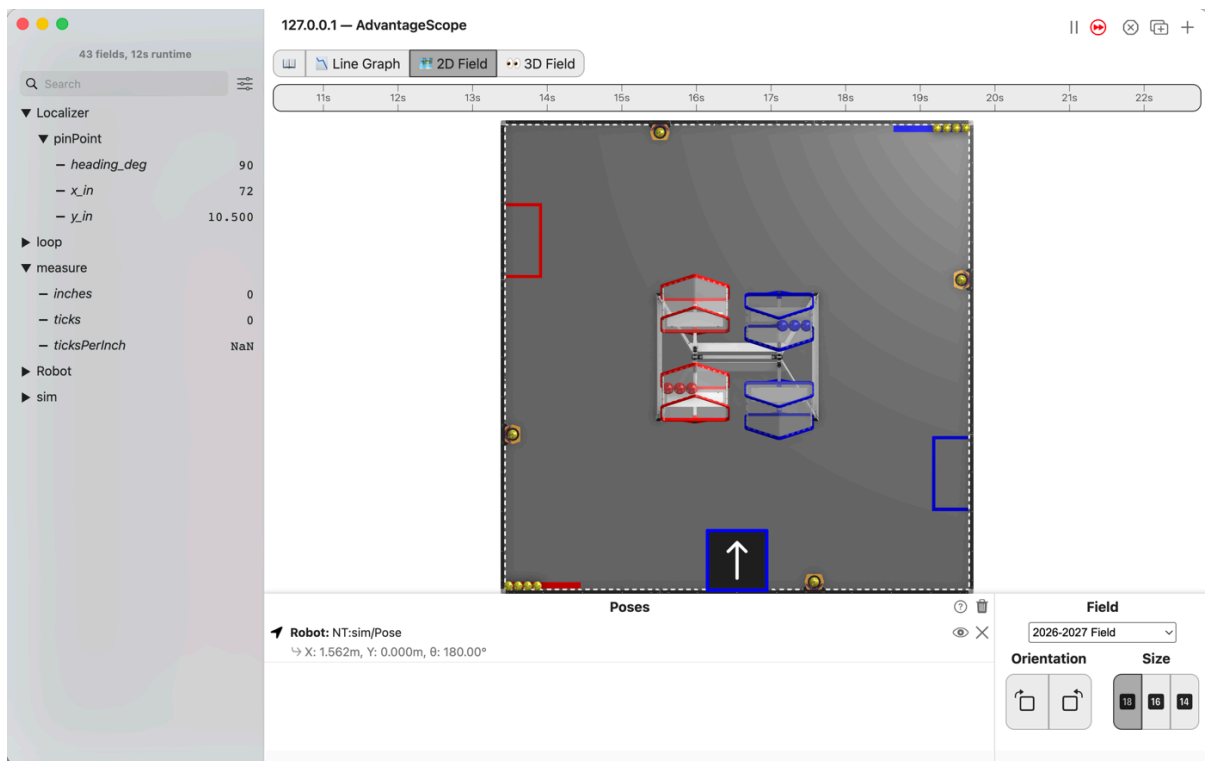


Figure 2.50. AdvantageScope's 2D Field with the robot at the bottom middle of the field, its arrow pointing up, and the list showing pinPoint at x 72, y 10.500 and heading 90, and measure with inches and ticks at 0 and ticksPerInch at NaN.

Two numbers in `Constants.java` that came off this robot instead of somebody else's: how many encoder ticks make an inch, and how far a wheel sits from the middle. The robot measures them while you push it and spin it by hand.

2.21.1 The steps

L195S010: copy the three starters into mytry

This lesson has three files. Two are `OpModes`, one for each measurement. The third is the drivetrain, which tells them what each wheel's encoder has counted.

Copy. In `lessons`, click `L195MeasureTicksPerInchOpMode` once. Hold Shift and click `L195TicksDriveTrain`, so all three are picked. Copy them, click `mytry` once, and paste. In the **Copy** box, click **OK**. If Android Studio asks whether to add the files to Git, click **Add**.

Copy all three at once. Copied one at a time, the `OpModes` still point at the drivetrain in `lessons`, and turn red.

Check. `mytry` holds `L195MeasureTicksPerInchOpMode`, `L195MeasureTurnRadiusOpMode` and `L195TicksDriveTrain`, and none of them shows red.

L195S020: switch both `OpModes` on

Change. In `L195MeasureTicksPerInchOpMode`, delete the line `@Disabled`, under the comment that names this step. Do the same in `L195MeasureTurnRadiusOpMode`.

Check. In both, under the step's comment, the next line reads `// public: the robot only`

runs an OpMode that is public.

L195S030: open a door to the encoders

Open L195TicksDriveTrain. A drivetrain sends four powers out to the motors. `wheelTicks()` is the same shape pointing the other way: it brings four numbers back.

Change. In `wheelTicks()`, find the comment that names this step. Under it, type:

```
// When on L195S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l195.html#l195s030
// for what to do here.
ticks[FL] = hardware.frontLeft.getCurrentPosition();
ticks[FR] = hardware.frontRight.getCurrentPosition();
ticks[BL] = hardware.backLeft.getCurrentPosition();
ticks[BR] = hardware.backRight.getCurrentPosition();
```

`getCurrentPosition()` is how many ticks a motor's encoder has counted since the robot started. FL, FR, BL and BR name the slots, as in L190.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L195MeasureTicksPerInchOpMode"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the measure folder.

Check. `inches` and `ticks` read 0, and `ticksPerInch` reads NaN, which means “not a number”. Nobody can push a robot in the simulator, so it stays at x 72 and y 10.5, and 0 divided by 0 is not a number. The measuring is for the robot.

L195S040: push the robot and measure an inch

The robot keeps two accounts of the same trip. The Pinpoint says how far it went in inches, since its own wheels measure the floor. The drive encoders say how many ticks they counted. One divided by the other is the number.

Send. Press Ctrl+C in the Terminal to stop the simulator. Send the code to the robot, as in L050. On the Driver Station, pick L195 Measure ticks per inch from the Lessons group. Press INIT, then PLAY.

Push. Push the robot slowly straight forward, at least 36 in, keeping it pointing the same way. Push slowly: a wheel that slips travels without turning, and the answer comes out short.

Check. The Driver Station shows `ticksPerInch` with a number after it. Do it three times. Three answers a tick or two apart are a measurement. Open `Constants`, in the `pedro` folder, and put the number in `ticksPerInch`, which is 45.0 until you do. Send the code again and run it once more to check.

L195S050: spin the robot and measure the radius

A wheel r inches from the middle travels r inches for every radian the robot turns. So the radius is the wheel's inches divided by the radians. This OpMode turns ticks into inches with `ticksPerInch`, so L195S040 comes first.

Send. On the Driver Station, pick L195 Measure turn radius. Press INIT, then PLAY.

Spin. Spin the robot on the spot by hand, at least two full turns, keeping it in about the same place.

Check. The Driver Station shows a turn radius. Measure the distance between two opposite wheels, corner to corner, with a tape measure, and halve it. The two agree. Put the number in `turnRadiusInches` in `Constants.java`, which is 8.0 until you do.

2.22 Pedro routes: a cheat sheet

Every way to say where the robot should go, and which ones to avoid. Measured against

`com.pedropathing:core:3.0.1`, the version this project pins, on 2026-09-29. Where the documentation and the jar disagree, the jar is what is written here.

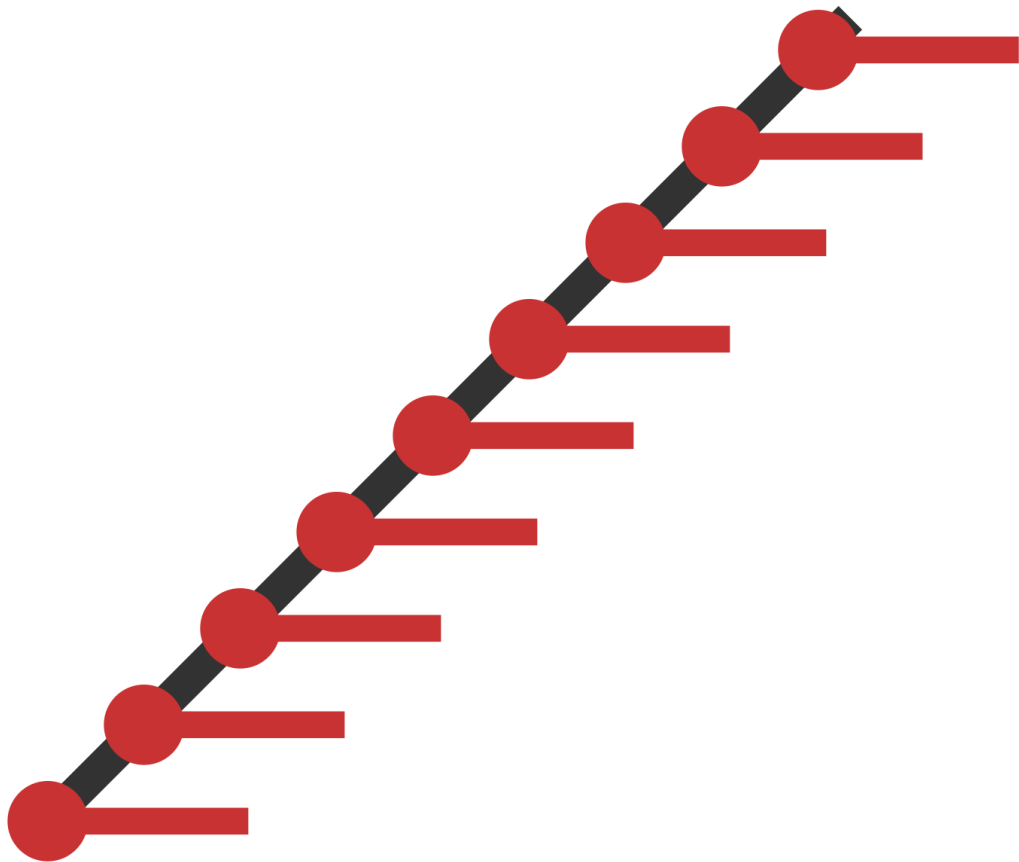
In the pictures the black line is where the robot drives and each red pin is its nose, sampled every eighth of the way along. Both come from `Path.get(t)` and `Path.heading(t)`, so they are the library's own output rather than a drawing of it.

2.22.1 The shapes

A route is a shape plus a heading rule. There are four shapes.

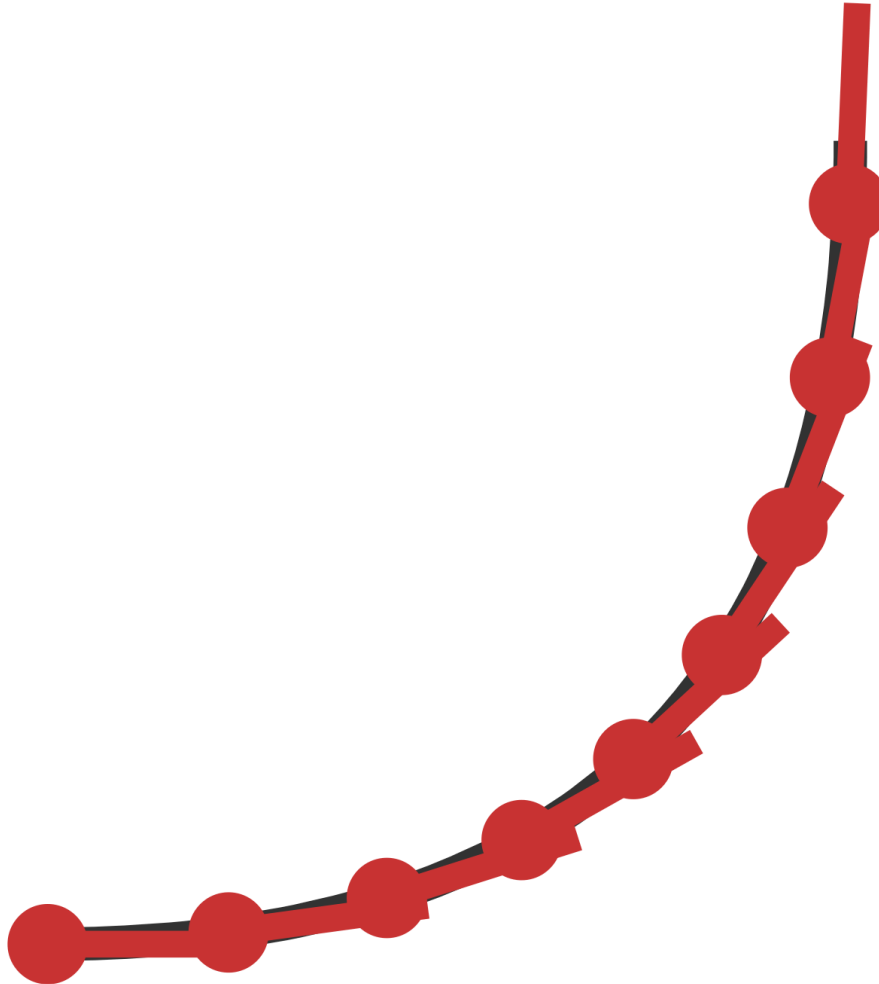
Shape	What it is
<code>Paths.line(start, end)</code>	A straight line between two poses.
<code>Paths.curve(start, control..., end)</code>	A Bézier curve. The first and last poses are the ends; every pose between them pulls the curve without being touched.
<code>Paths.through(poses...)</code>	A curve that passes through every pose given.
<code>Paths.path(paths...)</code>	Several paths run as one, in order.

line



Two poses, nothing between them. `line(start, end).constant(start)` is the plainest route there is and the one to reach for first.

curve



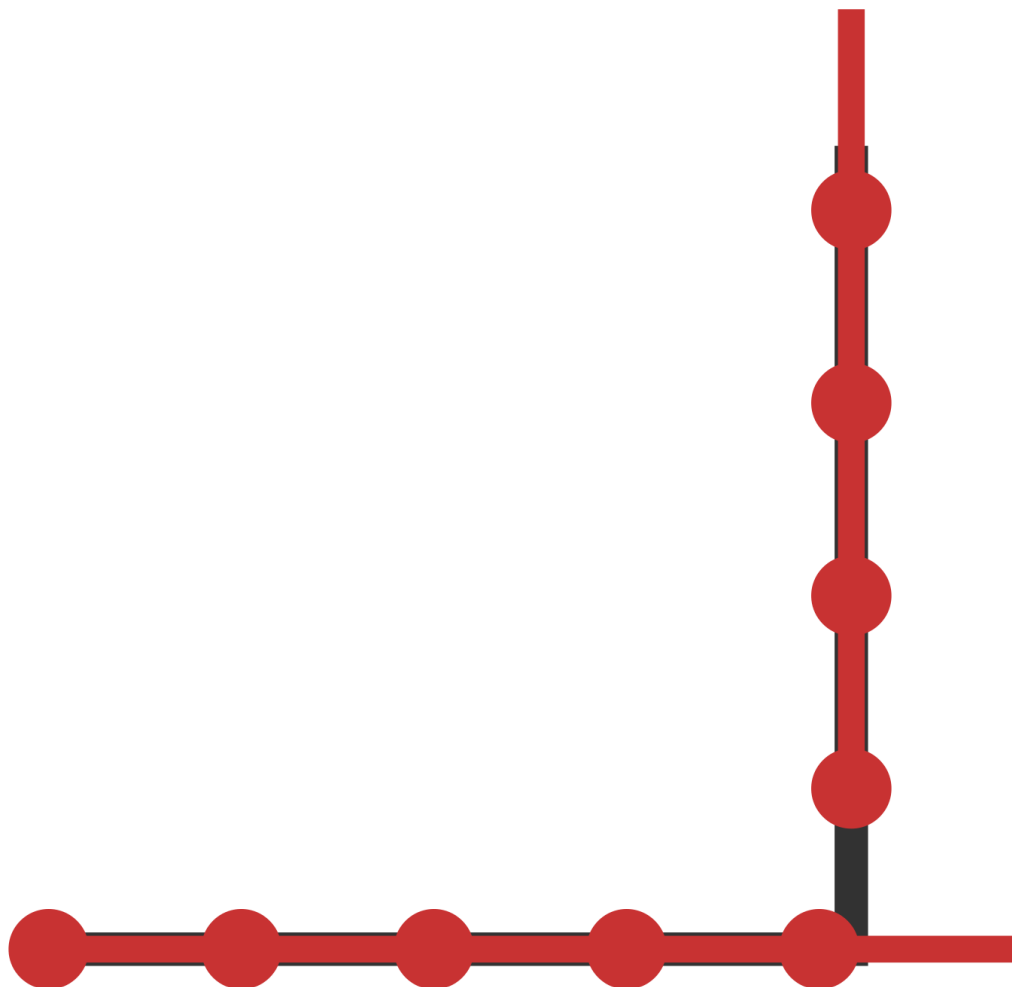
The control pose is a magnet, not a waypoint: the curve bends toward it and never reaches it. Its heading is ignored.

through



Every pose is on the route. The cost is at the ends: with `tangent()` the nose starts at -18.4 degrees and finishes at 108.4 degrees on the same three poses where `curve` runs 0 to 90, because the spline arrives at an angle. Check the end heading rather than assuming it.

path



Two legs of a route, joined. The nose snaps at the corner rather than easing round it: measured 0 degrees for the first leg and 90 for the second, with nothing in between.

2.22.2 The heading rules

Rule	What it does	Use it?
<code>constant (pose)</code>	Holds that pose's heading the whole way.	Yes.
<code>constant (radians)</code>	The same, from a number.	Careful: radians.
<code>tangent ()</code>	Nose follows the direction of travel.	Yes.
<code>reverseTangent ()</code>	Tangent plus 180 degrees, for driving backwards.	Yes.
<code>facingPoint (pose)</code>	Nose always aimed at one spot on the field.	Yes, with one edge.
<code>linear (a, b)</code>	Sweeps from one heading to another along the route.	Not on a line.

<code>longLinear(a, b)</code>	The same, taking the long way round.	No.
<code>Interpolator.pieceswise()</code>	A different rule on each stretch of the route.	Untested here.

There is no default

The documentation says `tangent()` “is the default interpolation type when you create a path”. It is not. A path with no heading rule throws as soon as anything asks for a heading:

```
java.lang.UnsupportedOperationException: No heading interpolator set.
```

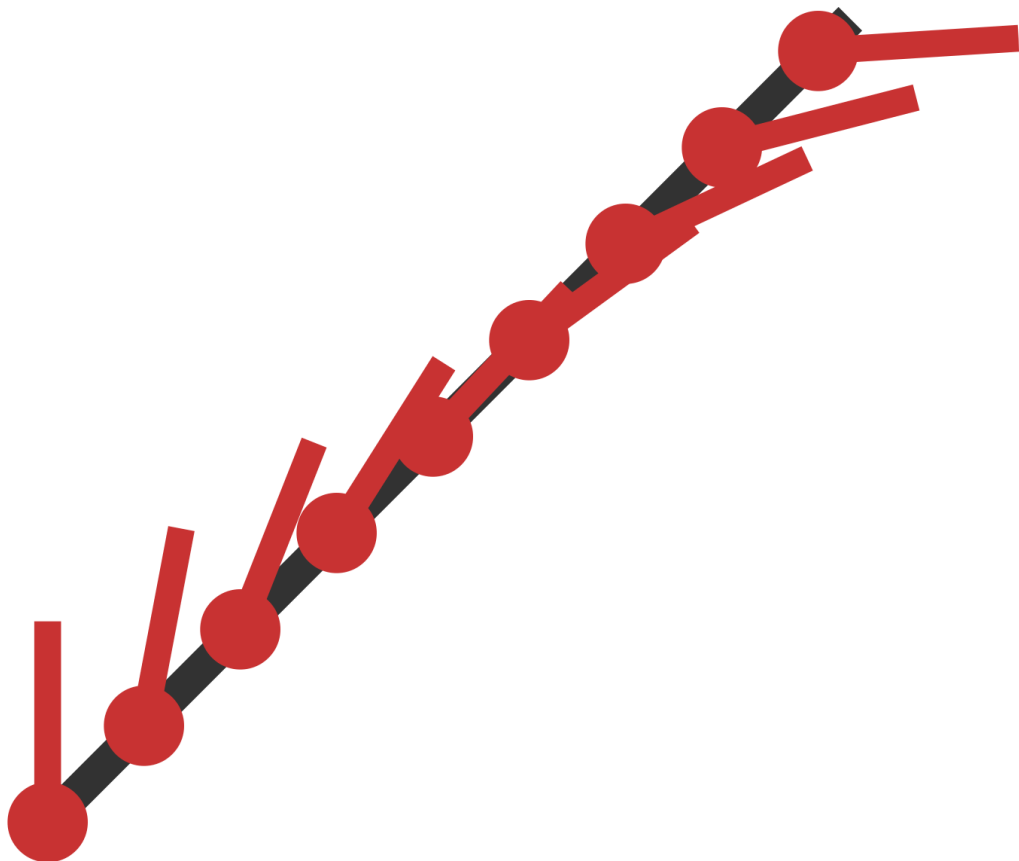
So every route needs a rule. It is not optional.

The number form is radians

`constant(90)` is not 90 degrees. It is 90 radians, which wraps to 116.6 degrees. `constant(0)` is safe only because zero is the same in both.

Two ways to avoid it. Pass a pose, `constant(start)`, and the heading comes from the pose. Or convert, `constant(Math.toRadians(90))`. A pose built by `PoseFactory.degrees()` converts for you; `constant` does not.

2.22.3 Do not use `linear()` on a line



Asked to sweep from 0 to 90 degrees along a line, 3.0.1 sweeps from 90 to 0. Measured:

<code>line(s, e).linear(0, toRadians(90))</code>	<code>t=0</code>	90.0	<code>t=0.5</code>	45.0	<code>t=1</code>	0.0
<code>line(s, e).linear(e, s)</code>	swapped	<code>t=0</code>	0.0	<code>t=0.5</code>	45.0	<code>t=1</code> 90.0

The end pose is wrong too: `endPose().heading()` reports the start heading.

This is [issue 176](#), with [180](#) and [181](#) reporting the same thing. The cause given there is `Curve.pathCompletion()` returning the fraction of the path remaining rather than the fraction completed. All three are closed, and 3.0.1 still does it.

The workaround in those issues is to swap the arguments, and it works, and it reads as a mistake to whoever finds it next. It also fails for a 180 degree turn, where the two ways round are the same distance.

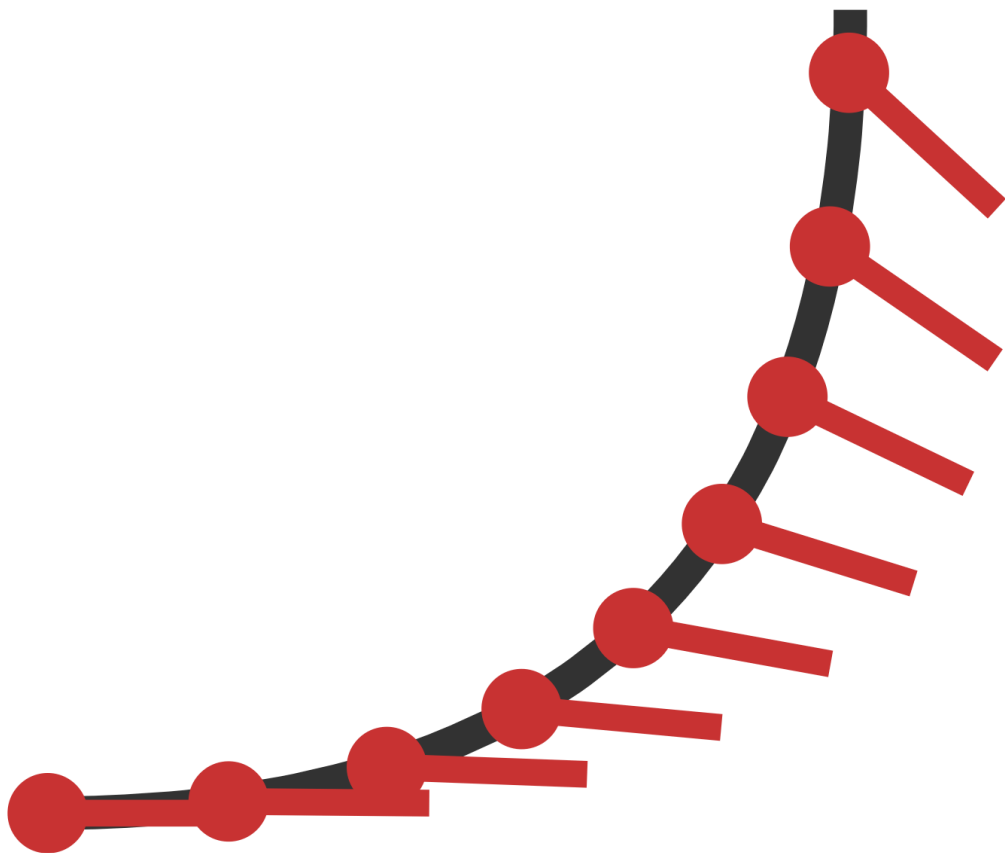
On a curve, `linear()` is correct. Measured on the same three poses: `curve(s, m, e).linear(0, toRadians(90))` runs 0, 24.6, 45, 90. So the bug is the line's, which is what [issue 176](#)'s title says: line and compound paths.

Two ways round it on a straight leg. Hold one heading with `constant` and put the turn on its own leg. Or make the leg a shallow curve, where `linear` behaves.

2.22.4 `longLinear()` is reversed the same way

`longLinear(0, toRadians(90))` runs 90, 157.5, 225, then 0 at the end. Neither end is what was asked for. It is undocumented as well: the documentation page lists `tangent`, `linear`, `constant`, `facingPoint`, `piecewise` and `custom`, and not this.

2.22.5 `facingPoint()` has one edge



It works: the nose tracks the point as the robot moves. The edge is a route that starts *at* the point it faces, where there is no direction to face. Measured on a line leaving the point, the heading reads 0 at $t=0$ and the correct -135 degrees from the next sample on, so the first instant is a discontinuity. [Issue 69](#) is about this interpolator; it is closed.

2.22.6 What this page does not cover

- `Interpolator.piecewise()`, which nothing here has run.

- Custom interpolators.
- Deceleration, path and end constraints, and foresight, each of which has its own reference page and changes how a route is driven rather than what it is.
- Whether any of this behaves on a floor. Everything above is the library's own arithmetic, read out of the jar. When the simulator and the robot disagree, the robot is right.

2.23 Glossary

Every word here is one the task pages use, and each entry names the page that introduces it. Where the guide could have invented a friendlier word and did not, the reason is that the code, Pedro's documentation and the Driver Station all use this one.

OpMode

One program the Driver Station can run. Each lesson is an OpMode, and its name in the list is the one in its @TeleOp or @Autonomous line. [L020](#)

TeleOp and Autonomous

The two kinds of OpMode. A TeleOp reads the gamepads; an autonomous runs on its own. [L125](#) and [L130](#) are the two autonomous lessons. [L125](#)

Driver Station

The phone or tablet that holds the gamepads and picks the OpMode. Lines printed with `Tracker.printToDs` show up there. [L050](#)

Panels

The web page that draws what the code published while it ran. Every `Tracker.publish` key shows up there, and [the cheat sheet](#) lists them. [L050](#)

Drivetrain

The four motors and the code that decides what power each one gets. [L040](#) sets the four powers itself. From [L110](#) on, a lesson's drivetrain is a class of its own. [L040](#)

Deadband

A band of stick readings near the middle that count as let go. Without one the robot creeps when nobody is touching it. [L060](#)

Squaring a stick

Making a small push mean a smaller power, by multiplying the stick by itself and putting the sign back. [L060](#)

Normalizing

Scaling four wheel powers down together when the biggest one is over 1. The robot still goes where it was asked. It only goes slower. [L080](#)

Mixing

Turning forward, sideways and turn into four wheel powers with four sums. [L090](#)

Mecanum

A wheel with rollers set at 45 degrees on its rim, which lets the robot drive sideways. The four wheels' rollers make an X seen from above. [L090](#)

Commanded wheels

Four powers a lesson has set itself. The path follower cannot overrule them. It gets the wheels back when the lesson hands them back. [L110](#)

Follower

Pedro's code that drives the robot along a path, or holds it at a pose. From [L110](#) on it holds the drivetrain and the lessons reach past it. [L110](#)

Pose

Where the robot is and which way it faces: x, y and a heading. A pose is a place on the field, not a distance from the robot. [L120](#)

Heading

Which way the robot is facing, as an angle. Counter-clockwise counts up, and the code keeps it in radians even where a page says degrees. [L120](#)

Localizer

Whatever works out the pose. A robot can run more than one at a time. That is what L120 is for. [L120](#)

Shadow localizer

A second localizer that is read and logged but never steers anything. It is how two answers get compared on one run. [L120](#)

Dead reckoning

Working out where you are by adding up every step you took, with nothing to check it against. The drive encoders do it; the Pinpoint does it better. [L120](#)

Pinpoint

The odometry computer. It has its own two measuring wheels, and it works out the pose. The follower believes this one. L195 measures against it. [L115](#)

Path

A line or curve the follower drives along, with a heading to hold while it does. [L125](#)

Hold

Telling the follower to sit at one pose and stay there. If something pushes the robot, the follower puts it back. [L130](#)

Field relative

Sticks that mean the field. Push away from you and the robot moves away from you, whichever way its nose points. [L140](#)

Robot relative

Sticks that mean the robot's own directions. Better for lining up against something. [L150](#)

Heading hold

Keeping the nose where the driver left it, and giving it straight back the moment the turn stick moves. [L160](#)

Instant command

A command that does its work and finishes in the same breath, rather than running for a while. A button press is usually one. [L170](#)

Feedforward

A guess at the power a wanted speed needs, made before anything is measured. Gets most of the way there at once. [L190](#)

Feedback

A correction worked out from two speeds: the one asked for and the one measured. It cleans up what the guess got wrong. [L190](#)

Ticks

What an encoder counts as a motor turns. A tick is a fixed fraction of a turn, so ticks become inches once you know how many make one. [L195](#)

Ticks per inch

How many ticks a wheel counts for an inch of travel. Measured, not looked up. [L195](#)

Turn radius

How far a wheel sits from the middle of the robot, along the diagonal it pushes. It turns radians per second into inches per second at the wheel. [L195](#)

Coast and brake

What a motor does at zero power. Braking holds the robot still; coasting lets the wheels roll, which is what a measurement you push by hand needs. [L195](#)

2.24 Cheat sheet

Every table here is read straight out of the lesson code, so it says what the code says.

2.24.1 What to run

One OpMode name for the Driver Station, one command for the laptop. Each command runs only that lesson's own tests, which is what makes a red one easy to read. A dash means no test names that lesson: its check is on the floor.

Lesson	On the robot	On the laptop
L020	L020 Sticks (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1020*'</code>
L030	L030 Speeds (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1030*'</code>
L040	L040 Tank (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1040*'</code>
L050	L050 Blocks (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1050*'</code>
L060	L060 Shaping (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1060*'</code>
L070	L070 Buttons (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1070*'</code>
L080	L080 Arcade (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1080*'</code>
L090	L090 Mecanum (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1090*'</code>
L110	L110 Follower (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1110*'</code>
L120	L120 Compare Localizers (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1120_theEncoderLocalizerRunsAlongsideAndIsLogged'</code> <code>--tests '*L120EncoderLocalizerTest*'</code>
L125	L125 Drive 24 (Autonomous)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1125*'</code>
L130	L130 Forward Then Strafe (Autonomous)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1130*'</code>

L140	L140 Field Relative (TeleOp)	./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l140*'
L150	L150 Robot Relative Button (TeleOp)	./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l150*'
L160	L160 Heading Hold (TeleOp)	./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l160*'
L170	L170 Drive To Pose (TeleOp)	./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l170*'
L180	L180 Combined (TeleOp)	./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l180*'
L190	L190 Velocity Drive (TeleOp)	./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l190*'
L195	L195 Measure ticks per inch (TeleOp) L195 Measure turn radius (TeleOp)	./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l195*'

2.24.2 What shows up in Panels

A key ending in ... has a name added to the end of it, one per wheel or per localizer.

Log key	Published by
Localizer/driveWheelEncoders /...	L120CompareLocalizersOpMode, L180CombinedOpMode
arcade/forward	L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode

arcade/strafe	L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode
arcade/turn	L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode
button/a	L070ButtonsOpMode, L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode
button/event	L070ButtonsOpMode, L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode
command/forward_ips	L190VelocityDriveOpMode
command/left_ips	L190VelocityDriveOpMode
command/turn_radps	L190VelocityDriveOpMode
drive/mode	L170DriveToPoseOpMode, L180CombinedOpMode
drive/robotRelative	L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode
heading/holding	L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode
heading/target_deg	L180CombinedOpMode
measure/inches	L195MeasureTicksPerInchOpMode
measure/radians	L195MeasureTurnRadiusOpMode
measure/ticks	L195MeasureTicksPerInchOpMode
measure/ticksPerInch	L195MeasureTicksPerInchOpMode
measure/turnRadius_in	L195MeasureTurnRadiusOpMode
measure/wheel_inches	L195MeasureTurnRadiusOpMode
power/backLeft	L090MecanumOpMode
power/backRight	L090MecanumOpMode
power/frontLeft	L090MecanumOpMode
power/frontRight	L090MecanumOpMode
power/left	L060ShapingOpMode, L070ButtonsOpMode, L080ArcadeOpMode
power/right	L060ShapingOpMode, L070ButtonsOpMode, L080ArcadeOpMode

speed/left	L030SpeedsOpMode, L040TankOpMode, L050BlocksOpMode, L060ShapingOpMode, L070ButtonsOpMode
speed/right	L030SpeedsOpMode, L040TankOpMode, L050BlocksOpMode, L060ShapingOpMode, L070ButtonsOpMode
stick/leftX	L070ButtonsOpMode, L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode, L190VelocityDriveOpMode
stick/leftY	L020SticksOpMode, L030SpeedsOpMode, L040TankOpMode, L050BlocksOpMode, L060ShapingOpMode, L070ButtonsOpMode, L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode, L190VelocityDriveOpMode
stick/rightX	L070ButtonsOpMode, L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode, L190VelocityDriveOpMode
stick/rightY	L020SticksOpMode, L030SpeedsOpMode, L040TankOpMode, L050BlocksOpMode, L060ShapingOpMode, L070ButtonsOpMode, L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode

Generated by applying solutions/ on solutions-1010, every lesson in order.

2.25 Where this came from

2.25.1 The guide's own terms

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The repository's own `LICENSE` is FIRST's notice over the FTC SDK. It covers the code, not these pages.

2.25.2 Game Manual 0

[Game Manual 0](#) is licensed [CC BY-NC 2.0 Generic](#). It is the best general FTC reference there is. This guide is not trying to replace it. These pages are about one robot and one set of lessons.

One page leans on it. [L090](#) explains why the mecanum rollers make an X, and links GM0's page on holonomic drivetrains, which shows it. No page here copies GM0's words. The guide's license is chosen to keep every restriction GM0's asks for anyway.

2.25.3 Pedro Pathing

[Pedro Pathing](#) is the path following library the lessons hand the drivetrain to. It is licensed BSD 3-Clause, and its name and logo are its own. This guide links Pedro's documentation rather than restating it, and the code the lessons write is ours.

2.25.4 The FTC SDK

The **FTC SDK** is FIRST's, under the BSD-style notice in the repository's root `LICENSE`. The Driver Station, the Control Hub and every `@TeleOp` in this guide are its.

2.25.5 AdvantageScope

AdvantageScope, from the Mechanical Advantage team, is what opens a flight log after a run.

2.25.6 Not affiliated with FIRST

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2.26 Answers

What each lesson changes, file by file, out of the solutions line. Use it when you are stuck, not instead of being stuck.

2.26.1 L020SticksOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying `solutions/` on `solutions-l010`:

```
TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry  
/L020SticksOpMode.java
```

L020

A change

Before L020:

```
(nothing)
```

After L020:

```
import org.firstinspires.ftc.teamcode.base.Tracker;
```

A change

Before L020:

(nothing)

After L020:

`double leftStickY = gamepad1.left_stick_y;`

A change

Before L020:

(nothing)

After L020:

`Tracker.publish("stick/leftY", leftStickY);`

A change

Before L020:

(nothing)

After L020:

`double rightStickY = gamepad1.right_stick_y;
Tracker.publish("stick/rightY", rightStickY);`

2.26.2 L030SpeedsOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L030SpeedsOpMode.java

L030

A change

Before L030:

(nothing)

After L030:

`double leftSpeed = -leftStickY;
Tracker.publish("speed/left", leftSpeed);`

A change

Before L030:

(nothing)

After L030:

`double rightSpeed = -rightStickY;
Tracker.publish("speed/right", rightSpeed);`

2.26.3 L040TankOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L040TankOpMode.java

L040*A change*

Before L040:

(nothing)

After L040:

```
hardware.frontLeft.setPower(leftSpeed);
hardware.backLeft.setPower(leftSpeed);
```

A change

Before L040:

(nothing)

After L040:

```
hardware.frontRight.setPower(rightSpeed);
hardware.backRight.setPower(rightSpeed);
```

2.26.4 L050BlocksOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

```
TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L050BlocksOpMode.java
```

L050*A change*

Before L050:

(nothing)

After L050:

```
hardware.frontLeft.setDirection(hardware.mecanumConfig.frontLeftDirection.get());
hardware.frontRight.setDirection(hardware.mecanumConfig.frontRightDirection.get());
hardware.backLeft.setDirection(hardware.mecanumConfig.backLeftDirection.get());
hardware.backRight.setDirection(hardware.mecanumConfig.backRightDirection.get());
```

2.26.5 L060ShapingOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

```
TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L060ShapingOpMode.java
```

L060*A change*

Before L060:

(nothing)

After L060:

```
double band = 0.05;
double leftPower = squared(deadband(leftSpeed, band));
Tracker.publish("power/left", leftPower);
double rightPower = squared(deadband(rightSpeed, band));
Tracker.publish("power/right", rightPower);
```

A change

Before L060:

```
hardware.frontLeft.setPower(leftSpeed);
hardware.backLeft.setPower(leftSpeed);
```

After L060:

```
hardware.frontLeft.setPower(leftPower);
hardware.backLeft.setPower(leftPower);
```

A change

Before L060:

```
hardware.frontRight.setPower(rightSpeed);
hardware.backRight.setPower(rightSpeed);
```

After L060:

```
hardware.frontRight.setPower(rightPower);
hardware.backRight.setPower(rightPower);
```

A change

Before L060:

(nothing)

After L060:

```
double deadband(double value, double band) {
    if (Math.abs(value) < band) {
        return 0;
    }
    return value;
}
```

A change

Before L060:

(nothing)

After L060:

```
double squared(double value) {
    return value * Math.abs(value);
}
```

2.26.6 L070ButtonsOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

```
TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L070ButtonsOpMode.java
```

L070*A change*

Before L070:

```
(nothing)
```

After L070:

```
boolean previousButtonA;
```

A change

Before L070:

```
(nothing)
```

After L070:

```
double leftStickX = gamepad1.left_stick_x;
Tracker.publish("stick/leftX", leftStickX);
double rightStickX = gamepad1.right_stick_x;
Tracker.publish("stick/rightX", rightStickX);
```

A change

Before L070:

```
(nothing)
```

After L070:

```
boolean buttonA = gamepad1.a;
Tracker.publish("button/a", buttonA);
```

A change

Before L070:

```
(nothing)
```

After L070:

```
if (buttonA != previousButtonA) {
    if (buttonA) {
        Tracker.publish("button/event", "A pressed");
    } else {
        Tracker.publish("button/event", "A released");
    }
}
previousButtonA = buttonA;
```

2.26.7 L080ArcadeOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

```
TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L080ArcadeOpMode.java
```

L080*A change*

Before L080:

```
(nothing)
```

After L080:

```
double band = 0.05;
double forward = squared(deadband(-leftStickY, band));
Tracker.publish("arcade/forward", forward);
double turn = squared(deadband(-rightStickX, band));
Tracker.publish("arcade/turn", turn);
```

A change

Before L080:

```
(nothing)
```

After L080:

```
double leftPower = forward - turn;
double rightPower = forward + turn;
double biggest = Math.max(1, Math.max(Math.abs(leftPower),
Math.abs(rightPower)));
leftPower = leftPower / biggest;
rightPower = rightPower / biggest;
Tracker.publish("power/left", leftPower);
Tracker.publish("power/right", rightPower);

hardware.frontLeft.setPower(leftPower);
hardware.backLeft.setPower(leftPower);
hardware.frontRight.setPower(rightPower);
hardware.backRight.setPower(rightPower);
```

2.26.8 L090MecanumOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

```
TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L090MecanumOpMode.java
```

L090

A change

Before L090:

```
(nothing)
```

After L090:

```
double strafe = squared(deadband(-leftStickX, band));
Tracker.publish("arcade/strafe", strafe);
```

A change

Before L090:

```
(nothing)
```

After L090:

```
double frontLeftPower = forward - strafe - turn;
double frontRightPower = forward + strafe + turn;
double backLeftPower = forward + strafe - turn;
double backRightPower = forward - strafe + turn;
double biggest = Math.max(1, Math.max(Math.max(Math.abs(frontLeftPower),
Math.abs(frontRightPower)),
Math.max(Math.abs(backLeftPower), Math.abs(backRightPower))));
```

```
frontLeftPower = frontLeftPower / biggest;
frontRightPower = frontRightPower / biggest;
backLeftPower = backLeftPower / biggest;
backRightPower = backRightPower / biggest;
Tracker.publish("power/frontLeft", frontLeftPower);
Tracker.publish("power/frontRight", frontRightPower);
Tracker.publish("power/backLeft", backLeftPower);
Tracker.publish("power/backRight", backRightPower);

hardware.frontLeft.setPower(frontLeftPower);
hardware.frontRight.setPower(frontRightPower);
hardware.backLeft.setPower(backLeftPower);
hardware.backRight.setPower(backRightPower);
```

2.26.9 L110FollowerDriveTrain

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L110FollowerDriveTrain.java

L110

A change

Before L110:

```
(nothing)
```

After L110:

```
wheels[FL] = forward - strafe - turn;
wheels[FR] = forward + strafe + turn;
wheels[BL] = forward + strafe - turn;
wheels[BR] = forward - strafe + turn;
```

2.26.10 L110FollowerOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L110FollowerOpMode.java

L110

A change

Before L110:

```
(nothing)
```

After L110:

```
L110FollowerDriveTrain drivetrain;
```

A change

Before L110:

```
(nothing)
```

After L110:

```
drivetrain = new L110FollowerDriveTrain(hardware);
```

A change

Before L110:

`initAfter();`

After L110:

`initAfter(drivetrain);`*A change*

Before L110:

`(nothing)`

After L110:

`drivetrain.sticks(forward, strafe, turn);`

2.26.11 L120CompareLocalizersOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L120CompareLocalizersOpMode.java

L120

A change

Before L120:

`(nothing)`

After L120:

`import org.firstinspires.ftc.teamcode.base.odometry.HardwareWheelSource;`*A change*

Before L120:

`(nothing)`

After L120:

`shadowLocalizers.add("driveWheelEncoders",
new L120EncoderLocalizer(new HardwareWheelSource(hardware)));`

2.26.12 L120EncoderLocalizer

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L120EncoderLocalizer.java

L120

A change

Before L120:

`(nothing)`

After L120:

```
if (lastWheels == null) {
    lastWheels = wheels.clone();
    lastRawHeading = rawHeading;
    return;
}

double frontLeft = wheels[0] - lastWheels[0];
double frontRight = wheels[1] - lastWheels[1];
double backLeft = wheels[2] - lastWheels[2];
double backRight = wheels[3] - lastWheels[3];

double forward = (frontLeft + frontRight + backLeft + backRight) / 4;
double strafe = (-frontLeft + frontRight + backLeft - backRight) / 4;
```

A change

Before L120:

```
(nothing)
```

After L120:

```
double heading = rawHeading + headingOffset;
x += forward * Math.cos(heading) - strafe * Math.sin(heading);
y += forward * Math.sin(heading) + strafe * Math.cos(heading);
```

2.26.13 L125Drive24OpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

```
TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L125Drive24OpMode.java
```

L125

A change

Before L125:

```
return Command.NOOP;
```

After L125:

```
return sequential(follow(follower, line(start, end).constant(start)));
```

2.26.14 L130ForwardThenStrafeOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

```
TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L130ForwardThenStrafeOpMode.java
```

L130

A change

Before L130:

```
(nothing)
```

After L130:

```
import static com.pedropathing.ivy.pedro.PedroCommands.hold;
```

A change

Before L130:

```
follow(follower, line(start, corner).constant(start))
```

After L130:

```
follow(follower, line(start, corner).constant(start)),
```

A change

Before L130:

```
(nothing)
```

After L130:

```
follow(follower, line(corner, end).constant(start)),
hold(follower, end)
```

2.26.15 L140FieldRelativeOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L140FieldRelativeOpMode.java

L140*A change*

Before L140:

```
drivetrain.sticks(forward, strafe, turn);
```

After L140:

```
double heading = follower.pose().heading();
double robotForward = forward * Math.cos(heading) + strafe *
Math.sin(heading);
double robotStrafe = -forward * Math.sin(heading) + strafe *
Math.cos(heading);
drivetrain.sticks(robotForward, robotStrafe, turn);
```

2.26.16 L150RobotRelativeButtonOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L150RobotRelativeButtonOpMode.java

L150*A change*

Before L150:

```
drivetrain.sticks(robotForward, robotStrafe, turn);
```

After L150:

```
boolean robotRelative = gamepad1.right_bumper;
Tracker.publish("drive/robotRelative", robotRelative);
if (robotRelative) {
```

```
        drivetrain.sticks(forward, strafe, turn);
    } else {
        drivetrain.sticks(robotForward, robotStrafe, turn);
    }
}
```

2.26.17 L160HeadingHoldOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L160HeadingHoldOpMode.java

L160

A change

Before L160:

```
(nothing)
```

After L160:

```
import org.firstinspires.ftc.teamcode.base.HeadingHold;
```

A change

Before L160:

```
(nothing)
```

After L160:

```
import org.firstinspires.ftc.teamcode.pedro.Constants;
```

A change

Before L160:

```
(nothing)
```

After L160:

```
    HeadingHold hold = new
HeadingHold(Constants.foresightConfig.headingFeedback.get());
```

A change

Before L160:

```
(nothing)
```

After L160:

```
    turn = hold.turn(follower, turn);
    Tracker.publish("heading/holding", hold.target() != null);
```

2.26.18 L170DriveToPoseOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L170DriveToPoseOpMode.java

L170*A change*

Before L170:

(nothing)

After L170:

```
import com.pedropathing.ivy.commands.Commands;
```

A change

Before L170:

(nothing)

After L170:

```
buttons.whenPressed(() -> gamepad1.y, Commands.instant(() -> {
    follower.hold(target);
    drivingItself = true;
}));
```

A change

Before L170:

(nothing)

After L170:

```
if (drivingItself) {
    if (!driverWantsControl) {
        Tracker.publish("drive/mode", "AUTO");
        drivetrain.releaseCommandedWheels();
        return;
    }
    drivingItself = false;
    follower.manual(0, 0, 0);
}
Tracker.publish("drive/mode", "DRIVER");
```

2.26.19 L180CombinedOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-1010:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L180CombinedOpMode.java

L180*A change*

Before L180:

(nothing)

After L180:

```
import org.firstinspires.ftc.teamcode.base.odometry.HardwareWheelSource;
```

A change

Before L180:

(nothing)

After L180:

```
shadowLocalizers.add("driveWheelEncoders",
    new L120EncoderLocalizer(new HardwareWheelSource(hardware)));
```

2.26.20 L190SpeedDriveTrain

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L190SpeedDriveTrain.java

L190

A change

Before L190:

(nothing)

After L190:

```
import org.firstinspires.ftc.teamcode.pedro.Constants;
```

A change

Before L190:

(nothing)

After L190:

```
double[] measured = measuredSpeeds.all();
double[] powers = new double[4];
for (int i = 0; i < 4; i++) {
    double feedforward = Constants.powerPerInchPerSecond * wanted[i];
    double feedback = VELOCITY_KP * (wanted[i] - measured[i]);
    powers[i] = clampToPower(feedforward + feedback);
}
setCommandedWheels(powers[FL], powers[FR], powers[BL], powers[BR]);
publishWheelSpeeds(wanted, measured, powers);
```

2.26.21 L190VelocityDriveOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L190VelocityDriveOpMode.java

L190

A change

Before L190:

(nothing)

After L190:

```
import org.firstinspires.ftc.teamcode.base.WheelTargets;
import org.firstinspires.ftc.teamcode.pedro.Constants;
```

A change

Before L190:

```
(nothing)
```

After L190:

```
L190SpeedDriveTrain drivetrain;
```

A change

Before L190:

```
(nothing)
```

After L190:

```
drivetrain = new L190SpeedDriveTrain(hardware);
```

A change

Before L190:

```
initAfter();
```

After L190:

```
initAfter(drivetrain);
```

A change

Before L190:

```
double forwardInPerS = 0;
double strafeInPerS = 0;
double turnRadPerS = 0;
```

After L190:

```
double forwardInPerS = deadband(-leftStickY, band) * MAX_IPS;
double strafeInPerS = deadband(-leftStickX, band) * MAX_IPS;
double turnRadPerS = deadband(-rightStickX, band) * MAX_TURN_RADPS;
```

A change

Before L190:

```
(nothing)
```

After L190:

```
double[] target = WheelTargets.forMecanum(forwardInPerS, strafeInPerS,
turnRadPerS, Constants.turnRadiusInches);
drivetrain.setCommandedWheelSpeeds(target[0], target[1], target[2],
target[3]);
```

2.26.22 L195TicksDriveTrain

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-1010:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L195TicksDriveTrain.java

L195

A change

Before L195:

```
(nothing)
```

After L195:

```
ticks[FL] = hardware.frontLeft.getCurrentPosition();  
ticks[FR] = hardware.frontRight.getCurrentPosition();  
ticks[BL] = hardware.backLeft.getCurrentPosition();  
ticks[BR] = hardware.backRight.getCurrentPosition();
```